

10:10:01, A Decision Science and ML Framework to navigate the Ride-Hailing dynamics of Mexico City

From Field Acquisition to Generative Simulation

Bernardo Lozano Wise

March 31, 2026

Abstract

This document serves as the chronological laboratory report and technical foundation for Project Pienza. It documents the evolution from $N = 1$ ground-truth data acquisition through rigorous data engineering, exploratory analysis, and supervised cloning, culminating in the deployment of Generative Adversarial Networks (GANs) for market simulation.

Disclaimer: Project **Pienza** and all associated materials, including this document and its analyses, are the product of an independent academic and research initiative. This work is intended solely for demonstration as a data science portfolio project. All source data has undergone a multi-layered anonymization protocol to protect privacy and obfuscate proprietary information. The content, models, and insights presented herein are for illustrative and educational purposes and are not intended for, nor should they be used for, any commercial, financial, or operational business activities.

Keywords: Behavioral Economics, Imitation Learning, XGBoost, Generative Adversarial Networks (GANs), Stochastic Modeling, Data Engineering, Optimal Stopping, Geospatial Telemetry, El semáforo de Palmas.

Contents

Project Strategy & Scope	5
Mission Statement	5
Scope & Constraints	5
Core Framework	5
Initial Hypothesis: The Payout Spread	5
Strategic Pivot to Classification	6
Project Roadmap	6
A Note on System Architecture & Evolution	6
Phase 1: Acquisition and Ground Truth	9
Engine 2: The “Total Offer” Stream (OCR Pipeline)	9
The Post-Session Reconciliation Protocol	10
Phase 1B: Data Staging and First Round of Feature Engineering	10
Core Entity: The <code>raw_offers</code> Schema	10
Target Engineering: <code>reason_primary</code>	11
Contextual Feature Extraction: <code>heuristic_flag</code>	11
Behavioral Targets and Operational Outcomes	12
Driver State Inference	12
Geospatial Enrichment	13
Acquisition Artifact: The Enriched <code>offers</code> Dataset	13

Phase 2: Data Engineering & Architecture	15
Phase 2A: Contextual Feature Synthesis	15
Incentive Deconstruction	15
Regex Parsing Architecture	15
Generative Variability and Remediation	15
Technical Retrospective	16
Chronology Features: Modeling Shift Progression	16
The Offer Window Constraint	16
Trajectory Intelligence: Localization Trials	16
Trial 1: External Telemetry (Google Location History)	16
Trial 2: Geometric Inference (Trilateration)	17
Trial 3: Linear Temporal Interpolation (Phase 2 Baseline)	17
Trial 4: Home Vector Alignment (Strategic Extension)	17
High-Precision Temporal Recovery	17
The Canonical Version of the offers Table	17
Phase 2B: Geospatial Bottleneck	18
Phase 2C: Stateful Intelligence	20
The Canonical Stateful Features	21
Phase 2D: Schema Design and ETL Pipeline	21
Integration of Official Ledgers	21
The Reconciliation Audit (Outer Join Logic)	22
Schema Design: The Star Schema Transition	22
Normalization Strategy & Relational Logic	23
Circular Avoidance: The Linear Financial Chain	23
The Idempotent ETL Pipeline	23
Tabula Rasa	23
The 2-Engine "Golden Link"	23
The SQL View Architecture	24
Computational Boundaries	24
Persistence & Cloud Synchronization	24
Phase 3: Exploratory Analysis & Causal Inference	27
Operational Baseline: Data Census	27
The Profitability Funnel	27
Traffic Index	28
The Physics of Selection	29
The Market Quality Index	29
Market Heterogeneity and Volatility	30
The Opportunity Oracle: Time as a Financial Derivative	31
Search Stream Sanitization (The Behavioral Clock)	31
The Strategic Quadrant Matrix	31
The Efficient Frontier (Optimal Stopping)	32
Net Expected Value (VEN) Analysis	33
The Efficient Frontier: Search Budget Decay	33
Phase 4: The Geometry of Chaos	42
Market Manifold Discovery: PCA	42
Yield Analysis Across Operational Tiers	47
Geo-spatial Clustering: HDBSCAN	49
Geospatial Constraints: The Dropoff Imperative	49
The Initial Clustering Attempt & The Coordinate Crisis	49
Geospatial Remediation: Root Cause Analysis	49
Diagnostic Engineering	50
Emergent Topologies	51

Final Feature Synthesis: The Silver Palette	53
Stateful Intelligence: The Volatility Suite	53
Phase 5 - Supervised Imitation Learning: From Sky to Psyche	56
Experimental Design: The Tri-League Architecture	56
Geospatial Feature Coalescence	56
Categorical Feature Selection	56
Algorithmic Justification: From First Principles to Ensemble Apex	58
Champion Selection: XGBoost	58
Hyperparameter Calibration: The Down-Tuning Strategy	58
The “State vs. Time” Dilemma	59
Model Performance: Bimodal Learning	60
Regime 1: Deterministic Operational Constraints	60
Regime 2: Conditional Strategic Nuance	61
The Gravitational Well Effect	61
Architectural Pivot: The Cognitive Cascade	61
Layer 1: The Balanced Bouncer Protocol	61
The Threshold Paradox: Street vs. Code	61
Priority-Based Decision Logic	61
Results: Filter Efficiency	62
Layer 2: The Strategist (Nuance Modeling)	63
Results: Strategic Performance Lift	64
Threshold Tuning and the Pareto Frontier	64
Strategic DNA: Feature Drivers for Layer 2	64
The Shadow Mode Trial: Behavioral Fidelity	66
Phase 5 Conclusion: Breaking the Static Barrier	67
Phase 6: Generative Moonshots	69
The Quest to $O(1)$ Inference: NLP Transformers	69
Linguistic Feature Engineering	70
Model Architectures: Custom vs. Pre-trained	71
Optimization and Training Constraints	71
Performance Benchmarking: Precision vs. Complexity	71
Generative Synthesis: The Pienza Manifold	72
Neural Architecture: The Generator Pipeline	73
Discriminator Architecture: The Adversarial Validator	74
Training Strategy and Optimization	75
Integrity Audit: Real vs Synth	76
Statistical Validation: KS Test and JS Divergence	79
Relational Integrity: Correlation Delta Analysis	80
Functional Validation: Predictive Parity (TRTR vs. TSTR)	81
Data Engineering 2.0: Cloud Infrastructure Migration	81
Network Graph Analysis: A Sandbox for Autonomous Vehicles	82
Base Topology: The Geofence Blueprint	83
Morphological Analysis: Small-World Dynamics	85
Centrality Assessment: The Operational Hierarchy	85
Vulnerability Analysis: Choke Points	86
The Mobility Tensor: Directed Flow and State Space	86
Market Capture and the Decoupling Index	87
Functional Morphology and Topological Classification	88
Micro-Dynamics of Capital: Edge Centrality and Vector Analysis	88
Hierarchical Trajectory Audit	89
Analysis of Vector Forces	89
Structural Resilience and Functional Robustness	89

Comparative Vulnerability Audit	89
Topological Robustness: The K-Connectivity Metric	90
Macro-Morphology and the Autocentric Singularity	90
Market Asymmetry: The Topological Trade Balance	90
Operational Implications of Flow Asymmetry	91
Stochastic Scaffolding: The Bridge to Markov	91
Structural Stability: Absorbing States and Total Retention	91
Strategic Valuation: The State-Value Function $V(s)$	92
Equilibrium and the Topological Premium	93
Operational Command: The Action-Value Matrix $Q(s, a)$	94
Multi-Agent Coordination: The Fleet Orchestrator	94
Demand Depletion and Swarm Neutralization	94
Tactical Deployment: The Five-Jump Mission Manifold	94
¿Será Posible el Sur?	96
References	97

Project Strategy & Scope

Mission Statement

The primary objective of **Project Pienza** is to model and optimize the decision-making policy of an expert agent within the ride-hailing ecosystem. The environment is formally defined as a *sequential, non-cooperative game* governed by intrinsic information asymmetry.

Unlike traditional "top-down" studies relying on aggregated data, Pienza employs a **"Bottom-Up"** methodology grounded in the operational reality of Mexico City. Through the longitudinal quantification of a single Expert Agent ($N = 1$), this research models the micro-economic physics of the Gig Economy.

Scope & Constraints

The project explicitly rejects the premise of reverse-engineering the proprietary pricing algorithms of ride-hailing platforms. Modeling a dynamic, global algorithm using a local boutique dataset is statistically unfeasible.

Instead, Pienza reorients the analytical lens—shifting away from the fare prediction models common in public repositories—to focus on the sole variable under absolute control: **The Agent's Decision**. By pivoting to Behavioral Economics and Decision Science, the project transforms a complex prediction task into a solvable policy optimization endeavor.

Core Framework

The project is architected across three strategic pillars that bridge the gap between tactical fieldwork and scalable autonomous intelligence:

1. **High-Fidelity Environment Digitization (The Foundation):** To overcome the *Survivorship Bias* inherent in ride-hailing data, a custom dual-engine ingestion pipeline was deployed to capture the **Total Offer Stream**. By utilizing OCR-based archival and geospatial triangulation, the system digitizes the "Negative Class" (rejected offers), providing the necessary ground truth to map the agent's exact decision boundary.
2. **Hierarchical Policy Cloning (The Engine):** The core analytical breakthrough involves the transition from monolithic modeling to a **Cognitive Cascade**. This hierarchical inference architecture utilizes XGBoost to replicate the agent's multi-tiered decision logic—separating deterministic operational "triage" (geospatial/economic filters) from nuanced opportunity-cost "gambles".
3. **Generative Synthesis & Topological Scaling (The Frontier):** The final pillar transforms the single-agent study into a scale-invariant simulation. Using **Conditional GANs (cGAN)**, the project synthesizes a 1,000,000-row data manifold that preserves the non-linear physics of the marketplace as encountered during the observation window.¹ This environment serves as the sandbox for **Network Graph Analysis**, where the city is modeled as a *Tensor of Potentials* to orchestrate autonomous fleet routing and maximize steady-state value.

Initial Hypothesis: The Payout Spread

Research began with a longitudinal field observation regarding fare elasticity. The hypothesis stated that the system applied an implicit efficiency penalty, observed as a discrepancy between the promised fare and realized earnings. A preliminary pilot study ($N \approx 150$, July-August 2025 cohort) revealed a **Payout Rate (Spread)** between 75% and 85% of the Base Fare. The distribution followed session-based moving averages, suggesting intra-block temporal stability driven by market conditions.

An initial model correlated this Spread with the **Time Delta** (the discrepancy between estimated and realized trip time). The goal was to validate whether time savings generated a fare penalty.

¹The synthetic marketplace functions as an infinite Keras generator of the specific market regime and "lived experience" captured in the dataset, rather than a generalizable, dynamic model of total city-wide liquidity.

Key Finding: A benchmark comparison showed that Simple Linear Regression outperformed base Machine Learning models. No significant non-linear relationship was identified; the correlation remained weak. While basic statistics resolved the regression problem for the pilot sample, scaling this to a robust ML model would require thousands of completed trips, representing an infeasible operational time cost for marginal analytical return.

Strategic Pivot to Classification

Based on logistical constraints and the low predictive value of the initial regression approach, the research objective was redefined:

- **From:** Price variance prediction (Regression), which required scarce completed trip data.
- **To:** Behavior modeling (Classification), which utilized the total offer stream.

This shift resolved the data scarcity issue by enabling the inclusion of the **Negative Class** (rejected offers). By modeling decision policy rather than price, the project applies high-dimensional Machine Learning (XGBoost) to capture the agent's non-linear decision boundary.

Project Roadmap

The project followed an iterative development lifecycle where feature engineering evolved in response to the analytical findings of each stage:

Phase 1: Acquisition & Ground Truth (Aug 22 – Oct 1, 2025)

High-fidelity manual capture of operational data via the bespoke *GTS Webapp*. Establishment of the target variable and the primary contextual ride attributes.

Phase 2: Data Engineering & Architecture (Oct 2 – Nov 20, 2025)

Transition to a relational database (`pienza.db`) and Star Schema design. Implementation of automated OCR pipelines and normalization protocols. Creation of the statefull `engineered_features` table.

Phase 3: Exploratory Analysis & Causal Inference (Nov 21 – Dec 3, 2025)

Diagnostic audit of marketplace physics and rational search time boundaries. Causal modeling of baseline heteroscedasticity and quantification of the platform's inelastic 'Integrity Buffer.'

Phase 4: Unsupervised Learning & Geo-Remediation (Dec 4, 2025 – Jan 2, 2026)

Application of HDBSCAN for zone discovery and surgical coordinate cleaning using hand-crafted heuristic polygons. Generation of the **Silver Palette** (geo-semantic attributes).

Phase 5: Supervised Imitation Learning (Jan 3 – Jan 12, 2026)

Model evaluation tournament and implementation of the *Cognitive Cascade* hierarchical architecture for the champion model.

Phase 6: Generative Moonshots (Jan 13, 2026 – March 31, 2026)²

$O(1)$ neural spatial inference and cGAN manifold synthesis. Formulation of the internal Mobility Tensor to establish the Markov Decision Process (MDP) baseline for autonomous routing.

A Note on System Architecture & Evolution

The rigorous architecture of the data ecosystem was established as a foundational objective. The infrastructure evolved through distinct operational states to match the complexity of the inquiry:

- **Phase 1 (Acquisition Staging):** Data ingestion from Engines 1 & 2 routed into *Google Sheets* for initial validation.

²The extended duration of this phase reflects a non-linear, iterative workflow; it was executed concurrently with the formal authoring of this manuscript, culminating in the simultaneous completion of Phase 6 and the first comprehensive draft of this paper.

- **Phase 2 (The Engineering Core):** Transition to a relational architecture. Execution of the idempotent ETL pipeline to forge the definitive **SQLite** asset (`pienza.db`).
- **Phases 3–5 (The Analytical Loop):** A file-based cloud architecture where Python Notebooks interacted with `pienza.db` as the immutable Single Source of Truth.
- **Phase 6 (Cloud Scaling):** Migration to **Google BigQuery** to support Generative AI workflows.
 - **pienza_mini:** A high-fidelity cloud replica of the ground-truth database, preserving the identical relational logic.
 - **pienza_big:** A 1,000,000-row synthetic dataset generated via GANs, persisted as Parquet artifacts and queried through BigQuery’s external table interface.
- **Interactive Deployment (Current State):** The project concludes with the transition from notebook-based exploration to an **Interactive White Paper** hosted on Streamlit. Leveraging *GitHub Codespaces* and an *orphan branch architecture*, this layer decouples the core research codebase from the operational interface, allowing for real-time model interrogation in a production-ready environment.

PHASE 1

Acquisition and Ground Truth

Tech Stack:	PWA (JS/Netlify), Google Apps Script, Gemini Pro Vision.
Core Challenge:	Market opacity and pervasive survivorship bias in standard platform exports.
Key Insight:	Rejection data (the negative class) contains more predictive signal for policy modeling than acceptance data alone.
Operational Outcome:	The successful deployment of a dual-engine telemetry system, yielding a consolidated ground-truth ledger of roughly 4,700 total market interactions.

Phase 1: Acquisition and Ground Truth

Project Pienza models the decision policy of an expert agent rather than a system in a learning phase. Prior to data collection, the subject had completed 24 months of field operations, concluding the *Exploration* stage of the Reinforcement Learning cycle.

Utilizing the framework of Hopfield Networks as explored by Anil Ananthaswamy in *Why Machines Learn*,^[1] the agent’s decision policy is modeled as a “cooled” system settled into a stable local minimum (*an attractor state*). This equilibrium represents an optimization contingent on the agent’s specific constraints: geographic preferences, risk tolerance, and physical endurance. While a theoretical global optimum for city-wide revenue may exist in the state space, the captured dataset represents a phase of pure *exploitation*. By prioritizing established, low-variance heuristics over high-energy experimental maneuvers, the resulting dataset provides a high-fidelity record of a converged policy.

Phase 1A: The 2-Engine Ingestion Infrastructure

Project Pienza utilizes a proprietary, dual-engine acquisition ecosystem to overcome the data sparsity inherent in third-party platform exports. The primary acquisition campaign was executed over a strict 6-week observation window (August 22 – October 1, 2025), digitizing the agent’s operational reality in real-time.

Engine 1: The Operational Telemetry (GTS Webapp)

To establish the ground truth of completed missions, a bespoke Progressive Web Application (PWA) designated as the **Geotimestamps (GTS) Webapp** was deployed. The interface was optimized for low-cognitive-load fieldwork, functioning as a “One-Touch Timestamping” instrument while maintaining resilience through a production-grade stack (Netlify Frontend, Google Sheets Backend).

Lifecycle Mapping Protocol (T0–T4). The system logs the five critical state transitions of a gig economy mission with geospatial precision:

- **T0 (Search):** Idle state; actively seeking offers.
- **T1 (Acceptance):** Offer accepted; en route to pickup. (*Captures quoted upfront fare*).
- **T2 (Arrival):** At pickup location; waiting for passenger.
- **T3 (In-Progress):** Mission started.
- **T4 (Completion):** Mission finalized. (*Captures realized net fare*).

Technical Stack & Evolution. The system was engineered to maintain 100% capture fidelity through two critical iterations:

- **Stateful Persistence (v4.0):** To prevent data loss during network intermittency, the app implements a *Stateful LocalStorage Manager* and an *Offline-First Queue System*.
- **Asynchronous Geospatial Enrichment:** To overcome the geographic blindness of simple timestamps, a client-side pipeline orchestrates the `navigator.geolocation` API with reverse-geocoding services, enriching every state transition with high-precision coordinates prior to persistence.

Output (trip_events): A verified timeline of physical state transitions and realized financial outcomes ($N \approx 350$ completed trips).

Engine 2: The “Total Offer” Stream (OCR Pipeline)

To neutralize **Survivorship Bias**, the system must capture the decision boundary—specifically the Rejected Offers (*The Negative Class*). As market infrastructure provides no native access to historical offer logs, a custom “Optical State Archival” architecture was deployed.

The acquisition interface utilized an iOS device configured with an **Assistive Touch Macro**. This enabled a single-gesture capture protocol (The “Manual Cookie”) that simultaneously acknowledged the offer cognitively and secured the raw visual data digitally.

To ensure statistical representativeness, three operational mandates were enforced:

1. **Full Spectrum Capture (Zero Filters):** All system-level destination filters and product category filters (Premium/Mid-tier/X) were disabled. This ensured the captured stream represented the unfiltered "True Market" liquidity, not a curated subset.
2. **Safety-Driven Random Sampling:** Offers occurring during high-cognitive-load maneuvers or complex passenger interactions were intentionally skipped. This data loss is classified as **Missing Completely at Random (MCAR)**, introducing no systemic bias into the profitability models.
3. **The "Driver-First" Learning Curve:** The dataset acknowledges a specific "Warm-Up" period (Week 1) where operational cadence took precedence over data capture, resulting in a small cluster of "Ghost Offers" (Completed missions without source image artifacts). These were programmatically handled in the Engineering Phase.

The pipeline utilizes the **Google Gemini Pro Vision API** to transform visual assets into structured data. The primary challenge involved isolating the *Foreground Signal* (the offer card) from *Background Noise* (active navigation and status bar metadata). To ensure data integrity, the system prompt enforces strict scope limitations, instructing the model to ignore metadata outside the central offer card.

Engine 2 Output (offers): This high-fidelity dataset captures the total universe of opportunities, including the critical Negative Class ($N \approx 4,700$ total offers).

At this stage, both engines exist as separate, unlinked entities. In the subsequent engineering phase, these disparate records were unified into a SQL-queryable relational schema.

The Post-Session Reconciliation Protocol

To maintain data integrity and preserve the agent’s decision context, a rigorous post-session protocol was enforced immediately upon the completion of each fieldwork shift:

1. **Telemetry Consolidation (GTS):** The raw, long-format event logs from the webapp were processed into a wide-format session ledger. This involved pruning duplicate entries and cancelled events, enabling the immediate calculation of session-level KPIs such as *Net Spread* and *Accumulated Deadhead*, metrics which would become engineered features in later stages.
2. **High-Fidelity Cognitive Backtagging:** The Agent manually reviewed and tagged every rejected offer from that session to populate the multiclass target variable (`reason_primary`). Executing this task same-day was imperative to capture the specific, contextual nuance of the decision before operational memory decay occurred.

Phase 1B: Data Staging and First Round of Feature Engineering

Executed concurrently within the strict acquisition window, Phase 1B focuses on the immediate semantic structuring and target definition of the incoming datastream.

Core Entity: The `raw_offers` Schema

The ingestion pipeline consolidated OCR outputs into a primary staging table. The raw schema captured the unstructured state of the visual data prior to normalization.

Column	Data Type	Description
ocr_id	Integer (PK)	Incremental sequential identifier assigned during ingestion
image_filename	String	Source asset filename stored in the archival vault.
time_taken	String	Raw time string (HH:MM) extracted from the device status bar.
ride_type	Categorical	Product class (e.g., <i>X</i> , <i>mid-Tier</i> , <i>Premium</i>).
upfront_fare	Numeric	The nominal fare amount presented in the offer card.
pickup_details	String (Raw)	Combined text field containing ETA and distance to pickup.
pickup_address	String (Raw)	Unstructured pickup location text (street, POI).
trip_details	String (Raw)	Combined text field containing estimated trip duration and distance.
dropoff_address	String (Raw)	Unstructured destination location text.
rider_rating	String (Raw)	Combined text string with Passenger Rating and Trip Count.
special_note	Text (Blob)	Unstructured text field capturing all incentive tags (<i>Surge</i> , <i>Turbo+</i>) and status flags (<i>Reservation</i> , <i>Priority</i>).

Table 1: Definition of the `raw_offers` schema post-ingestion.

Target Engineering: `reason_primary`

The multiclass target variable, `reason_primary`, is a hierarchical filtration system formalized as the **Cognitive Cascade**. To ensure a clean signal for supervised learning, each offer is assigned a single, mutually exclusive label representing the entry-point failure across a three-tiered triage: geospatial feasibility (Tier 1), economic viability (Tier 2), and strategic alignment (Tier 3). This prevents label contamination by focusing the model on the primary decision driver. The definitive taxonomy is outlined in Table 2.

Class Label	Operational Logic
<code>dropoff_non_operational</code>	Destination lies within a pre-defined zone outside the operational area.
<code>dropoff_proxy</code>	Destination is outside the primary zone but acceptable if aligned with a homecoming vector toward <i>Anzures</i> .
<code>low_profitability</code>	Offer fails baseline EPH requirements relative to estimated duration.
<code>long_pickup_time</code>	Uncompensated pickup time exceeds tolerance; threshold relaxes during extreme gridlock.
<code>strategic_mismatch</code>	High-value offer rejected due to unfavorable routing context (e.g., <i>Santa Fe</i> → <i>Polanco</i> during Friday’s peak gridlock).
<code>expected_value_gamble</code>	Viable offer rejected based on the probabilistic expectation of a superior imminent event.
NULL (Implicit)	Absence of objection signals an accepted offer.

Table 2: Lexicon of the Decision Target Variable (`reason_primary`).

Contextual Feature Extraction: `heuristic_flag`

The following table defines the categorical flags engineered to capture qualitative operational risks and strategic shifts that are not detectable through raw numerical features alone. These labels provide the model with the "Agent’s perspective" on market conditions.

Category	Flag Label	Operational Logic
Risk Mitigation	deadhead_risk	Short-duration offers with a high probability of immediate return to an idle search state.
	long_ride_risk	Trips > 45 min where high traffic volatility creates risk of uncompensated time.
	dropoff_uncertain	Destinations with high geospatial entropy/ambiguity.
Strategic Intent	obj_end_session	Intent to terminate the work shift; prioritizes homecoming vector alignment over intrinsic EPH.
	friday_gridlock	Friday-specific regime; prioritizes egress from high-friction density sectors (<i>Polanco</i>).
System/Market	system_error	Physically impossible platform time-estimates (e.g., 20 min predicted time from Vistahermosa to Polanco during morning traffic).
	market_anomaly	Non-standard offer physics (e.g., operational outliers near National Holidays).
	protest_anomaly	External socio-political shocks (marches, road closures) disrupting standard market flow.

Table 3: Categorical Lexicon of Heuristic Flags for Model Enrichment.

Behavioral Targets and Operational Outcomes

To differentiate between agent intent and realized performance, the schema implements two distinct categorical dimensions: `offer_action` (Binary Intent) and `outcome` (Realized State).

Target	Class Label	Strategic Context
Intent	Accept / Reject	Alternative binary target.
Outcome	completed	Transaction successfully finalized.
	rider_canceled	External termination by passenger post-acceptance.
	driver_canceled	Agent-initiated termination due to post-acceptance friction.
	system_failure	Offers intended for acceptance but lost due to operational capture latency. These are strategically mapped as ACCEPTED for model training to reflect the agent’s true policy.
	NULL	Implicit state for rejected offers; no operational outcome generated.

Table 4: Mapping of Behavioral Intent and Operational Outcome categories.

Driver State Inference

The variable `driver_state_at_request` was engineered to classify the agent’s operational context at the moment of an offer. This feature distinguishes between "Open" offers (received while *Idle*) and "Chained" offers (received while *On-Trip*), which is a primary driver of the agent’s acceptance threshold.

The inference process followed an evolutionary path from manual verification to hybrid automation:

1. **Verification Phase (Manual):** During the initial collection weeks, state assignment was performed via human-in-the-loop validation during the daily back-tagging protocol. The agent cross-referenced OCR timestamps against mission logs to establish a ground-truth baseline.
2. **Automated Phase (SQL + Google Apps Script):** The final population of the state variable was executed through a two-step process. First, the ETL pipeline established the relational join between the `offers` and `trip_events` tables. Second, a Google Apps Script (`code.gs`) was deployed to programmatically assign the *On-Trip* state to any offer with a timestamp falling between a mission’s T1 (Acceptance) and T4 (Completion) boundaries.

Geospatial Enrichment

To enable spatial clustering, the raw address strings required conversion to numerical coordinates. A serverless pipeline was architected using the **Google Geocoding API** within the Google Apps Script environment.

While the pipeline successfully populated the coordinate columns, a critical validation step was omitted: **Reverse Geocoding Verification**. The system naively accepted the API's top result as ground truth without auditing for semantic drift or "centroid defaults" (e.g., the API returning the center of a city for an ambiguous street name). This assumption of validity would later necessitate a massive "Deep Remediation" campaign during Phase 4, underscoring the risk of trusting black-box APIs without a human-in-the-loop audit.

Acquisition Artifact: The Enriched offers Dataset

By the conclusion of the acquisition campaign, the raw OCR stream had been enriched with manual behavioral labels, initial geospatial coordinates, and parsed operational metrics. The following schema represents the state of the dataset prior to the deep Data Engineering phase.

Feature Name	Data Type	Categories / Definition
<i>Source: OCR Extraction & Initial Parsing</i>		
time_taken	String	Raw HH:MM timestamp from device.
product_tier	Categorical	<i>X, mid-Tier, Premium.</i>
upfront_fare	Numeric	Nominal fare value (MXN).
special_note	String	Unstructured text (<i>Surge, Turbo, Priority</i>).
star_rating	Numeric	Passenger rating (1.0–5.0 scale).
trip_count	Integer	Lifetime trips completed by passenger.
time_to_pickup	Numeric	Parsed duration to pickup (seconds).
dist_to_pickup	Numeric	Parsed distance to pickup (km).
est_trip_time	Numeric	Parsed estimated trip duration (seconds).
est_trip_dist	Numeric	Parsed estimated trip distance (km).
<i>Source: Cognitive Backtagging & Logic</i>		
offer_action	Binary	<i>Accept, Reject.</i>
reason_primary	Categorical	<i>dropoff_non_operational, low_profitability, long_pickup, strategic_mismatch, expected_value_gamble.</i>
heuristic_flag	Categorical	<i>deadhead_risk, long_ride_risk, dropoff_uncertain, obj_end_session, friday_gridlock, system_error, market_anomaly, protest_anomaly.</i>
driver_state	Categorical	<i>Idle (Open), On-Trip (Chained).</i>
outcome	Categorical	<i>Completed, Rider_Cancel, Driver_Cancel, System_Failure.</i>
<i>Source: Google Geocoding API</i>		
pickup_lat/lon	Float	Initial coordinate estimation.
dropoff_lat/lon	Float	Initial coordinate estimation.

Table 5: Final Schema State of the Acquisition Phase ($N \approx 4,700$).

PHASE 2

Data Engineering & Architecture

Tech Stack:	Python (Pandas, SciPy), SQLite, Google Apps Script, OSMnx, Advanced Regex, Red Gate Data Modeler (Vertabelo).
Core Challenge:	Taming the high-entropy chaos of generative OCR outputs and systemic geospatial corruption to build a structurally sound, temporally-firewalled database.
Key Insight:	Decisions do not exist in a vacuum. To accurately clone an agent's policy, the data must capture the "internal state" of the session—accumulated sunk costs, shift progression, and fatigue—without allowing future data to leak into past decisions.
Operational Outcome:	The deployment of a "Tabula Rasa" idempotent ETL pipeline, forging the definitive "Golden Link" between fragmented visual telemetry and official bank ledgers to crystallize the <code>pienza.db</code> Golden Master.

Phase 2: Data Engineering & Architecture

Timeline: October 2 – November 20, 2025

The mission of Phase 2 was to transmute raw inputs into the definitive `pienza.db` warehouse. While retrospectively structured as a logical progression, the execution followed a complex, non-linear trajectory defined by recursive discovery loops.

The initial architecture (Contextual Features → ETL → DB) was stress-tested by a preliminary EDA, which exposed critical latent flaws—specifically systemic geospatial corruption and the absence of state-dependent context. This necessitated a "Rollback and Refine" protocol: retreating from exploratory analysis to execute deep geospatial remediation (Phase 2B) and advanced contextual engineering (Phase 2C).

Only after these intelligence layers were stabilized was the final, structural consolidation executed (Phase 2D). This iterative campaign culminated in the deployment of an **Idempotent ETL Pipeline**, capable of regenerating the entire database from source in a single, reproducible run.

Phase 2A: Contextual Feature Synthesis

Having officially transitioned from the Acquisition into the Engineering Phase, the primary mandate was to continue extracting high-fidelity signal from the raw OCR logs. These represent the second set of contextual features and are integrated directly into the `offers` table, completing the static snapshot of each decision event.

Incentive Deconstruction

The `special_note` field from the raw OCR ingestion arrived as a high-entropy text blob containing compounded operational metadata (e.g., *Surge*, *Turbo+*, *VIP status*, *Reservations*). To transform this into predictive features, a modular deconstruction engine was architected using a suite of Regular Expressions (Regex).

Regex Parsing Architecture

The engine was designed to isolate binary flags and extract continuous numerical values for specific incentive tiers. This process involved handling multilingual syntax (Spanish/English) and non-standard character encoding, such as the `CHAR(160)` non-breaking space, which frequently disrupted standard string splitting.

Category	Feature (Boolean / Amount)	Regex Logic (Simplified)
Incentives	<code>is_turbo_plus / amount</code>	<code>(?i)((?:Turbo\+.*?[\d\.]+) (?:[\d\.]+\+.*?turbo fare))</code>
	<code>is_surge / amount</code>	<code>(?i)(Surge.*?[\d\.]+ fare includes a surge...)</code>
	<code>is_priority / amount</code>	<code>(?i)((?:priority prioritario).*?[\d\.]+ ...priority trip start)</code>
	<code>is_reservation / amount</code>	<code>(?i)res.*?(\d+\.\d+ \d+)</code>
Trip Attributes	<code>is_long_trip</code>	<code>(?i)\b(long largo)\b</code>
	<code>is_multiple_destinations</code>	<code>(?i)\w*dest\w*</code>
	<code>is_teens</code>	<code>(?i)Teens</code>
Rider Metadata	<code>is_vip</code>	<code>(?i)(VIP Rider is VIP user is a VIP)</code>
	<code>is_exclusive</code>	<code>(?i)(Exclusivo Exclusive)</code>
	<code>is_identity_verified</code>	<code>(?i)(Identidad verificada Identity verified)</code>
	<code>rider_star_rating</code>	<code>(\d+\.\d+ \d+)</code>
	<code>rider_reviews</code>	<code>\((\d+)</code>

Table 6: Comprehensive deconstruction schema for the `special_note` high-entropy field.

Generative Variability and Remediation

During a subsequent EDA phase, a systemic inconsistency was identified regarding incentive classification. While the ingestion script was strictly sequential, the **generative extraction layer** exhibited non-deterministic

behavior. Varying levels of model “creativity” across processing batches led to the conflation of *Surge* and *Turbo+* labels.

For instance, the model occasionally produced compounded, hallucinated strings such as: “*Turbo+ of \$16.00 and Turbo+ of \$14.07 are included...*” These unstructured artifacts rendered standard Regex parsing insufficient. Because these discrepancies threatened the integrity of downstream profitability analysis, a second iteration of **Manual Remediation** was executed. The Agent performed a line-by-line audit, cross-referencing corrupted records against the original image assets to ensure 100% data purity for the final training set.

Technical Retrospective

This cleaning phase yielded critical engineering insights regarding the extraction of structured data from visual UI elements via LLMs:

- **Downstream Fragility:** Text parsing via Regex is highly susceptible to the non-deterministic variability of LLM-based OCR outputs. A single word change in the model’s response can break a production pipeline.
- **Upstream Visual Mapping:** A superior architecture involves *Visual Semantic Instruction* during the ingestion phase. Instructing the Vision model to distinguish incentives based on UI iconography (e.g., mapping the “Lightning Bolt” icon directly to a `surge_amount` key) and requiring a strictly typed JSON output eliminates the need for “artisanal” post-processing.

Chronology Features: Modeling Shift Progression

To capture the impact of endgame decision-making, features were engineered to model an offer’s position within a discrete work session.

The Offer Window Constraint

For analytical consistency, a work session is defined by its **Offer Window**—the duration between the inaugural offer and the terminal offer of a shift—rather than total physical operational time. To prevent **Label Contamination**, the window was programmatically truncated 15 minutes prior to the Agent’s arrival its home base. This ensures that the dataset minimizes “terminal rejections” where the agent had no real intent of accepting any ride whatsoever. Two primary features were calculated to quantify temporal positioning:

- **time_in_session_sec:** The elapsed time between the current offer and the session’s first recorded offer.
- **session_progress_ratio:** A normalized metric ranging from 0.0 (first offer) to 1.0 (final offer). This feature is the primary proxy for the agent’s psychological state during the “Endgame” phase. It allows the model to learn the collapsing acceptance threshold for non-homecoming vectors as the ratio approaches 1.0.

Trajectory Intelligence: Localization Trials

While the `offers` table successfully captured the coordinates for pickup and dropoff points, the Agent’s own physical location at the moment of the offer remained unrecorded. Although the Agent possessed subjective ground truth from field experience, this spatial context was missing from the structured dataset. Recognizing that the Agent’s relative position is a high-value predictor for decision modeling, a series of four localization experiments were conducted to infer this latent variable.

Trial 1: External Telemetry (Google Location History)

Status: Failed. Initial attempts were made to utilize the `location-history.json` export from Google Takeout. However, the device configuration during the acquisition phase was restricted to “While Using” rather than “Always,” resulting in a truncated and inconsistent dataset. The lack of continuous polling rendered this external source insufficient for high-precision temporal alignment.

Trial 2: Geometric Inference (Trilateration)

Status: Investigatory. This protocol attempted to infer location using only the data within the offer card (*Distance to Pickup*). By identifying clusters of three or more concurrent offers, the system utilized the `distance_to_pickup` as a radius for a trilateration optimization algorithm (`SciPy.optimize.minimize`) to find the agent's most probable coordinate. This approach was ultimately rejected due to platform-level address obfuscation. Without precise centroids for the circles, the optimization results deviated significantly from operational reality.

Trial 3: Linear Temporal Interpolation (Phase 2 Baseline)

Status: Ratified. A robust internal baseline was developed leveraging the verified coordinates of accepted trips ($T_0 \dots T_4$) as high-fidelity anchors. Initial implementations executed interpolation across a continuous timeline. This "session-blindness" caused offers occurring between distinct shifts (e.g., overnight gaps) to be interpolated across massive temporal distances, generating physically impossible vectors.

To ensure physical plausibility, the interpolation logic (`pandas.merge_asof`) was re-architected with a strict `by='session_id'` constraint. This mathematically guaranteed that the "before" and "after" anchors for any given offer were selected exclusively from within the same operational session, permanently eliminating the teleportation error.

Trial 4: Home Vector Alignment (Strategic Extension)

Status: Final Champion (Post-Phase 2). This methodology involved calculating the **Cosine Similarity** between the *Movement Vector* (Agent $\rightarrow$ Dropoff) and a fixed *Home Vector* (Agent $\rightarrow$ Anzures). While this feature eventually became the dominant spatial predictor, the project relied on the Trial 3 Interpolation baseline for the duration of the Phase 2 structural engineering. This feature successfully quantified the Agent's session-termination behavior.

High-Precision Temporal Recovery

Standard minute-level timestamps (HH:MM) proved insufficient for analyzing high-frequency market dynamics. To recover seconds-level precision, a custom Python pipeline was developed to extract internal metadata from the non-standard PNG chunks of the source assets.

This timing capability was identified during the localization trials, as the investigation initially sought to retrieve GPS metadata from the image assets. While geospatial data was absent, the discovery of second-level markers enabled a higher analytical resolution. Two primary architectural risks necessitated this precision: iOS filename recycling and system-level offer repetition. The system often resends identical offers, creating visually congruent artifacts. As these repetitions can occur within a single minute but at distinct second intervals, standard HH:MM granularity would result in data loss through erroneous duplicate detection. To ensure uniqueness, a primary key, `event_id`, was generated via a SHA-256 hash of a combined event signature:

$$event_id = \text{SHA-256}(\text{image_content_hash} + \text{high_precision_timestamp}) \quad (1)$$

This transformed the unstructured file repository into an audited relational ledger. The resulting high-fidelity temporal baseline unlocked the engineering of velocity-based features from Phase 2D.

The Canonical Version of the offers Table

The following table represents the canonical version of `offers` prior to the ETL process into `pienza.db`.

Logical Domain	Engineered Columns	Role
Identity	offer_id, session_id, image_content_hash	Traceability
Temporal	offer_timestamp, time_in_session_sec, session_progress_ratio	Chronology
Physics	upfront_fare, product_category time_to_pickup_sec, dist_to_pickup_km est_trip_time_sec, est_trip_dist_km	Financials Logistics (Pickup) Logistics (Trip)
Geospatial	pickup_address, dropoff_address pickup_lat/lon, dropoff_lat/lon inferred_agent_lat/lon/bearing/speed_mps interpolation_quality, is_imputed	Raw Geodata Absolute Location Agent Vector Data Integrity
Incentives	is_surge, surge_amount is_turbo_plus, turbo_plus_amount is_reservation, reservation_amount is_priority, priority_amount	Dynamic Pricing Strategic Bonus Scheduling Fee Demand Fee
Service Flags	is_exclusive, is_vip, is_identity_verified is_long_trip, is_multiple_destinations, is_teens	Client Tiering Operation Complexity
Rider Profile	rider_star_rating, rider_trip_count	Risk Metrics
Decision	offer_action, reason_primary (Target Variable) heuristic_flag, driver_state_at_request, outcome	Behavioral Data Strategic Context Operational State
Audit Layer	special_note_raw, comment_1, comment_2, record_status	Forensics

Table 7: Canonical offers schema state at the conclusion of the initial engineering phase.

Phase 2B: Geospatial Bottleneck

Initial ingestion revealed systemic coordinate errors affecting $\approx 12.8\%$ of the dataset. A visual audit identified two primary failure modes: **Out-of-Bounds (OOB) Errors**, where ambiguous API requests placed coordinates outside the country (e.g., *Santa Fe* resolving to New Mexico, or *Terminal 2, Col. Peñón de los Baños* resolving to Guanajuato instead of the AICM zone), and **Artificial Singularities**, where distinct addresses collapsed into default centroids by the API.

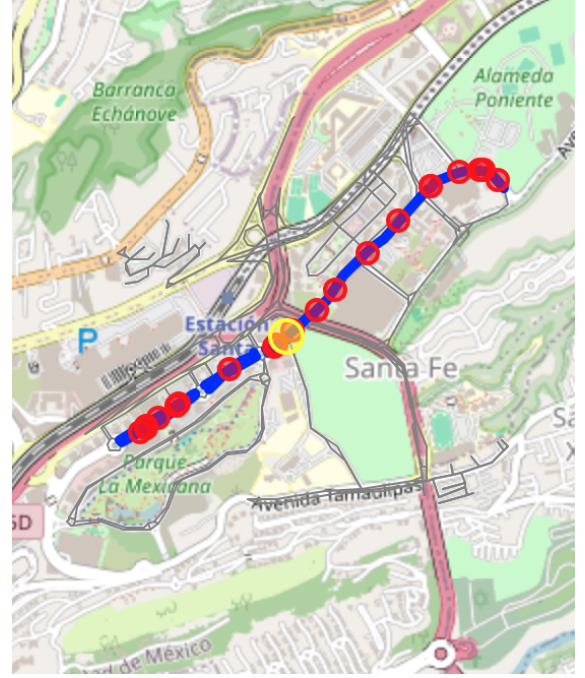
To correct the OOB errors, a containerized filtration script was executed specifically over the list of anomalous records, enforcing a strict bounding box for Mexico City (19.0, -99.5 to 19.8, -98.7). Subsequent geocoding requests were forced into this regional bias.

To resolve the singularities, a geometric engine was developed using the `osmnx` road network graph. The goal was to implement a linear jittering protocol to distribute points along the true geometry of target streets.

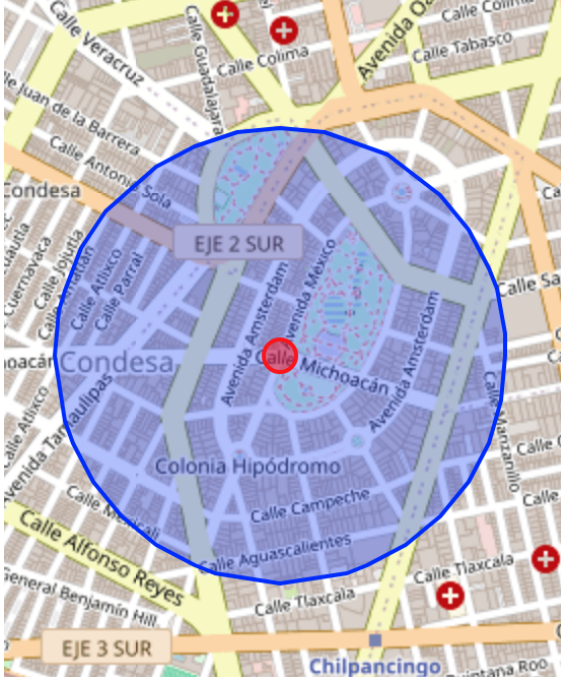
As shown in Figure 1, the engine isolated street-level *LineStrings* from their initial centroids. However, deployment confirmed that a fixed search radius was fundamentally contingent on **street length**. A small buffer effective for short segments proved insufficient for arterial routes or major avenues (e.g., *Masaryk*, *Homero*). This created a cycle of diminishing returns where optimizing the radius for one street-type introduced errors into others, requiring constant manual overrides.



(a) Ave. Santa Fe: Default Centroid



(b) Ave. Santa Fe: Linear Jittering



(c) Ave. Mexico: Default Centroid



(d) Ave. Mexico: Linear Jittering

Figure 1: Geospatial Reconciliation Results. The left column illustrates the default API centroids (Artificial Singularities). The right column represents the remediation protocol, randomizing coordinates across the true *LineString* geometry.

Upon identifying the limits of automated matching, the project prioritized analytical velocity over synthetic perfection. Since the *dropoff* (destination) is the primary driver of the decision policy and the ambiguous set within that quadrant was statistically small (≈ 310 observations), the existing noise was determined

to be acceptable. The project accepted the verified centroids for the contextual *pickup* data to maintain development speed.

Quadrant	Nature	Observations
dropoff_unambiguous	High-Precision Ground Truth	$\approx 4,325$
dropoff_ambiguous	Managed Operational Noise	≈ 310
pickup_ambiguous	Contextual Noise (Sufficient Precision)	$\approx 4,265$
pickup_unambiguous	Standard Operational Signal	≈ 500

Table 8: Geospatial integrity census post-remediation.

Outcome: The dataset was geographically bounded to the operating area of Mexico City. By managing the noise floor, the project preserved the integrity of mission-critical destination features for the subsequent analytical phases.

Phase 2C: Stateful Intelligence

While the physical parameters of an individual offer define a potential transaction, they fail to capture the broader operational context that governs an agent's decision boundary. An agent's risk tolerance and profitability thresholds are dynamic functions of the current session's history—including accumulated fatigue, recent financial yield, and immediate market volatility. To bridge this gap, the project engineered a suite of stateful, session-dependent features.

These features were implemented through a custom Google Apps Script (`code.gs`) designed to operate as a sequential state machine. This engine iterates through the chronologically sorted dataset, maintaining a persistent "memory" of the agent's state across five critical dimensions:

1. **Temporal Integrity (The No Look-Ahead Rule):** To prevent *Data Leakage*, the engine strictly adheres to a temporal firewall. Features modeling past performance—such as the `cycle_rolling_avg_spread`—only incorporate data from missions officially completed *prior* to the current offer's timestamp.
2. **Total Uncompensated Load:** The engine calculates `total_accumulated_deadhead_sec`, aggregating the full non-revenue burden of the current cycle: *Looking for rides* (T0) + *Driving to Pickup* (T1) + *Waiting for Passenger* (T2). This metric quantifies the accumulating "sunk cost" pressure on the agent's decision boundary.
3. **Market Physics & Dynamics:** Utilizing high-fidelity timestamps, the script calculates high-velocity signals such as `offer_density`. A continuous **Traffic Index** is derived by normalizing trip duration against a baseline velocity of 120s/km, providing a scalar metric of congestion friction.
4. **North Star Convergence:** The engine implements a **\$200 MXN** realized Earnings Per Hour (EPH) baseline as the Agent's "North Star." A suite of normalized indices (`eph*_index`) quantifies how each offer deviates from this target, allowing the model to learn the agent's strategy of selecting rides that force the session mean to converge on this value despite individual variance. This enables the model to detect how the agent's multidimensional decision logic aligns towards forcing long-term convergence on this objective goal, even when individual sessions or offers are suboptimal.
5. **The Profitability Funnel (Yield Analysis):** To quantify the delta between the platform's presentation and operational reality, EPH is calculated at four levels of granularity:
 - **Direct:** Upfront Fare / Est. Trip Time. (The raw platform signal).
 - **Operational:** Adjusted for `Time_to_Pickup`.
 - **Realized:** Realized Fare / Est. Trip Time. (Accounting for financial spread).

- **Complete:** Realized Fare / (Total_Accumulated_Deadhead + Est. Trip Time). (The holistic yield).

This funnel quantifies the Information Asymmetry gap where high-nominal offers may resolve as low-yield operations.

Technical Note on Model Integrity (_ML vs _EDA): A rigorous distinction is maintained between suffixes. Features intended for Supervised Learning (_ML) are engineered using only historical averages available at the moment of decision. Features incorporating actual outcomes (_EDA) are firewalled for exploratory analysis only, ensuring zero data leakage in the predictive engine.

The Canonical Stateful Features

These features were engineered within the `engineered_features` table to provide the model with a stateful representation of the market and the agent's internal progress.

Dimension	Feature Name	Engineering Logic & Strategic Proxy
Market Pressure	time_since_last_offer	Seconds elapsed since previous request. Proxy for market "silence."
	offer_density_[10-180]sec consecutive_rejects	Rolling counts of offers across 10, 30, 60, 180s windows. Stateful counter resetting on acceptance. Proxy for frustration/patience.
Supply & Traffic	traffic_index_base_120	Normalized congestion metric (Baseline: 120s/km). Values > 1.0 indicate friction.
	cycle_avg_dtp_km	Mean Distance-to-Pickup in current cycle. High values = Low local supply.
	cycle_std_dtp_km	Standard deviation of DTP. Measures supply volatility.
	cycle_ttp_dtp_ratio	Ratio of Pickup Time to Pickup Distance. Localized traffic proxy.
	dispatch_lead_time_sec	Time remaining on active trip when next offer is received (Chained Logic).
Internal State	total_acc_deadhead_sec	Cumulative unpaid seconds (Search + Pickup + Waiting) in current cycle.
	cycle_rolling_avg_spread	Temporally-safe rolling average of the Upfront/Realized fare delta.
	cycle_cum_net_earnings	Cumulative realized income for the session.
Base Profitability	eph_direct	Raw EPH (Upfront Fare / Est Trip Time).
	eph_direct_index/label	Normalized Score and Categorical Label (> 1.0 = Premium; else, Discount).
	eph_operational	True EPH adding Pickup Time. The first reality check.
	eph_operational_index/label	Normalized Score and Categorical Label (> 1.0 = Premium; else, Discount).
	is_operational_downgrade	Bool: True if offer flips from Premium (Direct) to Discount (Op).
Predictive (ML)	eph_realized_ML	Predicted EPH adjusting for historical Spread (No future leakage).
	eph_realized_index/label_ML	Normalized Score and Categorical Label (> 1.0 = Premium; else, Discount).
	is_spread_downgrade_ML	Bool: True if Spread adjustment kills North Star EPH.
	eph_complete_ML	Holistic EPH: (Predicted Fare / Total Cycle Time). The Agent's true yield.
	eph_complete_index/label_ML	Normalized Score and Categorical Label (> 1.0 = Premium; else, Discount).
	is_total_cycle_downgrade_ML	Bool: True if the final outcome was a downgrade vs EPH realized.
Exploratory (EDA)	eph_realized_EDA	Actual EPH using finalized Realized Fare (Post-facto).
	eph_realized_index/label_EDA	Normalized Score and Categorical Label (> 1.0 = Premium; else, Discount).
	is_spread_downgrade_EDA	Bool: True if the actual platform payment lowered yield.
	eph_complete_EDA	Absolute Truth EPH: Actual Fare / Actual Total Time.
	eph_complete_index/label_EDA	Normalized Score and Categorical Label (> 1.0 = Premium; else, Discount)..
	is_total_cycle_downgrade_EDA	Bool: True if the final outcome was a downgrade vs EPH realized.
Strategic Context	home_vector_alignment	Cosine Similarity score (-1 to 1) relative to <i>Home Base</i> .
	pickup/dropoff_ambiguity	Binary flags for low-confidence geospatial coordinates.
	day_type/time_block	Semantic temporal segments (<i>Weekday, Morning, Friday</i>).

Table 9: Comprehensive stateful feature inventory.

Phase 2D: Schema Design and ETL Pipeline

Once the offer stream was enriched with context, it was necessary to anchor these decision events to the "Official Truth" of the platform's ledgers. This required a rigorous reconciliation campaign to validate financial outcomes and account for discrepancies in transaction status.

Integration of Official Ledgers

Two new tables were architected to ingest the platform's official data exports:

Normalization Strategy & Relational Logic

The schema utilizes strict normalization to ensure data integrity and consistent labeling for machine learning.

- **Traceability (1:1 Mirroring):** The `raw_offers_ocr` table serves as an immutable staging mirror. It maintains a strictly enforced 1:1 relationship with the cleaned `offers` table, ensuring that every engineered record can be traced back to its raw OCR source string for forensic auditing.
- **Categorical Dimensions:** Low-cardinality strings were extracted into lookup tables linked via Foreign Keys (suffix `_fk`). While this degree of normalization is typically optimized for transaction processing (OLTP), it was retained in this static analytical (OLAP) context to enforce a rigid vocabulary for the model features.
- **Heuristic Flags (Many-to-Many vs. Pragmatism):** The schema includes a bridge table (`heuristic_flag_offers`) architected to support a many-to-many relationship, allowing multiple flags per offer. However, for pragmatic simplicity during the manual backtagging phase, a "Single Flag" protocol was enforced (one dominant heuristic per offer), though the architecture remains extensible.
- **Atomic Design:** The system was architected for maximum atomicity in its storage layer. While engineered features are deliberately non-atomic to provide session context, the telemetry data is stored in its rawest state. Although wide-format ledgers with pre-calculated KPIs existed during the spreadsheet phase, the database was designed to ingest the raw, "long" event stream into the `trip_events` table. This design offloads mission reconstruction to the SQL view layer, ensuring that durations and efficiency metrics are calculated programmatically rather than persisted as static, pre-aggregated values.

Circular Avoidance: The Linear Financial Chain

The relationship between internal logs and official platform data is architected to prevent circular references. The flow of truth is strictly linear:

$$\text{offers} \xrightarrow{1:1} \text{trip_events} \xrightarrow{1:0..1} \text{lifetime_trips} \xrightarrow{1:1} \text{activity_earnings} \quad (2)$$

This chain ensures that financial data flows from the definitive bank settlement (`activity_earnings`) back to the behavioral trigger (`offers`) without ambiguity.

The Idempotent ETL Pipeline

The engineering phase concluded with the consolidation of all transformation logic into a single, idempotent ETL pipeline; executing the end-to-end transformation cycle—from raw Google Sheets to the final population of `pienza.db`—guaranteeing 100% reproducibility. The pipeline rejects standard "Incremental Load" patterns in favor of a "Big Bang" reconstruction philosophy to eliminate state risk.

Tabula Rasa

The pipeline initiates with a clean slate command. The script detects the existence of the binary database file and executes a hard deletion (`os.remove`). This enforces a stateless execution environment, ensuring that no "ghost data" or schema drift from previous iterations can survive to contaminate the current analysis. The database schema is then freshly instantiated from the `schema.sql` definition file.

The 2-Engine "Golden Link"

This transformation involved reconciling data from (`trip_events`) with (`offers`). A two-stage strategy was employed to forge this link:

1. An optimistic matching algorithm attempted to link records using a hierarchy of logic:
 - *Tier 1:* Exact Match (Date + Exact Upfront Fare).
 - *Tier 2:* Fuzzy Match (Date + Rounded Integer Fare).
 - *Tier 3:* Orphan Match (Unique Fare within 24h window).

2. The cascade suffered from "Stolen Identity" errors; to resolve this, a remediation layer was deployed, where the Agent manually overrode specific edge cases.

Following the establishment of the primary link between `offers` and `trip_events`, the remaining entities in the financial chain were integrated via deterministic 1:1 joins. By using high-precision timestamps and fare values as anchors, the system successfully synchronized agent telemetry with the official platform ledgers (`lifetime_trips` and `activity_earnings`), ensuring 100% referential integrity across the entire transaction lifecycle.

The SQL View Architecture

While the base tables maintain a normalized 3NF structure for referential integrity, they are abstracted into a series of SQL Views that serve as the primary interface for analysis. These views transform the relational schema into an active intelligence engine:

<code>v_trip_funnel_wide</code>	Utilizing conditional aggregation (<code>MAX(CASE WHEN ...)</code>), this view flattens the vertical, sequential stream of telemetry logs into a single horizontal lifecycle. This is the foundational step for time-series reconstruction.
<code>v_trip_final_kpis</code>	Built upon the pivoted funnel, this view calculates the "Physics of Profitability." It derives durations via Julian Day conversions and computes critical metrics such as <i>Spread Percentage</i> and EPH.
<code>v_mission_dossier</code>	Integrates the final mission KPIs directly with the originating <code>offer_id</code> . This view serves as the ultimate proof of the "Golden Link," anchoring every financial outcome to the specific algorithmic incentive that triggered it.
<code>v_broche_fks</code>	A high-level audit view that traces foreign key lineage across all six layers of the architecture (<code>OCR</code> → <code>Offers</code> → <code>Features</code> → <code>Events</code> → <code>Lifetime Trips</code> → <code>Earnings</code>). It validates the integrity of the end-to-end data lineage.
<code>v_offers_human</code>	The primary interface for EDA. It denormalizes the eight core dimension tables into human-readable labels and joins the engineered features, providing a context-rich narrative of every offer.
<code>v_lifecycle_audit</code>	An alignment view that compares five dimensions of truth. By measuring the <i>Delta Timestamp</i> between manual agent logs and corporate server timestamps, it statistically validates the integrity of the $N = 1$ dataset.

Computational Boundaries

The ETL pipeline enforces strict operational boundaries, offloading expensive perception tasks and probabilistic modeling to external systems:

- **Decoupled Perception (OCR):** The transformation of raw image assets into text is treated as an upstream dependency. By assuming the existence of the structured Parquet staging files, the ETL remains fast and deterministic, allowing for dozens of schema-design iterations without incurring the high API latency or cost of reprocessing visual assets.
- **Decoupled Inference (ML):** The pipeline terminates at the creation of the feature store (`v_ML_Supervised`). This enforces a clean separation between data engineering and data science, preventing "logic creep" where model weights or hyperparameters could inadvertently contaminate the data definitions.

Persistence & Cloud Synchronization

To address the risk of silent data corruption, a hardened termination sequence ensured the database moves from a cached state to a persistent artifact:

1. **WAL Checkpointing:** The script executes a `PRAGMA wal_checkpoint(TRUNCATE)` command. This explicitly forces SQLite to merge the Write-Ahead Log (sidecar file) into the main binary, ensuring 100% of the session's transactions are persisted.
2. **OS Buffer Flush:** The pipeline utilizes the `os.fsync` system call to bypass standard file-system buffers and force the operating system to write the data to the physical disk descriptor.
3. **Sync Buffer:** A deliberate 10-second pause is introduced to allow the cloud synchronization daemon to detect the binary change and complete the upload before the runtime environment is destroyed.

The conclusion of Phase 2 resulted in the certification of `pienza.db` as a **Golden Master**. The system has successfully transitioned from loose operational files to a structured, relational engine. The data is now clean, reconciled, and feature-rich, providing the bedrock for the subsequent Exploratory Data Analysis.

PHASE 3

Exploratory Analysis & Causal Inference

Tech Stack:	Optimal Stopping Theory, Net Expected Value (VEN), OLS & Polynomial Regression, LOWESS Smoothing, Heteroscedasticity Analysis.
Core Challenge:	Defining a mathematically rational search time via behavioral economics, and modeling the inelastic threshold of platform payouts through causal inference.
Key Insight:	Search time is not merely friction; it is a depreciating financial derivative with a strict 9-minute rationality cap. Concurrently, the platform's pricing algorithm enforces an inelastic "Integrity Buffer" to explicitly neutralize the moral hazard of intentional driver delay.
Operational Outcome:	The derivation of the dimensionless Market Quality Index (MQI) and the Efficient Frontier, establishing the hard mathematical boundaries of agent rationality required for the transition into Machine Learning.

Phase 3: Exploratory Analysis & Causal Inference

Following the relational stabilization of the dataset in Phase 2, the project’s analytical focus pivots toward inference and causal modeling. The Data Analysis phase is structured as a **Diagnostic Audit** to ensure the dataset is a precise digital twin of the agent’s operational reality.

Operational Baseline: Data Census

The primary objective of the univariate audit was to establish the numerical context of the $N \approx 4,765$ offers, defining the inherent market frequency and structural biases before feature analysis.

Decision Action	Count	Rate (%)	Product Tier	Rate (%)
Reject	4,419	92.74	UberX	75.93
Accepted	346	7.26	Comfort	17.17
			Business_Comfort	3.69
			Envíos_Uber	1.49
			Black	0.82
			Uber_Planet	0.80
			Uber_Pet	0.10

Table 10: Decision and Product Mix Census (Total Offers $N \approx 4,765$).

Primary Target Label	Count	Rate (%)	Mission Outcome	Count	Rate (%)
dropoff_non_operational	2,366	49.65	Completed	250	72.0%
low_profitability	838	17.59	System Failure	43	12.4%
long_pickup_time	366	7.68	Rider Canceled	24	8.1%
ACCEPTED (Implicit/NaN)	346	7.26	Driver Canceled	26	7.5%
expected_value_gamble	330	6.93			
dropoff_strategic_mismatch	275	5.77			
dropoff_proxy	239	5.02			
system_logic_failure	5	0.10			
Total Observations	4,765	100.00	Total Accepted Missions	347	100.0%

(a) Distribution of the Primary Target Variable.

(b) Distribution of Mission Outcomes.

Figure 3: Data Census: Structural imbalance of the decision target vs. the realized outcome of accepted missions.

Key Insight: The census confirms a severe structural imbalance in the target variable, where nearly 50% of all events are dedicated to filtering non-operational offers. This necessity justifies the subsequent development of a **Hierarchical Classification** model, explicitly designed to manage this class imbalance. Model validation will strictly utilize **Stratified K-Fold Cross-Validation** to ensure all classes are proportionally represented in each training fold.

The Profitability Funnel

To evaluate market viability, four sequential Earnings Per Hour (EPH) metrics were established. These metrics quantify the transition from the system’s upfront quote to a holistic operational reality:

1. **EPH Direct:** Derived from the *Upfront Fare* and *Estimated Trip Duration*.
2. **EPH Operational:** Incorporates the uncompensated *Time to Pickup* (DTP).
3. **EPH Realized:** Accounts for the *Realized Fare* post-adjustment.

4. **EPH Complete:**³ The final metric including *Total Deadhead* (search time, pickup, and passenger wait times).

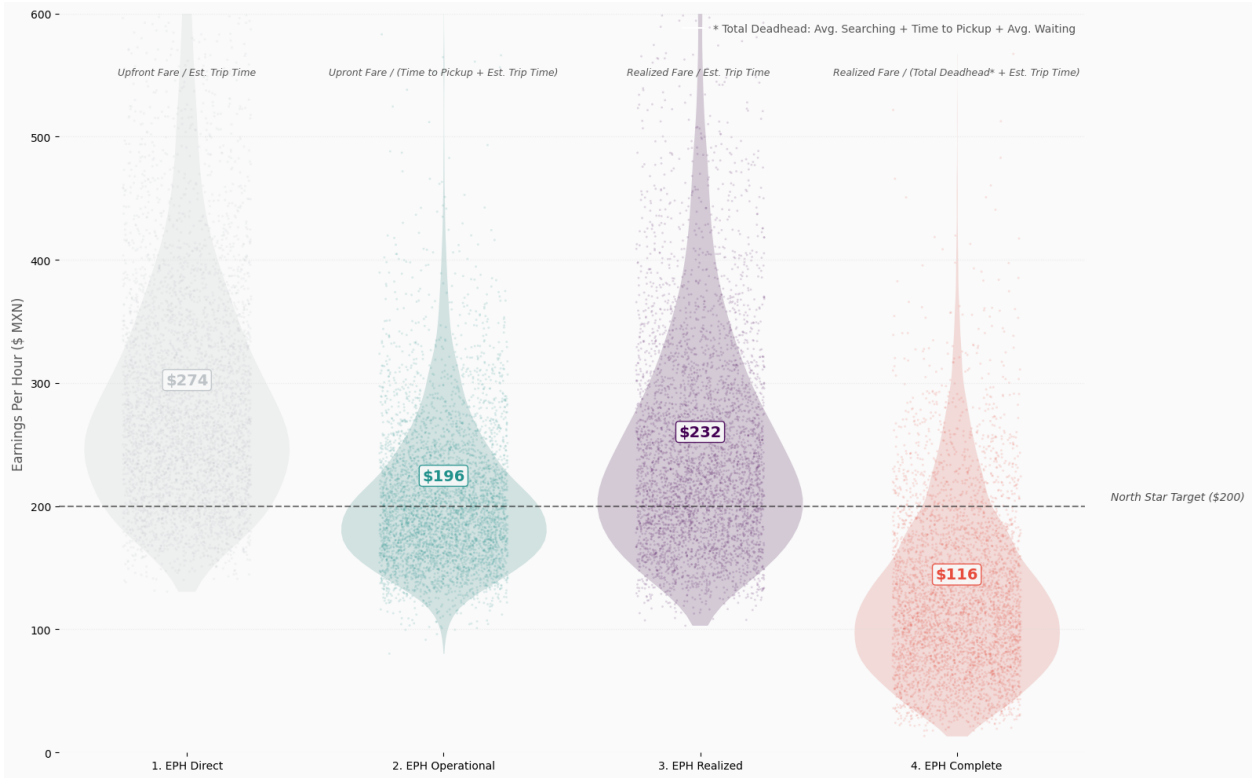


Figure 4: Phase III: Yield metrics from initial quote to holistic outcome. Median values drop from a quoted **\$274/hr** to a realized **\$116/hr**.

Applying a **\$200/hr North Star** threshold to the **EPH_Complete** metric reveals that 83.45% of total offers fall below the sustainability baseline, while only 13.48% exceed the target. This result indicates that viable market opportunities constitute a specific minority of the environment, confirming that high-precision filtering is required to isolate these profitable events from general market noise.

Traffic Index

To quantify the non-linear relationship between traffic friction and potential yield, the analysis engineered a **Time-to-Distance Index** based on the expected ratio relative to a fluid market baseline (2 min/km). This index serves to segment the operational environment into distinct states (*Fluid* to *Gridlock*).

³For non-completed offers, values were imputed using session-specific rolling averages, adhering to a strict *no-look-ahead* protocol to prevent temporal data leakage. Full formula implementations are available in the `code.gs` script: [\[GitHub Repository\]](#).

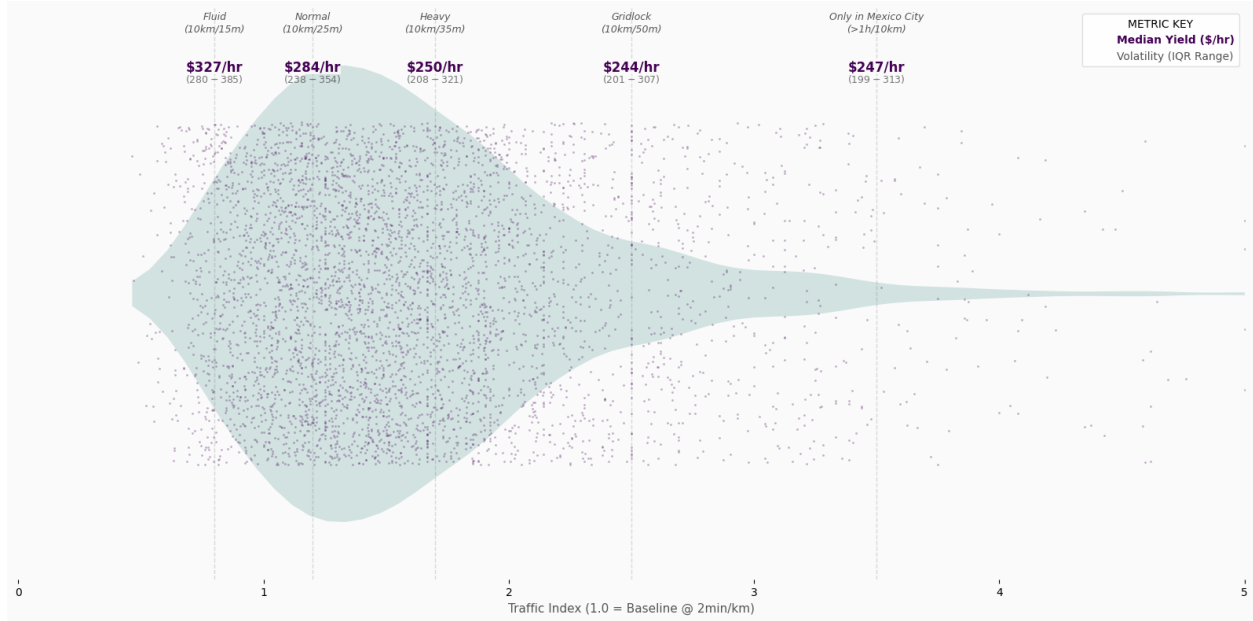


Figure 5: Phase III: Yield Performance Across the Friction Spectrum. The plot illustrates the distribution of Median **EPH Direct** across defined traffic states, establishing the precise financial compensation presented relative to increasing logistical complexity.

While the segmentation confirms the structural relationship between market friction and quoted yield, the analysis shows that the EPH performance exhibits a plateau at high friction levels, confirming that the system's compensation mechanism accounts for expected time overruns, but places a hard cap on maximum session return.

However, there is a key limitation to this observation, Traffic Index only captures **Expected Friction**. The actual financial impact is driven by realized time duration. High-friction states (Index > 2.0) introduce significant non-linearity, where the difference between expected and realized travel time can result in an uncompensated cost that does not scale linearly with the quoted fare. This discovery will be quantified in during the causal inference study, where the initial hypothesis is revisited.

The Physics of Selection

Following the relational stabilization of the dataset in Phase 2, the analytical focus shifts to the deconstruction of the Agent's decision boundary. The following investigations move beyond descriptive statistics to model the underlying "physics" of the marketplace and the agent's response to volatility.

The Market Quality Index

The fundamental challenge in modeling the Gig Economy is the **Mixture Problem**. Raw performance metrics are often contaminated by unproductive market regimes where even an optimal policy cannot reach the global revenue baseline. A model trained on raw revenue would fail to distinguish between a strategic error and a stochastically poor opportunity set. This approach dissociates the quality of a selection from the absolute value of the currency to reveal the underlying decision boundary.

To render heterogeneous product tiers comparable and ensure the framework scales across diverse environments, a **Global Anchor** (μ) was established for each product category (*Economy*, *Mid-Tier*, *Premium*) based on the historical median EPH of the entire operational history. By dividing the specific EPH of an offer by its category anchor, the absolute currency was transformed into a dimensionless **Quality Index (QI)**:

$$QI = \frac{EPH_{offer}}{EPH_{anchor}} \quad (3)$$

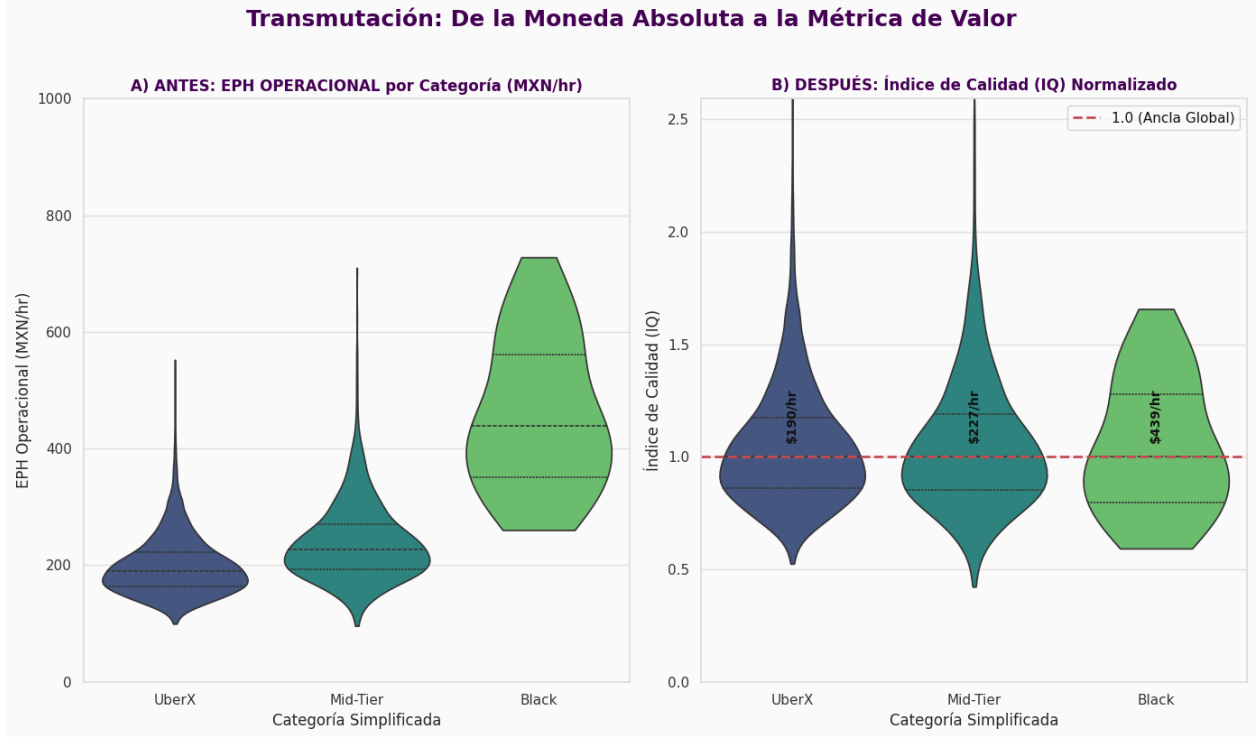


Figure 6: Transmutation from Absolute Currency to Value Metric. Plot A illustrates the structural variance between product tiers. Plot B shows the normalized distribution where all tiers are anchored at 1.0, enabling the objective identification of outliers regardless of category.

As illustrated in Figure 6, an QI of 1.0 represents market parity. An QI of 1.5 identifies a 50% excess value relative to the specific asset class.

Market Heterogeneity and Volatility

By monitoring the distribution of offer quality over the chronological progression of the campaign, the marketplace exhibits extreme variance across discrete work sessions.

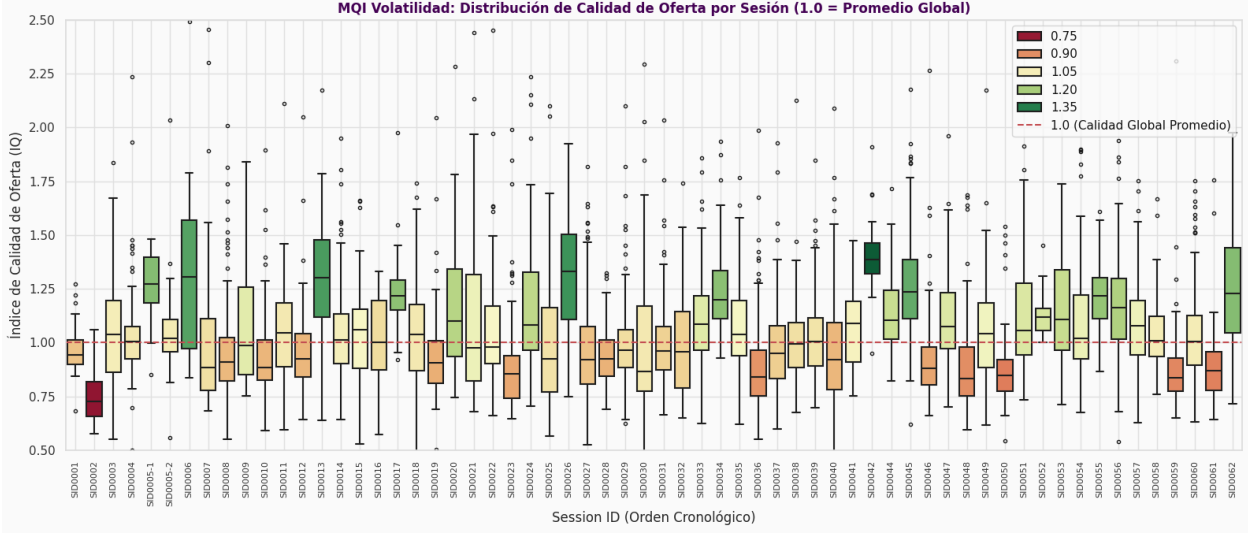


Figure 7: Market Quality Index (QI) Volatility by Session. The boxplots illustrate the distribution of offer quality for each session. The horizontal baseline at 1.0 represents the global market average. Color-coding (Red to Green) indicates the session-level mean quality (*MQI*).

As shown in Figure 7, the agent operates within a non-stationary environment. Certain sessions (indicated in green) represent high-quality market states, while others (in red) signify economic scarcity.⁴

The Opportunity Oracle: Time as a Financial Derivative

While the *MQI* focuses on *what* can be selected, the **Opportunity Oracle** framework quantifies *when* to select it. In this model, "Search Time" is not categorized merely as operational friction (deadhead), but as the **Investment Capital** required to purchase a probabilistic option on future high-yield offers.

Search Stream Sanitization (The Behavioral Clock)

To accurately measure market velocity, the search stream underwent a strict temporal sanitization protocol. The analysis decoupled the physical duration of a trip from the decision intervals by **resetting the behavioral clock** ($\Delta t = 0$) immediately following every acceptance event. This isolation reveals the true "Heartbeat" of the marketplace—the frequency at which opportunities are presented—independent of the ride execution phase.

The Strategic Quadrant Matrix

To analyze how environmental variance dictates performance, the operational environment was mapped into a four-quadrant phase space (Figure 8). This framework cross-references **Market Quality** (*MQI*) on the y-axis against **Market Velocity** (Median inter-offer latency) on the x-axis. Although *MQI* and velocity can vary throughout a single shift, this model utilizes a *static session-level aggregate* to categorize the dominant market regime for each period of activity.

⁴**Analytical Case Study:** For a granular, session-by-session audit of the agent's selection behavior relative to the *MQI*—including forensic analysis of high-performance events and sub-optimal sessions—readers are invited to explore the interactive case reports in the canonical Jupyter Notebook: [\[GitHub: Project Pienza - Phase 3 EDA - MQI\]](#).

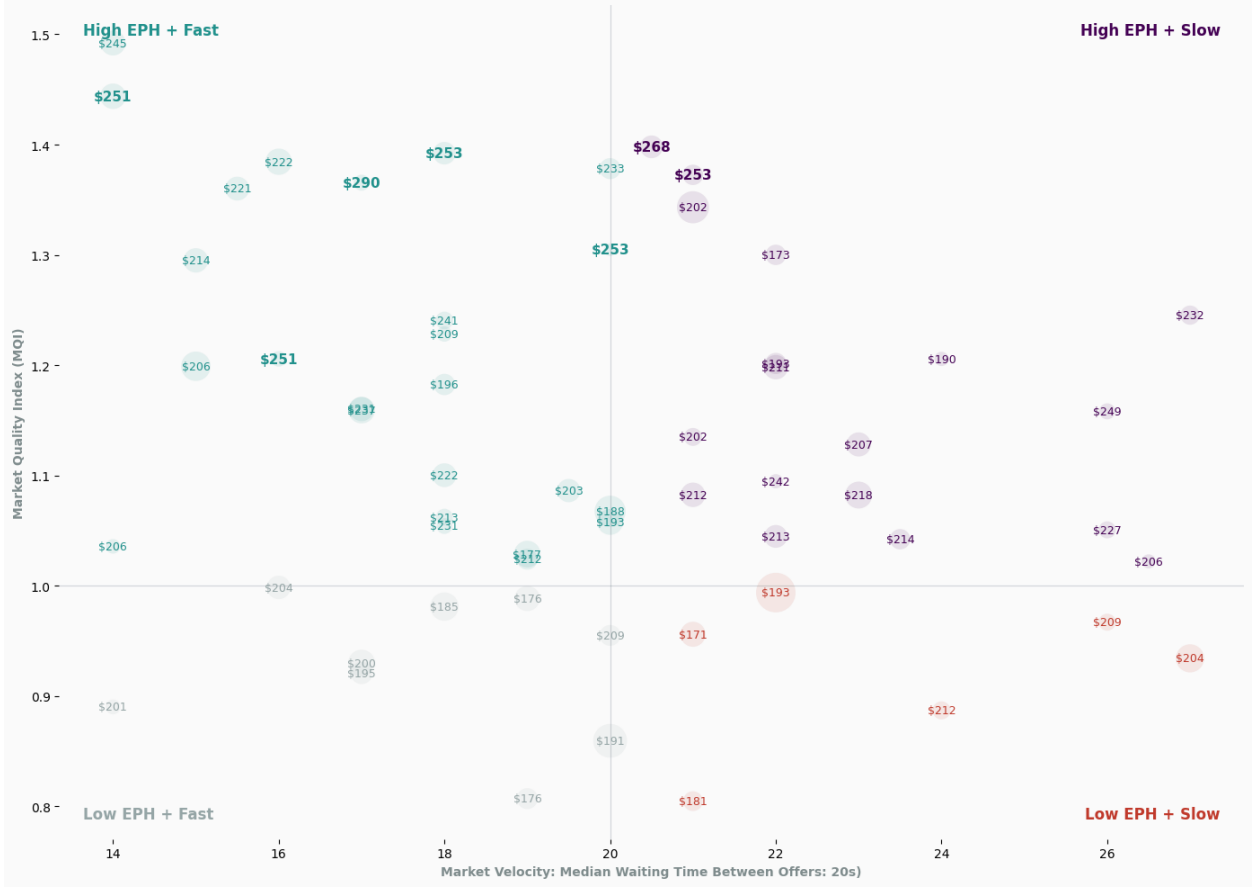


Figure 8: Strategic Quadrant Analysis: Market Quality vs. Velocity. Each observation represents a discrete work session. Data labels indicate the realized session-level Average `eph_operational`. The vertical boundary at 20s denotes the median market tempo observed during the campaign.

The performance metrics in Table 11 confirm that while sustainability is maintained in "Rich" regimes, "Poor" quadrants represent a state of mandatory yield erosion regardless of market velocity. This mapping serves as the basis for calculating the **Efficient Frontier** in the following section.

Market Quadrant	Average Operational EPH
Rich / Fast	\$223.98 / hr
Rich / Slow	\$217.32 / hr
Poor / Slow	\$194.96 / hr
Poor / Fast	\$192.93 / hr

Table 11: Operational yield categorized by environmental quadrant.

The Efficient Frontier (Optimal Stopping)

The final exploratory objective was the determination of the **Optimal Stopping** threshold. Using a vectorized look-ahead algorithm, the study calculated the **Net Expected Value (VEN)** of rejecting a baseline offer to

wait for a superior target ($QI \geq 1.5$). This analysis transforms the abstract concept of "patience" into a deterministic, time-based rule.

Net Expected Value (VEN) Analysis

The decision to reject a viable offer is modeled as a probabilistic bet. The project calculates the **Net Expected Value (VEN)** by subtracting the deterministic search-time cost (*Accumulated Deadhead*) from the weighted expected payoff of future offers.

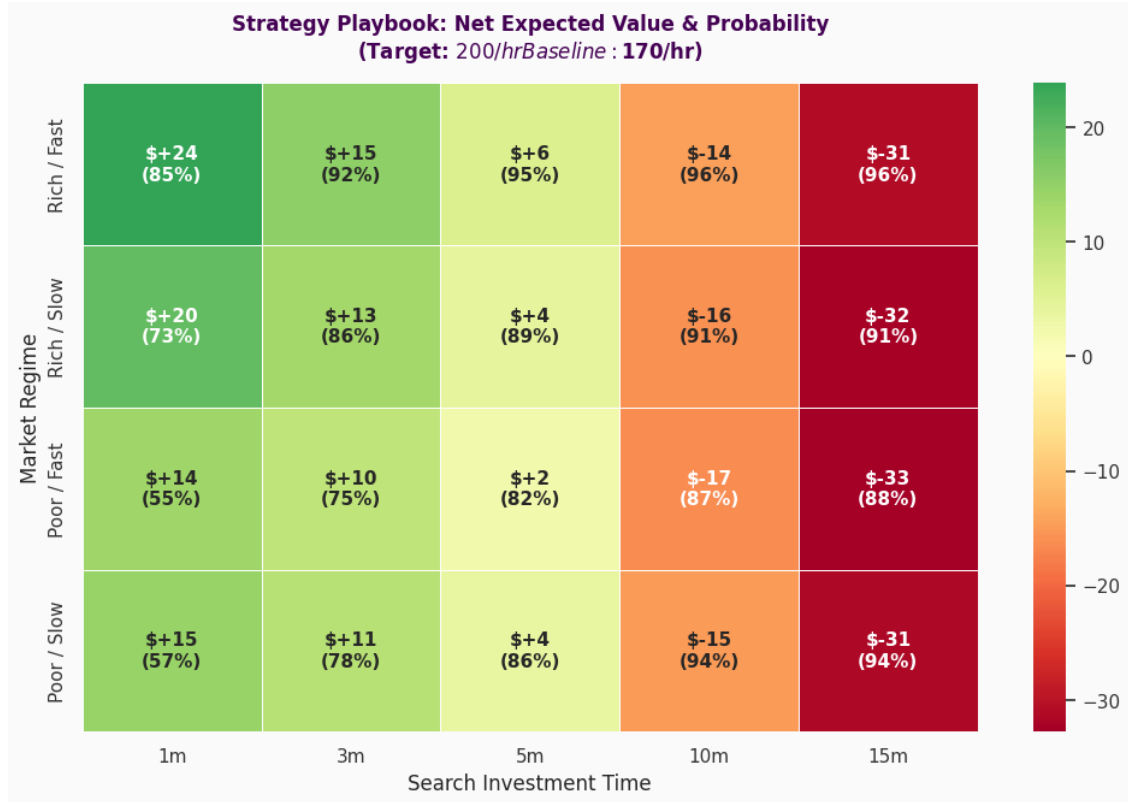


Figure 9: The \$200 Strategy Playbook: VEN and Success Probability. Each cell displays the projected financial delta (MXN/hr) and, in parentheses, the empirical probability of encountering a mission $\geq \$200/hr$ within the search investment time. Green regions identify a rational search window; red regions signal yield erosion.

As illustrated in Figure 9, the “Rationality Window” for a \$200/hr target is strictly limited to search horizons of less than 5 minutes. A critical finding is the **Profitability Paradox**: beyond the 10-minute horizon, although the probability of encountering a target offer is extremely high ($> 90\%$), the resulting VEN is negative across all regimes (e.g., $-\$31/hr$ in the *Rich/Fast* quadrant). This confirms that in a high-friction stochastic market, excessive patience ensures yield degradation despite eventual mission success.

The Efficient Frontier: Search Budget Decay

By simulating the break-even point ($VEN = 0$) across a range of profitability targets, the project synthesized the **Efficient Frontier**—the curve representing the maximum rational search time for any given level of ambition.

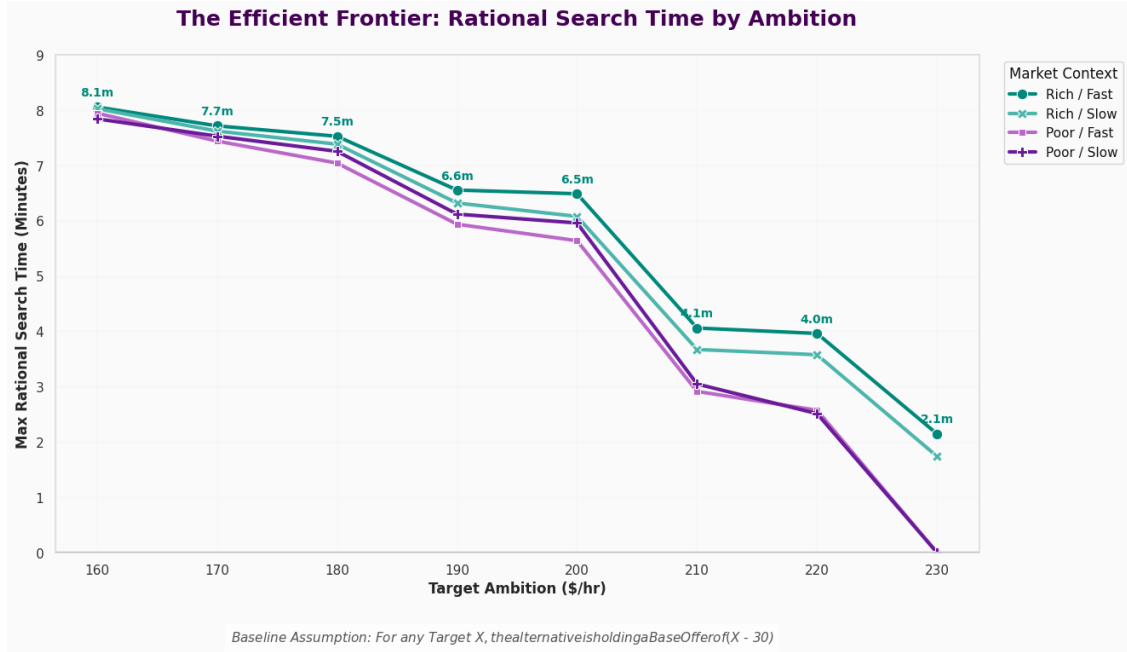


Figure 10: The Efficient Frontier: Rational Search Time by Ambition. The plot illustrates the search budget for targets ranging from \$160/hr to \$230/hr across the four market quadrants.

The analysis, detailed in Figure 10, reveals three primary findings:

1. **Search Budget Decay:** There is an inverse relationship between target EPH and rational search time. To capture a \$230/hr mission, the agent can justify a search window of only ≈ 2.1 minutes.
2. **Market State Resilience:** The "Rich / Fast" quadrant consistently provides the widest search budget, allowing for higher selectivity compared to "Poor / Slow" regimes.
3. **The 9-Minute Absolute Cap:** Regardless of the target or market quadrant, the rational search budget effectively hits a ceiling at 9 minutes. Beyond this temporal horizon, the probability of yield improvement becomes statistically negligible.

Outcome: This framework provides the hard constraints for the Agent's Digital Clone. By identifying the 9-minute search cap, the project ensures that the simulation engine operates within the boundaries of mathematical rationality.

Causal Inference: Revisiting the Initial Hypothesis

Following the acquisition phase, the initial hypothesis regarding the Payout Spread was formally tested using the verified ground-truth of completed missions ($N = 249$, August 22 – October 1 window). The longitudinal data confirms that the "structural haircut" identified during the pilot study remains a consistent property of the marketplace architecture throughout the entire campaign.

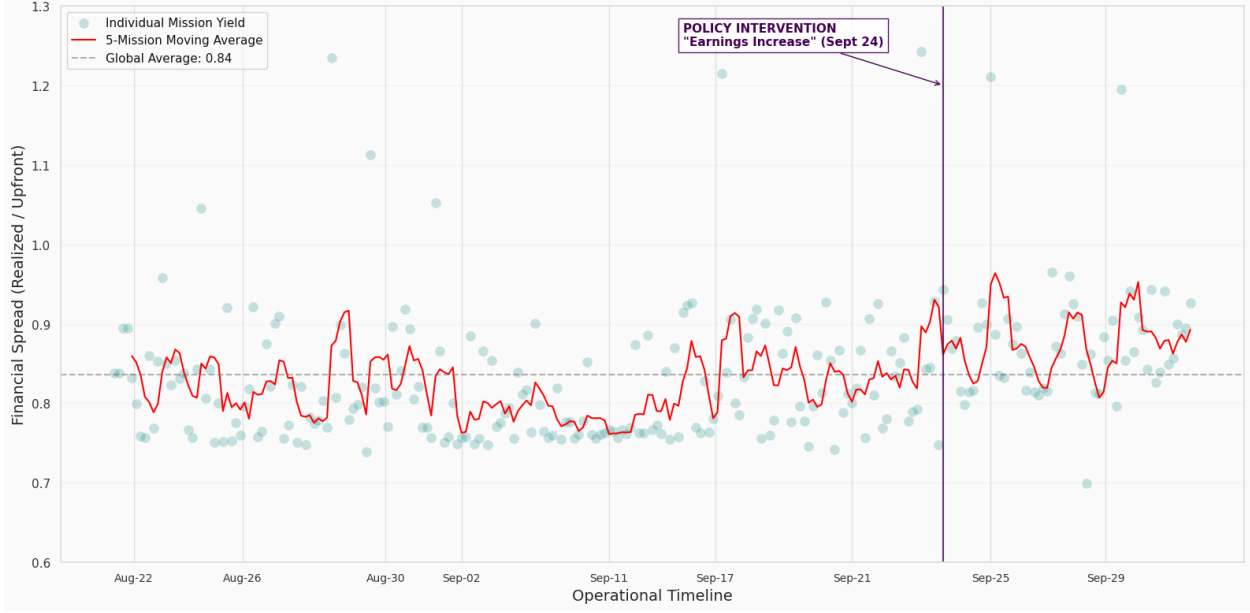


Figure 11: Phase 1: Financial Stability and Policy Intervention Audit. The 5-mission moving average (red line) shows an anchored operational regime around the 0.84 baseline.

The analysis reveals a highly stable financial equilibrium. As illustrated in Figure 11, while individual mission yields exhibit significant dispersion, the 5-mission moving average remains tightly anchored at a global mean of **0.8361**. This confirms that the platform operates under a consistent payout target for the majority of the operational window.

A critical exogenous event occurred on September 24, when the platform issued a formal "Policy Intervention" message promising an increase in earnings. The empirical data confirms the validity of this signal:

- **Mean Spread (Pre-Intervention):** 0.8229
- **Mean Spread (Post-Intervention):** 0.8798
- **Realized Yield Lift:** 6.91%

This finding proves that while the system is anchored, it is not static, and formal policy communications result in a measurable shift in the underlying financial physics of the marketplace.

The Baseline Model: Financial Predictability

To transition from longitudinal monitoring to formal inference, a baseline Ordinary Least Squares (OLS) model was architected to quantify the structural relationship between the quoted fare and operational reality. The estimated regression equation is defined as:

$$\widehat{\text{realized_fare}} = 6.31 + 0.79 \cdot \text{upfront_fare} \quad (4)$$

The model results, summarized in Table 12, establish the global financial "physics" of the marketplace.

Table 12: Baseline OLS Metrics: Financial Predictability Audit

Metric	Value	Diagnostic	Value
R-squared (R^2)	0.930	Prob (F-statistic)	1.00×10^{-144}
Intercept (β_0)	6.3081	Skewness	2.638
Slope (β_1)	0.7906	Kurtosis	13.054

The model exhibits a high coefficient of determination ($R^2 = 0.930$), confirming that 93% of the variance in realized earnings is explained by the initial offer. This demonstrates a robust *Predictive Synchrony* across the system.

However, the high explanatory power masks severe structural instability. The extreme **Kurtosis (13.05)** and **positive Skewness (2.64)** serve as immediate analytical red flags. These metrics indicate a highly non-normal error distribution with "heavy tails," suggesting that the model's predictability is not uniform. The presence of these outliers and the significant deviation from normality point toward a violation of the homoscedasticity assumption, necessitating a deeper investigation: residual analysis.

Heteroscedasticity: The Cone of Uncertainty

To validate the stability of the baseline model, an analysis on residuals was conducted. The objective was to determine if the prediction error remained constant across the financial spectrum or if it scaled with the magnitude of the transaction.

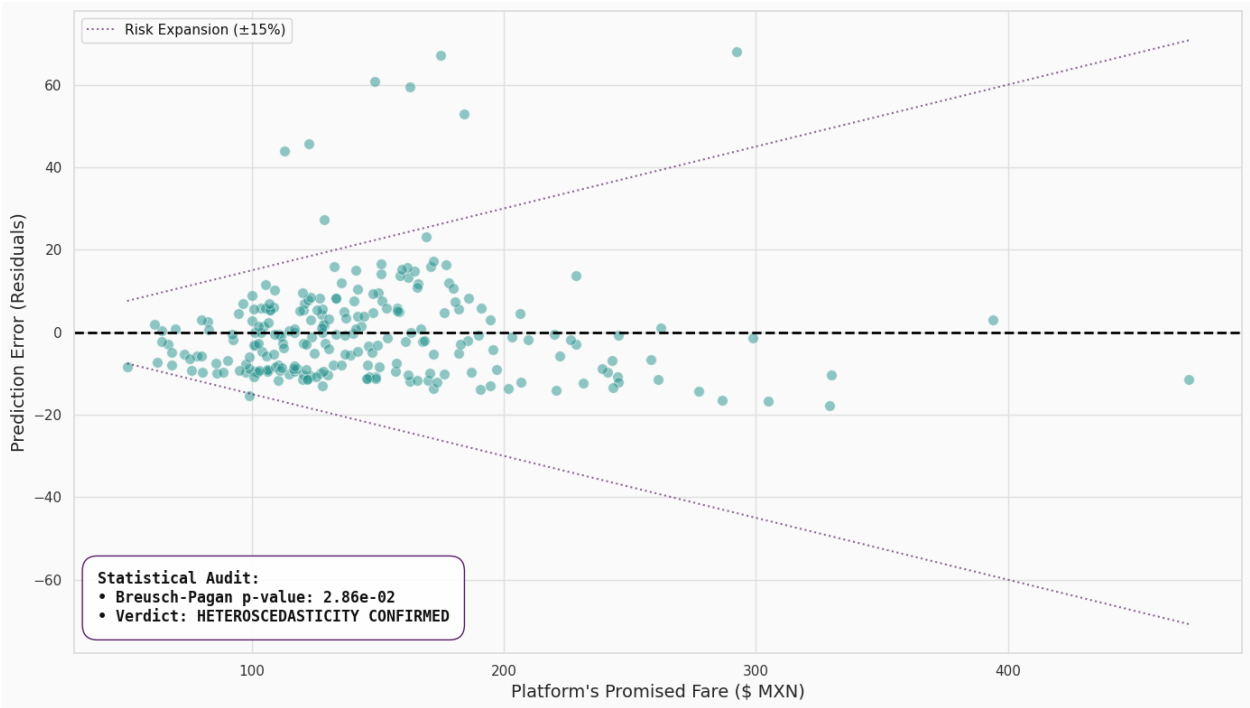


Figure 12: Phase 2: Heteroscedasticity Audit. The fanning pattern of the residuals identifies a *Cone of Uncertainty*, where prediction error expands proportionally with the quoted fare.

The visual inspection in Figure 12 identifies a classic fanning pattern, or *Cone of Uncertainty*. While the model is highly precise for low-value missions, the variance of the residuals expands significantly as the **Quoted Fare** increases. To confirm this observation with statistical rigor, a Breusch-Pagan test was executed,

yielding a p-value of 0.0286. This result formally rejects the null hypothesis of homoscedasticity, confirming that the model’s error variance is non-constant ($p < 0.05$).

The 15% Risk Expansion Band confirms that absolute financial uncertainty scales proportionally with the **Quoted Fare**. While this diagnostic provided a clear measure of historical variance, the causal inquiry was temporarily paused once it became evident that the model lacked the explanatory power to determine *why* this uncertainty existed. It was only during subsequent multivariate interaction analysis on the completed missions cohort that a superior predictive signal was identified: the **Temporal Variance** (or *Time Spread*).

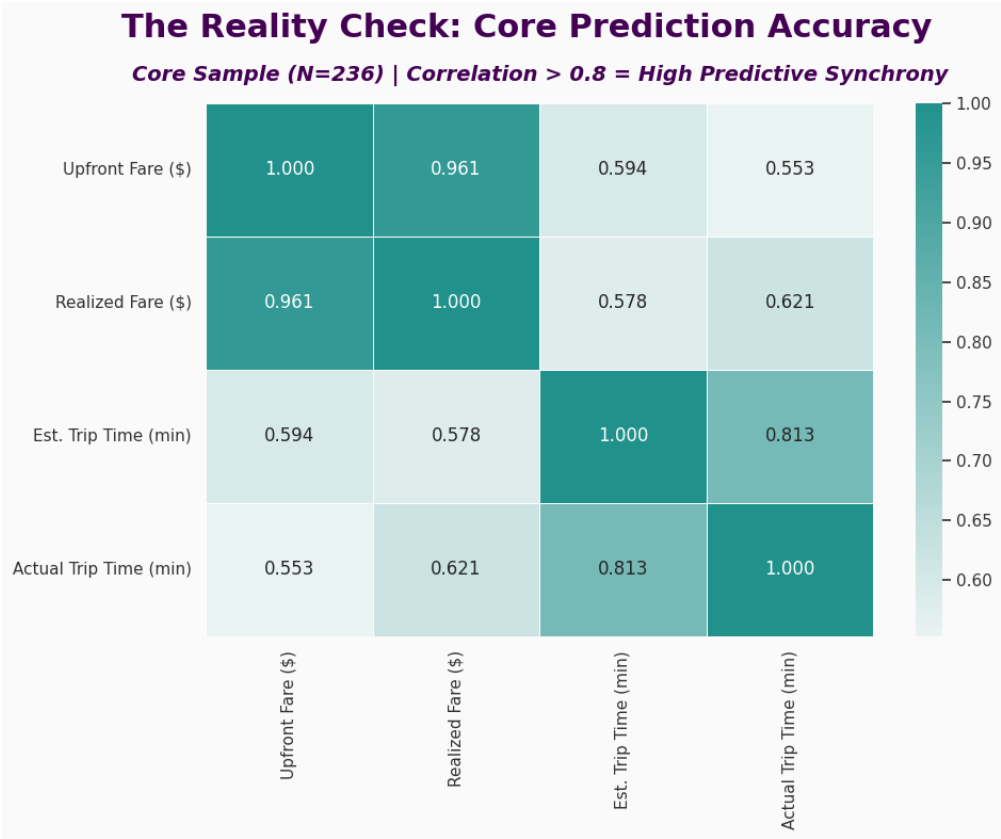


Figure 13: The Reality Check Matrix. A significant disparity exists between the high financial synchrony ($r = 0.961$) and the lower operational precision ($r = 0.813$).

The correlation matrix reveals a systemic **Hierarchy of Predictability** that prioritizes price stability over operational variance. While the correlation between the Upfront Fare and Realized Fare is nearly perfect ($r = 0.961$), the Operational Synchrony for trip duration is significantly lower ($r = 0.813$). The most critical evidence of this decoupling is the correlation between Realized Fare and Actual Trip Time, which stands at only **0.621**. In an unbuffered marketplace, the final price would be highly sensitive to the actual time spent on the road ($r > 0.90$); however, the observed 0.621 correlation proves that the platform deliberately insulates the financial outcome from temporal fluctuations. The system isn’t merely predicting the fare; it is enforcing it.

This decoupling suggests a *Fraud Prevention Mechanism* in action. By maintaining a 0.961 financial synchrony while allowing for a 0.813 temporal variance, the algorithm acts as a **Strategic Buffer** that suppresses the marginal financial gain for additional minutes. This might be specifically designed to neutralize moral hazard by ensuring that the financial incentive for a driver to **intentionally take additional minutes for their own profit** is effectively eliminated. Within this window, moderate operational noise or deliberate padding is filtered out of the final transaction rather than being passed on to the passenger. To quantify the exact

boundaries of this buffer and identify the threshold where the platform's response shifts from inelasticity to legitimate compensation for exogenous shocks, a **Fraud Prevention Response Curve** has been architected.

The Response Curve: Quantifying Inelasticity

To visualize the strategic buffer in action, the relationship between temporal variance and financial outcome was modeled using a Locally Weighted Scatterplot Smoothing (LOWESS) regression. The resulting **Fraud Prevention Response Curve** (Figure 14) identifies the empirical boundaries of the platform's risk-sharing architecture.

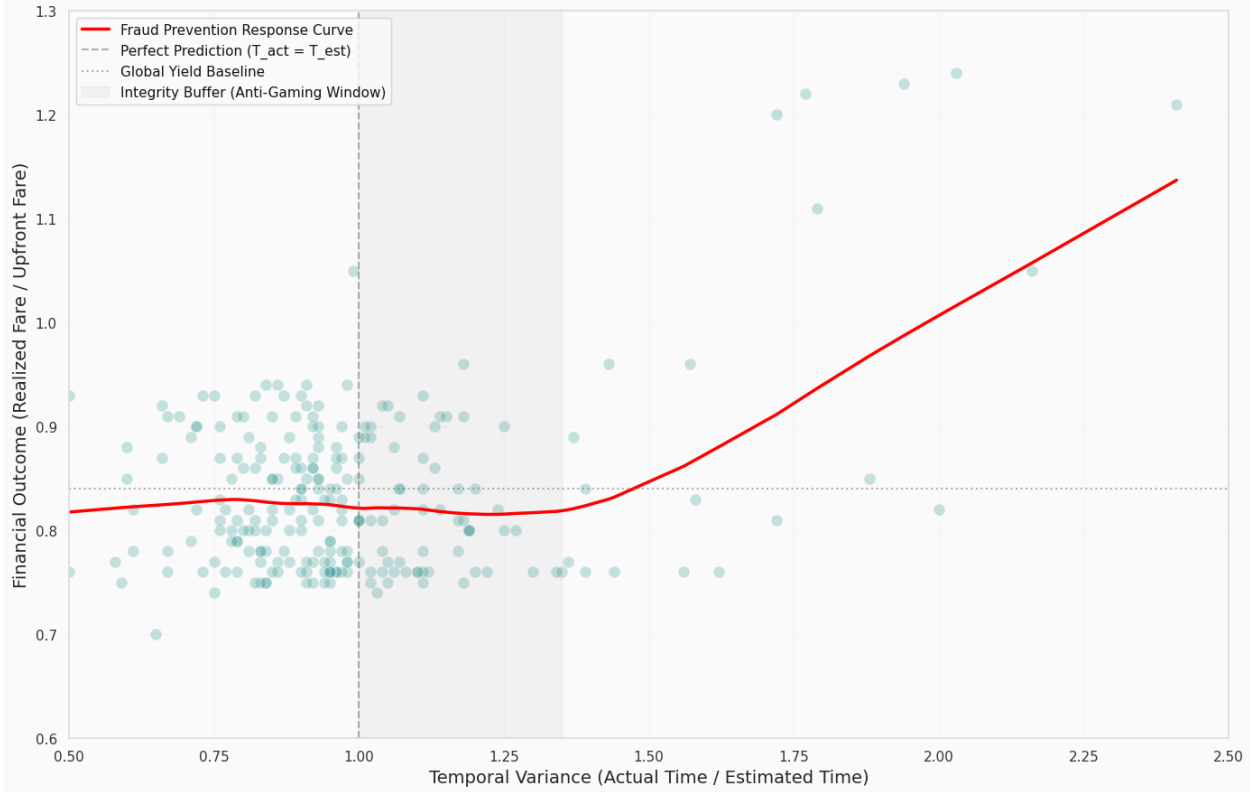


Figure 14: Phase 3: The Fraud Prevention Mechanism Audit. The response curve remains inelastic within the *Integrity Buffer* (1.0x – 1.35x), decoupling moderate temporal variance from financial compensation.

The visualization identifies two distinct operational regimes. Within the **Integrity Buffer** (indicated by the shaded region between 1.0x and 1.35x), the response curve exhibits near-total inelasticity. In this window, an increase in trip duration does not result in a proportional increase in realized fare. By maintaining a flat payout profile during moderate delays, the mechanism ensures that the financial burden of standard operational noise is absorbed by the operational margin, effectively removing the incentive for intentional duration padding.

Beyond the **1.35x threshold**, the curve reaches an inflection point and shifts to a sharp positive slope. This transition indicates that the algorithm distinguishes between driver-induced inefficiency and exogenous systemic shocks (e.g., major accidents or extreme rush-hour volatility). Once the delay is clearly beyond the driver's control envelope, the platform triggers a compensatory adjustment, realigning the financial outcome with the actual labor performed. This non-linear architecture confirms that the system is expertly calibrated to prioritize price integrity and fraud prevention while maintaining a safety net for legitimate market disruptions.

Mathematical Formalization: The Inelasticity Threshold

To provide a precise mathematical anchor for the observed response curve, a quadratic polynomial regression was fitted to the dataset ($N = 236$). This model quantifies the exact point where the platform's *Fraud Prevention Mechanism* reaches its maximum inelasticity before transitioning into compensatory adjustment. The estimated quadratic equation is defined as:

$$\widehat{Yield} = 0.9915 - 0.3645(\Delta T) + 0.1945(\Delta T^2) \quad (5)$$

Where ΔT represents the **Temporal Variance** (Actual/Estimated Time). The model is statistically robust, with an F-statistic of 46.80 and a highly significant p-value (8.22×10^{-18}), confirming that the non-linear architecture is a structural property of the system rather than a stochastic artifact.

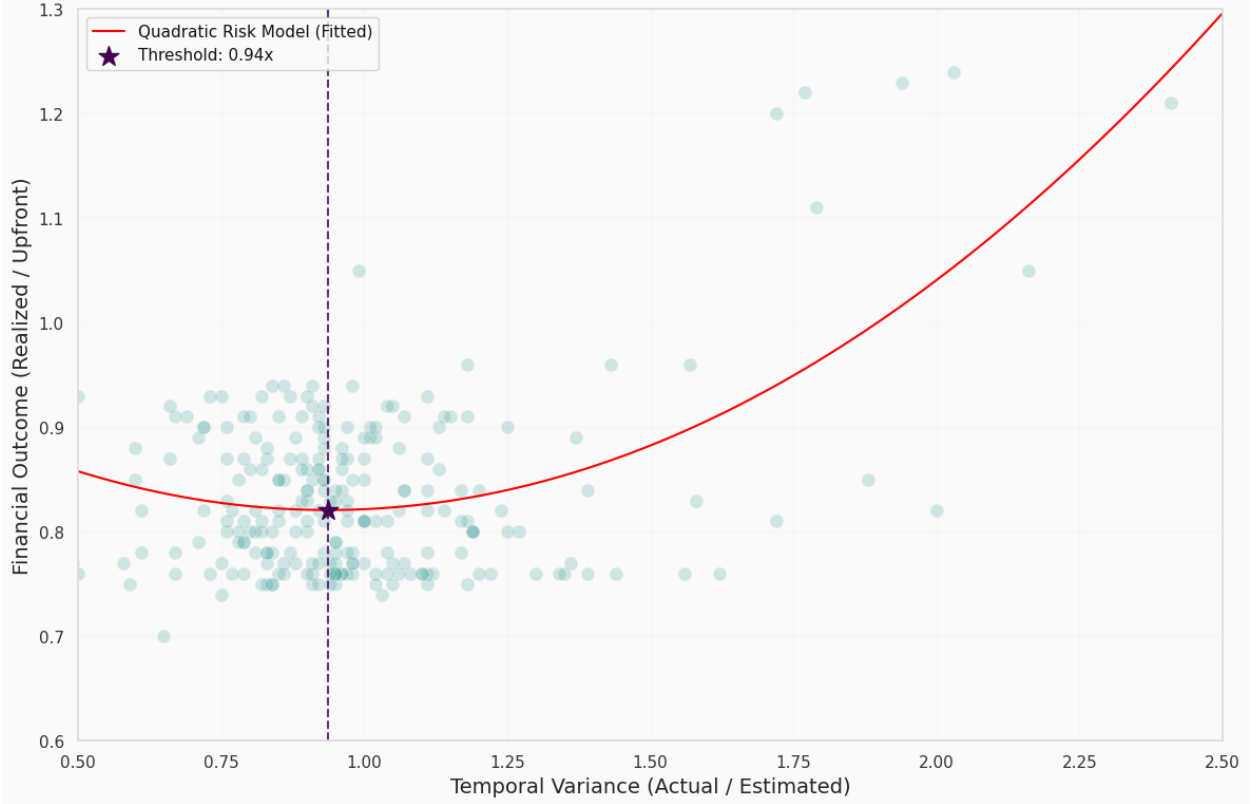


Figure 15: The Mathematical Threshold of Operational Inelasticity. The lowest point of the parabola identifies the *Tipping Point* at 0.94x, marking the structural floor of the compensation model.

The analysis identifies the Inelasticity Threshold at 0.94x, as illustrated in Figure 15. This finding is strategically significant: it proves that the "floor" of financial yield is reached even before the trip is officially categorized as delayed (1.0x).

The system exhibits total indifference to temporal fluctuations within the 0.90x to 1.30x range. By anchoring the payout at this 0.94x floor, the algorithm effectively decouples marginal labor time from marginal compensation. This architectural choice ensures that price integrity is maintained for the passenger while guaranteeing that the Agent is fully paid even if the mission is completed faster than expected. The sharp upward trajectory beyond 1.4x confirms that the system incorporates a high-threshold safety net for catastrophic prediction errors, further reinforcing the platform's focus on mitigating moral hazard through strategic inelasticity.

Phase 3 Conclusion: From Inference to Predictive Intelligence

The progression from the initial pilot study back in August to the formal causal audit in late November represents more than a chronological evolution; it marks the total resolution of the project’s foundational inquiry. The analytical depth achieved during this phase has provided answers to structural questions that were not even apparent at the project’s inception. Through a rigorous diagnostic audit of the total offer stream, the research has established the **Market Quality Index (MQI)**, defined the **Efficient Frontier** of labor selection, and mapped the **Strategic Quadrant Matrix**.

The final causal deconstruction of the *Fraud Prevention Mechanism* serves as the definitive empirical anchor for the Agent’s decision policy. Following the principle of parsimony, this research has utilized classical statistics to address every possible business question within its reach. Having exhausted the explanatory power of inferential models, Project Pienza is now uniquely positioned to transition into the domain of high-dimensional Machine Learning. The subsequent phases will employ Unsupervised Clustering for market segmentation and Supervised Imitation Learning for behavioral cloning—tasks that require the non-linear complexity of algorithmic modeling to solve what classical statistics has so effectively identified.

PHASE 4

Unsupervised Learning & Geo-Remediation

Tech Stack:	PCA, Lasso Regression (L1), K-Means, HDBSCAN, Uber H3 Indexing, SQL, Human-in-the-Loop (HITL) Auditing.
Core Challenge:	Overcoming a massive "Coordinate Crisis" caused by API hallucinations, and translating the opaque, high-dimensional math of unsupervised clustering into actionable business logic.
Key Insight:	Algorithms are blind to operational reality. Purely mathematical clustering will merge zones separated by impassable ravines; therefore, machine-discovered topology must be ruthlessly constrained by expert human heuristics.
Operational Outcome:	The synthesis of the "Silver Palette" feature store—mapping 6 core economic archetypes and 72 hand-crafted operational polygons—finalizing the topological intelligence layer prior to behavioral cloning.

Phase 4: The Geometry of Chaos

Timeline: December 4, 2025 – January 2, 2026

Following the causal audits of Phase 3, the project shifted from analyzing individual variables to discovering the **latent structures** that govern the Mexico City marketplace. This phase represents the critical convergence of the Agent's tacit domain knowledge and unsupervised machine learning, moving beyond standard administrative boundaries to identify organic "Neighborhoods of Profit."

The mission of Phase 4 was twofold: first, to deconstruct the economic and topological archetypes of the market; and second, to execute a high-stakes remediation of the coordinate data. This period was characterized by intense "Human-in-the-Loop" intervention, where automated discovery was audited against operational reality. The campaign culminated in a month-long data surgery—spanning the terminal days of 2025—to reconcile systemic semantic drift and geocoding entropy.

The Clustering Phase of Project Pienza followed a four-stage progression:

- **Market Manifold Discovery:** Utilizing PCA to interrogate the high-dimensional feature space and identify the principal axes of market variance.
- **Economic Archetypes:** Deployment of K-Means to categorize the "DNA" of mission types based on yield and operational effort.
- **Geo-Spatial Clustering:** Application of HDBSCAN to discover the organic density-based zones of the city.
- **Final Feature Synthesis:** The formalization of the Silver Palette, integrating the topological intelligence layer together with the final round of stateful features.

Market Manifold Discovery: PCA

To isolate the pure operational signal, the initial 67-dimensional vector space underwent a rigorous filtration protocol, distilling it down to 54 candidate variables. This preparatory phase enforced several strict constraints:

1. **Variance Purge:** Quasi-constant features ($> 99\%$ identical values), such as `is_teens` and `is_imputed`, were removed to eliminate computational dead weight.
2. **Redundancy Resolution:** High-redundancy pairs ($r > 0.90$) were resolved through strategic selection. The pipeline prioritized **Scale-Invariant Indexes** over raw currency and **Accumulated Earnings** over absolute clock time, identifying the former as stronger behavioral drivers. Geospatial coordinates were retained as indispensable geometric inputs.
3. **Categorical Exclusion:** Product category labels were intentionally omitted. This forced the model to identify the underlying economic and logistical causes of a decision rather than relying on superficial categorical proxies.
4. **Leakage Prevention:** All features containing post-offer statuses, mission outcomes, or the `_EDA` suffix were purged to maintain a strictly predictive, forward-looking environment.

To ensure model stability and prevent the curse of dimensionality, the 54-feature matrix was distilled through a competitive selection tournament using **Lasso Regression (L1 Penalty)**. The L1 regularization acted as a strict mathematical filter, aggressively forcing the coefficients of irrelevant or noisy features to exactly zero.

While the algorithm efficiently identified primary predictors, a critical **Human-in-the-Loop Domain Override** was applied to the surviving features to align the machine with the Agent's operational reality. For instance, while the model temporarily selected distance as a strong predictor, the Agent explicitly overrode this in favor of *Time* metrics (e.g., `est_trip_time_sec`). For an Electric Vehicle (EV) operator navigating heavy urban gridlock, time is the absolute operational constraint, whereas distance is merely a flawed proxy.

Finally, before establishing the final matrix, physics-based variables (*fare*, *time*, *density*, *friction*) underwent **Log-Normal Transformations**. This corrected for distributional skewness, compressing extreme outliers to ensure that subsequent distance-based algorithms were not biased.

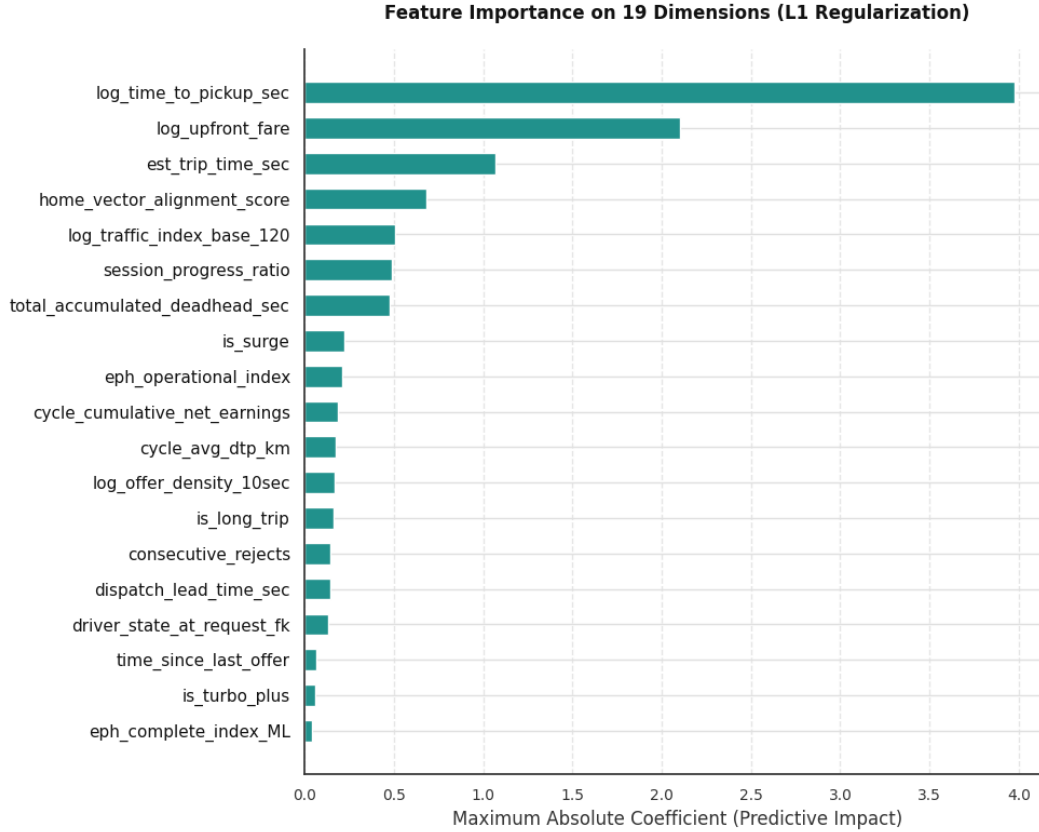


Figure 16: Feature Importance on the 19 Master Dimensions (L1 Regularization). The plot illustrates the absolute magnitude of the coefficients surviving the Lasso penalty. The clear dominance of `log_time_to_pickup_sec` and `log_upfront_fare` confirms the Agent’s primary sensitivity to immediate operational cost and financial reward.

As evidenced in Figure 16, the selection and intervention protocol resulted in a **19-feature whitelist** that perfectly captures the interplay between market physics and agent psychology. The hierarchy establishes *Logistical Friction* and *Magnitude* as the primary anchors of the decision boundary, while state-dependent features like `home_vector_alignment_score` and `session_progress_ratio` provide the necessary strategic context for shift-termination modeling.

Following feature curation, this purified 19-dimensional vector space was processed using **Principal Component Analysis (PCA)**. While subsequent clustering phases utilize the raw, scaled features to maintain direct interpretability, PCA serves a critical methodological function at this stage: it maps the system into an **orthogonal** feature space where every dimension exerts an independent gravitational pull, removing the warping effect of correlated variables and revealing the true latent structure of the market.

The PCA engine leverages the properties of the **Symmetric Covariance Matrix** (C). By applying the **Spectral Theorem**, the matrix is diagonalized ($C = Q\Lambda Q^T$), where Λ is a diagonal matrix of **eigenvalues** and Q is an orthogonal matrix of **eigenvectors**. This ensures that the resulting Principal Components are perfectly perpendicular (uncorrelated), and the market’s total variance is cleanly sorted along the diagonal.

Dimensionality reduction was modest, requiring 13 components to preserve 90% of the variance from the original 19-feature set (Figure 17). While standard PCA applications often exhibit heavy dominance in the first two components, Mexico City’s ecosystem reveals a **distributed eigenvalue structure**. The absence of a dominant principal component confirms the system’s high dimensionality; the agent’s decision logic is driven by the interplay of multiple orthogonal factors rather than a single primary variable.

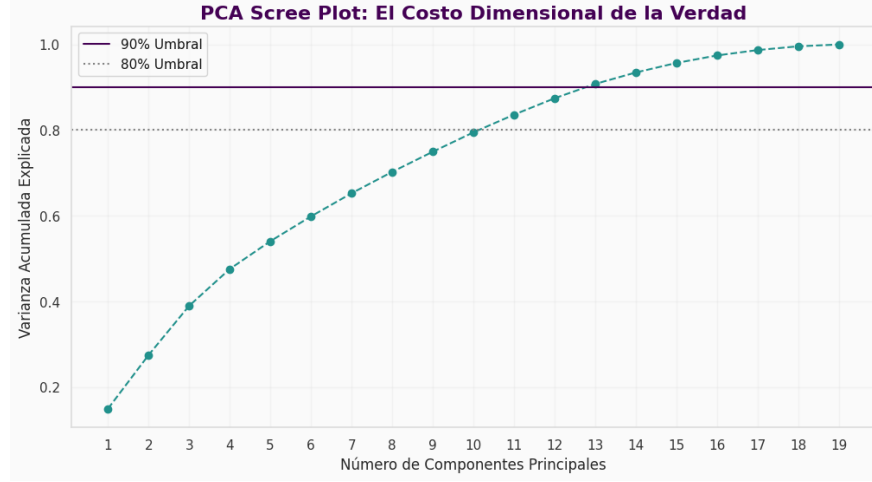


Figure 17: PCA Scree Plot. The gradual slope confirms a high-dimensional system where no single component explains more than 15% of the total variance.

The first three components, explaining $\approx 40\%$ of the variance, define the universal mathematical structure of the total offer stream:

- **PC1: Time in Session:** Weighted by `session_progress`, `deadhead`, and `cumulative_earnings`. It separates early-session, low-friction states from high-search-accumulation segments toward the conclusion of a shift.
- **PC2: Volumetric Scale:** Weighted by `upfront_fare` and *estimated trip duration*. This axis distinguishes the physical magnitude of the mission, separating high-fare, long-duration missions from low-fare, high-turnover events.
- **PC3: Logistical Yield:** Weighted by `dispatch_lead_time` and the *EPH Index*. It identifies opportunities where the financial compensation is high relative to the required operational time investment.

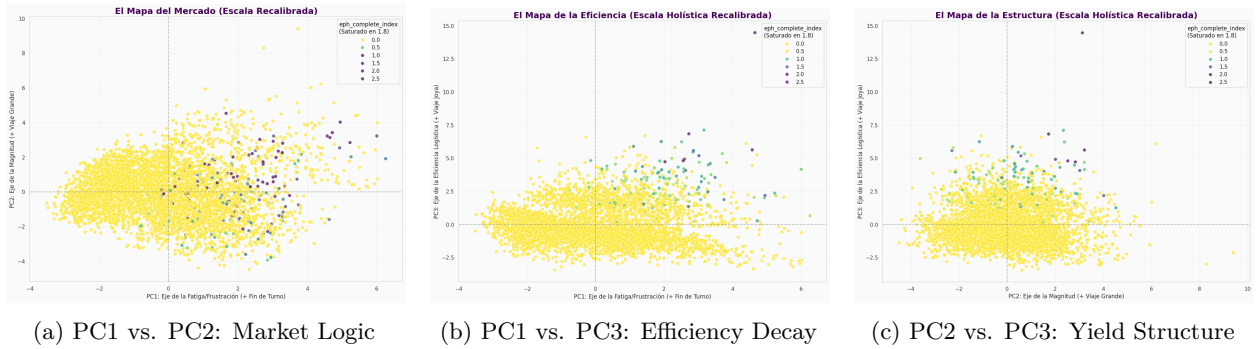


Figure 18: Strategic Projection: Latent Market Manifolds. Color intensity represents the `eph_complete_index`. The maps deconstruct the interplay between session state, physical scale, and realized yield.

The projection of the feature space onto the primary principal components identifies the density regions where efficiency is maximized (Figure 18):

1. **PC1 vs. PC2 (Market Logic):** Identifies how the available mission types shift as the agent accumulates operational time.
2. **PC1 vs. PC3 (Efficiency Decay):** Indicates a correlation between fresh session states and peak efficiency.
3. **PC2 vs. PC3 (Yield Structure):** Identifies that high-magnitude missions (High PC2) tend toward average yield, while high-efficiency outliers are primarily found in low-magnitude, high-turnover segments.

This structural analysis confirms that peak performance is not uniformly distributed but occupies a specific coordinate space within the market manifold. These findings provide the objective boundaries for the **K-Means Strategic Archetypes** in the subsequent stage.

Economic Archetypes: K-Means

An exploratory K-Means clustering algorithm was executed on the latent dimensions (PC_1, PC_2, PC_3). While the algorithm successfully partitioned the data across multiple tested k -universes ($k = 2, 3, 4, 6$), the cluster profiles proved fundamentally uninterpretable for practical business strategy. Because each Principal Component is a linear combination of all 19 underlying features, the resulting clusters became opaque “mixtures of mixtures.” Ultimately, clustering on latent dimensions created a “black box,” making it impossible to reverse-engineer clear behavioral policies or business rules from abstract eigenvectors.⁵

To resolve this interpretability barrier, a deliberate methodological pivot was executed to embrace a “Crystal Box” approach. PCA was formally abandoned as the input matrix for the clustering phase, with the methodology shifting to execute K-Means directly on the scaled raw features.

An initial execution on the full 19-variable set identified a secondary interpretability barrier. While individual variables were transparent, defining mission archetypes based on 19 simultaneous dimensions resulted in a second “black box” outcome where high-level strategic signals were masked by marginal fluctuations in state and demand variables.

To resolve this, an **Incremental Clustering** strategy was adopted. The model was initially constrained to the “Naked Physics” of the transaction: `upfront_fare`, `est_trip_time_sec`, and `est_trip_dist_km`, excluding all algorithmic incentives or agent-state features. Applying the principle of **Occam’s Razor**, the project identified that the market’s fundamental structure is almost entirely dictated by these spatio-temporal magnitudes.

A comparative analysis between the **Elbow Method** and **Silhouette Score** was conducted to determine the optimal cluster count (k) for Mexico City’s marketplace (Figure 19).

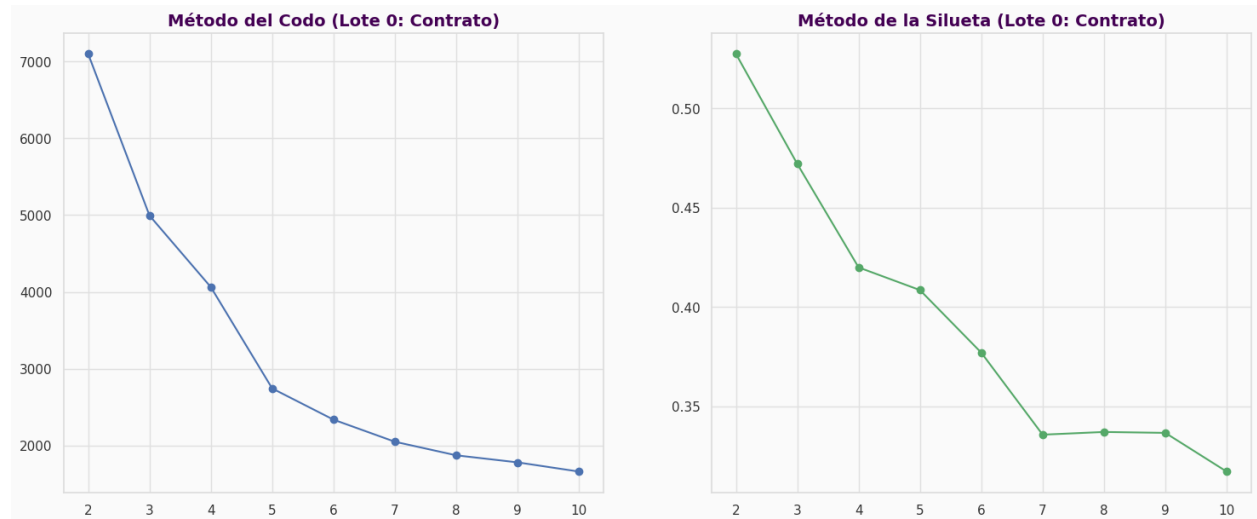


Figure 19: Hyperparameter Tournament (k): Elbow Method and Silhouette Score analysis for the “Naked Physics” feature set.

- **Mathematical Baseline ($k = 2$):** The Silhouette Score indicated a peak at $k = 2$. While geometrically optimal, this trivial binary split (Short vs. Long missions) was rejected for its failure to capture the nuanced operational regimes that dictate profitability.

⁵The evaluation of PCA-based clustering performance and the subsequent methodological pivot are documented in the Phase 4.1 notebook: [\[GitHub: Project Pienza - K-Means Trials\]](#).

- **Expert-Led Resolution ($k = 6$):** A visual inspection of the Elbow Plot identified a significant reduction in inertia occurring at $k = 6$. This higher-resolution setting was ratified as the optimal choice, as it provided sufficient granularity to decode the transition between distinct states in the agent’s policy.

The analysis revealed that the marketplace fractures into six distinct physical tiers based on temporal magnitude and logistical scale (Table 13).

Designation	Median Duration ⁶
Tier 1	12 minutes
Tier 2	25 minutes
Tier 3	40 minutes
Tier 4	56 minutes
Tier 5	80 minutes
Tier 6	224+ minutes

Table 13: Mexico City market archetypes based on $K = 6$ clustering.

Cross-referencing the Agent’s historical rejections against these six tiers provides empirical evidence of a **non-stationary decision policy**. The primary decision driver shifts dynamically as the physical scale of the offer increases:

- **Low-Magnitude Missions (Tiers 1–3):** Rejections are primarily governed by the **Economic Layer**. For "Sprints" (Tier 1), the dominant friction point is **low_profitability** ($\approx 44\%$ of rejections). The Agent’s priority is yield-per-minute efficiency.⁷
- **High-Magnitude Missions (Tiers 4–6):** Rejections are governed strictly by the **Geospatial Layer**. As trip duration approaches the one-hour threshold, economic metrics are ignored. For Tier 5 missions, **dropoff_non_operational** accounts for $> 98\%$ of rejections, reflecting a pure destination-based strategy.

A review of the base fare structure across these tiers reveals two distinct algorithmic laws governing the market (Figure 20).⁸

⁶Median duration is derived from multi-dimensional K -Means centroids rather than deterministic binning. Consequently, observations with identical durations may be assigned to different tiers based on the specific variance in fare magnitude and trip distance.

⁷Although Tier 1 missions exhibit the highest quoted EPH, the agent utilized historical operational knowledge identifying these as low-yield events once uncompensated search and pickup overheads are factored in.

⁸Base fare is isolated by removing all incentives from the upfront quote to ensure a normalized comparison of structural pricing.

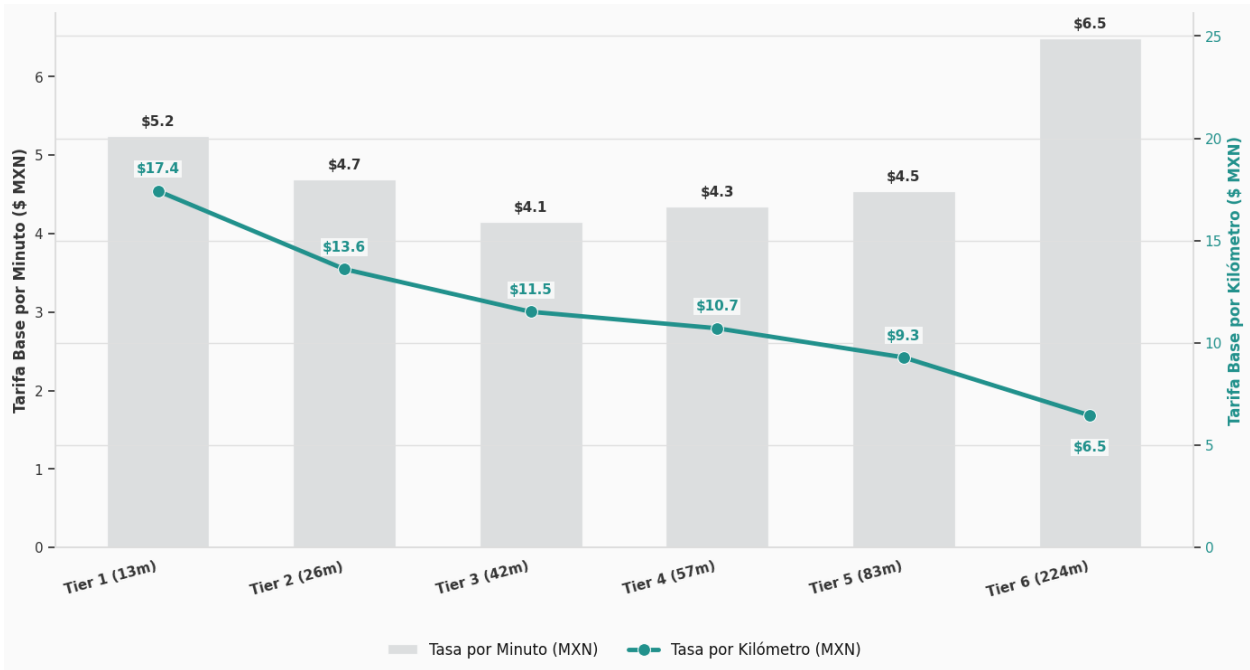


Figure 20: The Physics of Pricing: Mean yield per minute and kilometer across mission tiers. The dual-axis plot illustrates the systematic decay of distance-based rewards and the U-curve of temporal compensation.

1. **Compensation per kilometer:** exhibits an inverse relationship with mission scale. Yield begins at **\$17.4 MXN/km** for Tier 1 and collapses to **\$6.5 MXN/km** for Tier 6.
2. **Compensation per minute:** follows a non-linear "U" structure. Rates are high for Sprints (**\$5.2 MXN/min**) to offset fixed search costs, reach a minimum for mid-range urban missions (**\$4.1/min**), and increase again for Tier 5 and Tier 6.

Yield Analysis Across Operational Tiers

To evaluate the economic viability of the discovered archetypes, the analysis measured the performance of each tier against the **\$200/hr North Star** benchmark. As shown in Figure 21, the transition from quoted potential to holistic outcome identifies how trip duration dictates final efficiency.

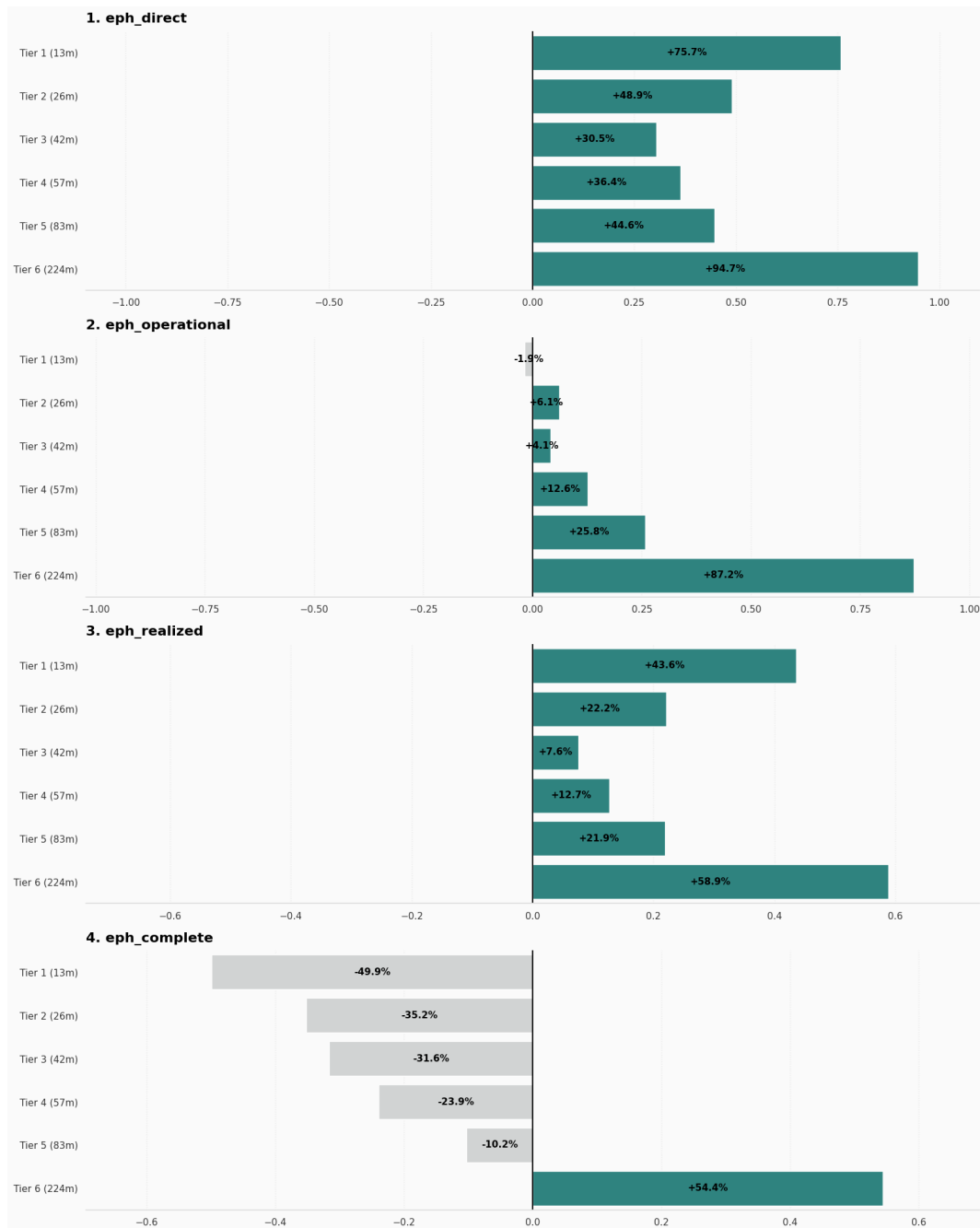


Figure 21: Yield Variance by Mission Tier. Bars represent the percentage deviation relative to the \$200/hr target. Negative values indicate a yield below the North Star threshold rather than a net financial loss.

The result identifies the **Amortization of Deadhead** as the primary driver of mission efficiency:

- **Search-Intensive Sprints (Tier 1):** Although quoted at +75.7% above the target, the realized yield collapses to -49.9%. In these short-duration missions, the uncompensated temporal investment (searching and pickup) outweighs the revenue-generating phase.
- **Scale Resilience (Tiers 4-6):** Long-duration missions demonstrate the highest capacity to meet the sustainability benchmark, successfully diluting fixed search costs across a larger revenue base. However, the yield for Tier 6 represents rare interstate statistical anomalies. This specific metric excludes the

massive time required to return back to the city, which would halve the realized yield in a full-cycle context.

Having established the fundamental economic archetypes, the study proceeds to the discovery of the geospatial dimension to identify the organic operational zones of the city.

Geo-spatial Clustering: HDBSCAN

The geospatial objective of this phase was to transform $\approx 4,700$ raw coordinates into a **Topological Operational Model**. This mapping converts raw spatial distributions into a high-value categorical feature (`zone_id`), enabling the downstream predictive pipeline to recognize geographic intent and operational risk.

Geospatial Constraints: The Dropoff Imperative

A fundamental architectural decision in this topology was the **exclusive focus on dropoff coordinates**. Pickup locations were intentionally omitted from the feature engineering phase for two primary reasons:

1. **Data Integrity:** Over 90% of raw pickup address strings exhibited extreme ambiguity, introducing high-risk noise into the dataset.
2. **Decision Theory:** Since the Agent is geographically indifferent to pickups; the decision to accept an offer is driven by *Time-to-Pickup* (TTP) rather than the specific neighborhood of origin. Conversely, the destination (dropoff) dictates the Agent's future geographic position, making it the primary determinant of subsequent operational risk and return.

The Initial Clustering Attempt & The Coordinate Crisis

To discover these organic dropoff zones, **DBSCAN** was initially deployed. However, it resulted in data fragmentation that lacked operational business logic. The architecture was subsequently upgraded to **HDBSCAN** (Hierarchical DBSCAN), which dynamically extracts clusters across multiple density scales, proving much more capable of capturing Mexico City's complex operational reality.

In an attempt to achieve 100% feature coverage, a **K-Nearest Neighbors (KNN)** classifier was deployed alongside HDBSCAN to re-assign unclustered noise points to their nearest cluster. This algorithmic fallback was quickly identified as a functional failure: by forcing orphaned points into the nearest core based strictly on Euclidean distance, the resulting map lost all operational relevance.

More critically, while the HDBSCAN model was mathematically sound, a deep validation audit cross-referencing the coordinate clusters against the raw `dropoff_address` text strings revealed a systemic data integrity failure. Inspection of the *AICM* (Airport) cluster, for example, revealed textual addresses belonging to distant parts of the city, such as Cuajimalpa.

While the algorithm had correctly identified density clusters, the underlying geocoded coordinates had become completely decoupled from their true textual addresses during the upstream ingestion phase. This "Garbage In, Garbage Out" (GIGO) corruption entirely invalidated the initial unsupervised topology and necessitated a major remediation campaign before any machine learning could proceed.

Geospatial Remediation: Root Cause Analysis

The discovery of decoupled coordinates within the initial clustering results triggered a deep **Root Cause Analysis (RCA)**. The investigation identified three compounding failure vectors responsible for the data corruption:

1. **Validation Gap:** The initial Phase 1 data acquisition utilized forward geocoding without a reverse-validation step, allowing API hallucinations to persist undetected in the dataset.
2. **Semantic Entropy:** Contradictory neighborhood labels within the platform's raw address strings (e.g., "*Calle Insurgentes Tacubaya, Condesa/Roma*") caused the black-box routing API to default coordinates to entirely incorrect boroughs.

3. **Propagation Error:** A logical bug during the Phase 2B jittering stage caused hundreds of distinct textual addresses to mistakenly collapse into a single, identical coordinate point.

Diagnostic Engineering

Before the dataset could be repaired, the corrupted records needed to be isolated without compromising the healthy data. To achieve this, a diagnostic SQL query was engineered based on a strict relational rule: **Fixed Coordinates + Changing Text = Error**. By grouping identical coordinates and counting distinct address strings, the pipeline successfully flagged the specific anomalies requiring intervention.

```
-- Logic: If one exact coordinate belongs to multiple distinct text addresses, it
is corrupted.
SELECT offer_id,
       CASE WHEN c.num_pickups > 1 THEN 'PICKUP_ERROR'
            WHEN c.num_dropoffs > 1 THEN 'DROPOFF_ERROR'
       END as diagnostic_flag
FROM offers o
JOIN (
  SELECT pickup_lat, pickup_lon, dropoff_lat, dropoff_lon,
         COUNT(DISTINCT pickup_address) as num_pickups,
         COUNT(DISTINCT dropoff_address) as num_dropoffs
  FROM offers
  GROUP BY pickup_lat, pickup_lon, dropoff_lat, dropoff_lon
  HAVING COUNT(DISTINCT pickup_address) > 1 OR COUNT(DISTINCT dropoff_address) >
         1
) c ON o.pickup_lat = c.pickup_lat AND o.dropoff_lat = c.dropoff_lat;
```

Listing 1: Diagnostic Query: Isolating Jitter Propagation Errors

In addition to the Jitter Propagation Errors, a six-stage **Human-in-the-Loop (HITL)** reconciliation pipeline was executed across all 9,530 coordinate pairs:

1. **Topological Audit:** The initial HDBSCAN output was cross-referenced against raw address strings to identify first-layer corruption, flagging offers where the semantic text contradicted the assigned geospatial cluster.
2. **Polygon Synthesis:** The agent manually architected 72 custom heuristic polygons (Figure 22) to serve as operational “Indifference Zones”—geographic areas defining the operational reality.
3. **Heuristic Audit (Polygons):** A secondary manual audit verified point-in-polygon alignment, flagging offers whose string addresses did not belong to the polygon zone.
4. **Cloud Re-geocoding:** In order to clean flagged offers and to detect anomalies invisible to visual inspection, the entire dataset was re-geocoded, both forward and reverse.
5. **Geospatial Delta Check:** By calculating the spatial variance (in meters) between the original coordinates and the secondary baseline, records exceeding a severe geographic deviation threshold were isolated for intervention.
6. **Deterministic Overwrite:** A bespoke ETL tool was engineered to manage the final manual corrections. The system staged flagged conflicts alongside verified API context, enabling the Agent to surgically overwrite corrupted coordinate pairs in the source tables while preventing manual entry errors.

Outcome: This intervention successfully corrected over 1,500 corrupted rows—nearly one-third of the operational history. The canonical Source of Truth was updated with this purified dataset, enabling the re-execution of the clustering pipeline with high-fidelity results.

— PLACEHOLDER TO INTRODUCE THE MAP —

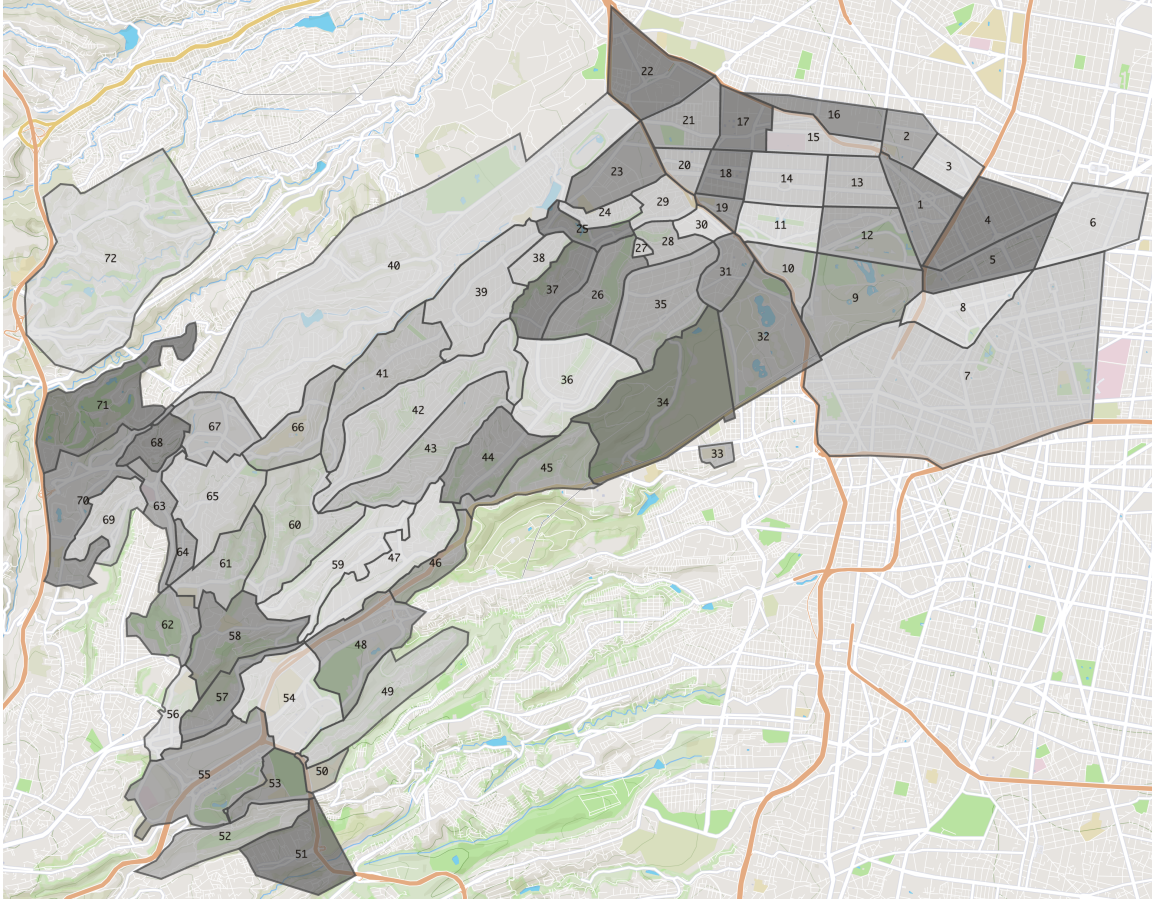


Figure 22: Strategic Ground Truth: 72 hand-crafted heuristic polygons defining the Agent’s operational zones.

ID	Zone Name	ID	Zone Name	ID	Zone Name	ID	Zone Name
1	anzures	19	polanco_grupo_mexico	37	nodo_monte_libano	55	santa_fe_centro_comercial
2	anahuac_1	20	polanco_uber_hq	38	tecamachalco	56	cruce_echanove
3	bahias	21	irrigacion	39	de_las_fuentes	57	carretera_libre
4	rios	22	sotelo	40	herradura_conscripto	58	vistahermosa
5	juarez_rosa	23	sedena	41	de_los_bosques	59	tamarindos
6	juarez_soho_house	24	reforma_social	42	ahuehuetes_norte	60	bosques_pabellon
7	roma_condesa_2	25	fuentes_casino	43	ahuehuetes_sur	61	loma_de_la_palma
8	roma_condesa_1	26	lomas_barrilaco	44	reforma_regina	62	carretera_al_olivo
9	bosque_1	27	lomas_olimpo	45	lomas_altas	63	vialidad_de_la_barranca
10	campo_marte	28	lomas_prado_norte	46	reforma_bnp	64	el_olivo
11	polanco_parque_lincoln	29	palmas_jp_morgan	47	agwa_bezares	65	blvrd_anahuac
12	polanco_gandhi	30	lomas_fc_cuernavaca	48	santa_fe_patio	66	universidad_anahuac
13	polanco_5	31	lomas_virreyes	49	santa_fe_quintana	67	interlomas_magnocentro
14	polanco_parroquia	32	bosque_2	50	santa_fe_tec	68	ave_club_de_golf_lomas
15	lagos	33	bondojito_asf	51	santa_fe_cumbres_de	69	interlomas_haciendas
16	frontera_polanco	34	bosque_3	52	santa_fe_bosques_de	70	jesus_del_monte
17	carso_antara_miyana	35	lomas_trastevere	53	santa_fe_colegios	71	lomas_country_club
18	polanco_palacio	36	nodo_reforma_palmas	54	santa_fe_ibero	72	bosque_real

Table 14: Geospatial Dictionary: Registry of the 72 operational polygons.

Emergent Topologies

By running HDBSCAN on the rectified dataset, the algorithm identified 44 granular, high-resolution micro-clusters (Figure 23).

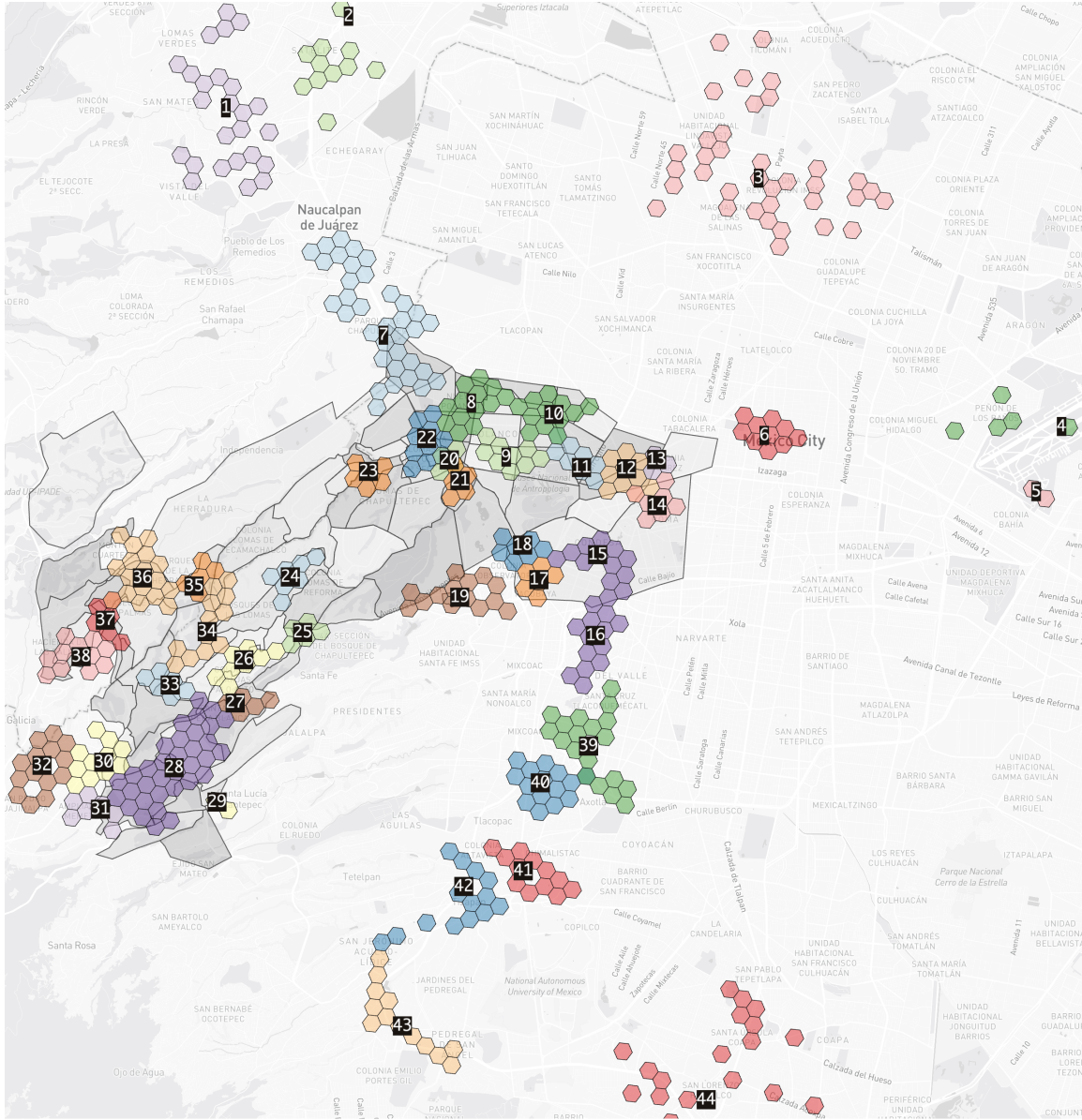


Figure 23: Topological Machine Truth: HDBSCAN clusters aggregated via H3 hexagons, revealing the organic "Gravity Wells" of the marketplace.

ID	Hub Name	ID	Hub Name	ID	Hub Name	ID	Hub Name
1	lomas_verdes	12	la_diana	23	cofre_de_perote	34	chamizal
2	plaza_satélite_mundo_e	13	col_juárez	24	duraznos	35	lomas_anáhuac
3	central_del_norte	14	roma_norte	25	tamarindos	36	vialidad_de_la_barranca
4	terminal_1_aicm	15	condesa	26	nodo_constituyentes_reforma	37	interlomas_magnocentro
5	terminal_2_aicm	16	nápoles_wtc	27	santa_fe_patio	38	haciendas_san_fernando
6	centro_histórico	17	san_miguel_chapultepec	28	santa_fe_core	39	felix_cuevas
7	sotelo_san_esteban	18	tacubaya	29	santa_fe_itesm	40	barranca_del_muerto
8	carso_antara_miyana	19	observatorio	30	cruce_echánove	41	san_ángel_altavista
9	campos_elíseos	20	fc_cuernavaca	31	santa_fe_abc	42	san_jerónimo
10	polanco_trebol	21	virreyes	32	cuajimalpa_centro	43	pedregal
11	anzures_reforma	22	el_semáforo_de_palmas	33	pabellón_bosques	44	viaducto_tlalpan

Table 15: Registry of the 44 machine-discovered operational hubs.

This output produced several key operational discoveries that were entirely invisible to standard geocoding APIs:

- **The Tacubaya Emergence:** A distinct, high-volume operational hub was discovered in *Tacubaya* only after the coordinate surgery. Prior to remediation, these offers were being incorrectly misallocated to the *Juárez* neighborhood.
- **Geometric Density vs. Operational Reality:** Relying purely on spatial density, the algorithm merged the *Haciendas* and *San Fernando* zones into a single cluster. While geographically adjacent on a map, these areas are operationally divided due to severe terrain and infrastructure constraints. This limitation confirmed that unsupervised models inherently lack operational context, underscoring the absolute necessity of Human-in-the-Loop (HITL) heuristics. Nevertheless, this specific algorithmic merger was intentionally preserved in the final output to respect the organic, data-driven integrity of the unsupervised model.

It is critical to note that the resulting map does not attempt to model the total ride-hailing universe of Mexico City. It is a strictly constrained mirror of the **Agent’s Operational Reality**. To maintain policy fidelity, the Agent never altered his heuristics simply to capture more data; consequently, reflecting only the specific times and zones of operations.

Deploying 72 granular polygons to classify a boutique dataset ($N \approx 4,700$) inherently introduced a high risk of model overfitting. To mitigate this risk, structural constraints were placed on the downstream predictive models; techniques such as *Feature Coalescence* and algorithmic regularization were implemented to ensure the model learned broad strategic principles rather than memorizing highly specific operational zones.

Final Feature Synthesis: The Silver Palette

To maintain the project’s directive of ensuring all analytical notebooks ingest data exclusively from a centralized SQL engine—a final feature table was materialized within `pienza.db` prior to the supervised learning phase. Unlike the primary schema established in Phase 2, this table was programmatically injected into the terminal stage of the ETL pipeline to consolidate the topological discoveries and stateful market signals.

Stateful Intelligence: The Volatility Suite

The causal analysis in Phase 3 identified a critical informational gap: the system’s *ex-ante* traffic index is a poor proxy for operational risk. The true determinant of yield is the **Prediction Error**⁹—the stochastic delta between estimated and realized durations. To quantify this uncertainty, a suite of four stateful features was engineered using a 2 min/km market baseline:

- **Realized Traffic Index (`realized_traffic_index`):** The post-facto ground truth of mission friction. It is calculated as the ratio of realized duration to the quoted expectation. This metric is strictly reserved for mission-completion audits.
- **Historical Traffic Context (`hist_avg_traffic_index`):** A session-level rolling average of previously realized indices. This feature serves as the technical proxy for the Agent’s “contextual memory” of recent urban congestion at the moment of the request.
- **Traffic Volatility ML (`traffic_volatility_index_ML`):** The primary predictive signal, calculated as the *Historical Traffic Context* minus the *Current Offer’s Expected Index*.
- **Traffic Volatility EDA (`traffic_volatility_index_EDA`):** Measures the absolute deviation between the realized duration and the expected quote; only for completed missions.

⁹While localized congestion is the dominant driver of duration variance, the prediction error also incorporates secondary operational factors such as trip-profile modifications or early drop-off requests. These edge cases were preserved within the analytical set to ensure the model captures the full spectrum of real-world operational volatility.

cid	Feature Name	Data Type
1	pickup_polygon_id	INTEGER
2	pickup_h3_hex_id	TEXT
3	dropoff_polygon_id	INTEGER
4	dropoff_h3_hex_id	TEXT
5	dropoff_hdbscan_id	INTEGER
6	realized_traffic_index	REAL
7	hist_avg_traffic_index	REAL
8	traffic_volatility_index_ML	REAL
9	traffic_volatility_index_EDA	REAL

Table 16: Structure of the `silver_palette` table: the final layer of features for Phase 5.

The completion of the *Silver Palette* marks the conclusion of the foundational data lifecycle. This journey progressed from high-fidelity acquisition (Phase 1) and structural normalization (Phase 2) to causal discovery (Phase 3) and the final topological mapping of the environment (Phase 4).

The project’s governing philosophy throughout these stages was to engineer an expansive **universe of all possibilities**—capturing the total dimensionality of the Agent’s reality. Recognizing that not all features are created equal, this high-resolution foundation was built specifically to afford the luxury of rigorous curation. By generating a dense field of 65+ potential signals, the project ensured that subsequent selection protocols could ruthlessly isolate only those variables with the highest predictive value. With the intelligence layer fully forged, the study now transitions to **Phase 5: Supervised Learning**, where this distilled signal is used to clone the Agent’s decision boundary.

PHASE 5

Supervised Imitation Learning

Tech Stack:	XGBoost, Naive Bayes, Logistic Regression, SHAP (Shapley Additive exPlanations), Mutual Information (MI), Chi-Squared (χ^2), Stratified K-Fold CV.
Core Challenge:	Overcoming the "Gravitational Well" of highly imbalanced data, where deterministic majority-class noise (obvious rejections) cannibalizes the machine's ability to learn nuanced, high-level strategy.
Key Insight:	The agent's decision-making is hierarchical, not flat. By deploying a "Cognitive Cascade" that triages basic logistical failures first, a secondary model can dedicate its full capacity to mapping complex behavioral mechanics like the Sunk Cost Effect and Income Targeting.
Operational Outcome:	The successful deployment of a hierarchical XGBoost clone that matched the human agent's strategic efficiency while capturing a 5.7% volume surplus by eliminating cognitive fatigue.

Phase 5 - Supervised Imitation Learning: From Sky to Psyche

Timeline: January 3 – January 12, 2026

The objective of Phase 5 was to synthesize a predictive model capable of replicating the Agent’s decision policy with high fidelity. This phase documents the transition from descriptive discovery to the engineering of a predictive inference engine via a comparative algorithmic tournament.

1. Experimental Design: The Tri-League Architecture 2. Categorical Features Selection 3. Model Tournament

Experimental Design: The Tri-League Architecture

To evaluate the interaction between data representation and model performance, the project utilized three distinct feature sets designated as “Leagues.” This framework isolates the impact of expert-led curation versus automated abstraction across different algorithmic families.

League	Basis	Dims.	Technical Objective & Alignment
League A	Wide Set (PCA)	19 PCs	Control Group: Retains 90% variance of the original 41-feature set to validate that L1-Lasso pruning did not destroy latent signals.
League B	Curated (Raw)	20 Feat.	Interpretability Mandate: Utilizes raw scaled variables to enable SHAP-based forensics and non-linear geometric splits for tree-based models.
League C	Curated (PCA)	12 PCs	Mathematical Purity: Eliminates multicollinearity via orthogonal components to provide a frictionless environment for linear and Bayesian estimators.

Table 17: Tri-League data representations for the algorithmic tournament ($N \approx 4,760$).

Geospatial Feature Coalescence

To minimize cardinality and prevent signal fragmentation, the 73 manual polygons and 45 HDBSCAN clusters were consolidated into a unified spatial taxonomy. Micro-polygons were semantically concatenated into **42 Macro-Zones** to ensure sufficient observation density (N) per category. Final feature assignment (`final_zone_id`) followed a deterministic three-tier cascade:

- **Tier 1 (Manual Priority):** Human-defined Macro-Zone boundaries override density clusters to enforce strategic intent.
- **Tier 2 (Machine Fallback):** HDBSCAN clusters capture hyper-dense infrastructural nodes (*e.g.*, *airport terminals*) located outside manual zones.
- **Tier 3 (Unassigned):** Residual coordinates are explicitly categorized as noise.

///// agregar que esto resuelve el problema del hdbscan, al forzar poligonos como el del itesm o el de haciendas que no incluyan zonas próximas (las aguilas; san fernando) que fueron incluidas en hdbscan porque excluidas por el coalesce ///7

Categorical Feature Selection

To finalize the feature matrix for supervised learning, a statistical audit was conducted to measure the predictive utility of all candidate categorical variables. The project utilized **Mutual Information (MI)** to identify non-linear dependencies and the **Chi-Squared (χ^2)** statistic to evaluate independence between features and the decision target.

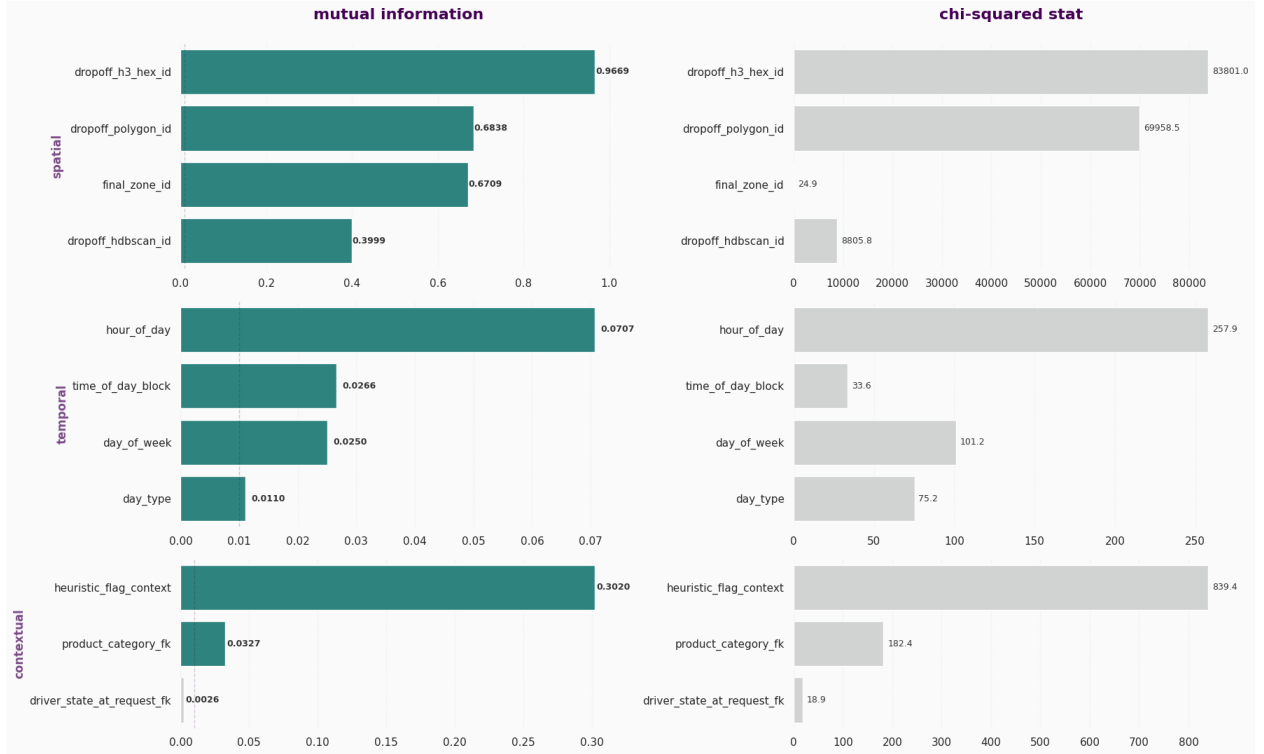


Figure 24: Categorical Feature Importance Audit: Mutual Information (Left) and Chi-Squared (Right) scores. The audit exposes the dominance of geospatial topology and the failure of several human-engineered abstractions.

The audit identifies `final_zone_id` (MI: 0.671) as the optimal spatial predictor. While high-cardinality grids like H3 with resolution 9 (MI: 0.967) achieve higher raw scores, they risk memorizing the training set. Consolidating 73 polygons into 42 Macro-Zones successfully filtered geometric noise while preserving the variable’s predictive power.

The results confirm that raw clock resolution significantly outperforms semantic abstractions. High-resolution features such as `hour_of_day` (MI: 0.071) and `day_of_week` (MI: 0.025) were retained as dominant drivers. Conversely, human-engineered features like `time_of_day_block` were purged, as their rigid definitions masked critical market variance.

The engineered `heuristic_flag_context` emerged as the primary non-spatial signal (MI: 0.302), validating the "Human-in-the-Loop" extraction phase. In contrast, the Agent’s immediate operational state (*Idle vs. On-Trip*) was identified as statistical noise (MI: 0.002) and removed to optimize model focus.

Domain	Feature	MI Score	Technical Verdict
Spatial	<code>final_zone_id</code>	0.6709	Keep: High-resolution zones derived from macro-polygon and cluster coalescence.
	<code>hour_of_day</code>	0.0707	Keep: Captures critical high-frequency demand cycles.
Temporal	<code>day_of_week</code>	0.0250	Keep: Preserves unique daily market physics.
	<code>heuristic_flag_context</code>	0.3020	Keep: Primary driver; encapsulates strategic intent.
Contextual	<code>product_category_fk</code>	0.0327	Keep: Retained to model service-tier behavioral shifts.
	<code>driver_state_at_request_fk</code>	0.0026	

Table 18: Final Categorical Feature Set advanced to Supervised Learning ($N \approx 4,760$).

The high-signal categorical features identified in this audit were integrated into each of the three experimental Data Leagues. This finalized feature space provides the standardized foundation for the subsequent **Algorithmic Tournament**, where various model architectures are evaluated across these refined data regimes.

Algorithmic Justification: From First Principles to Ensemble Apex

To identify the optimal architecture for behavioral cloning, a competitive tournament was executed across five developmental trials. Models were evaluated using the **F1-Macro** score to ensure predictive balance across the 92% class imbalance.

A priori selection of high-complexity models was rejected in favor of an iterative tournament designed to establish the statistical performance floor and quantify the non-linear complexity of the Agent's decision policy. This sequence serves as the formal justification for the eventual deployment of gradient-boosted ensembles.

- **Trial 1: Establishing the Statistical Floor (Naive Bayes).** Gaussian Naïve Bayes was deployed to measure performance under the assumption of feature independence. This trial established the baseline "failure rate" for probabilistic models in a high-imbalance, stateful environment.
- **Trial 2: Quantifying the Linear Ceiling (Logistic Regression).** By testing linear decision boundaries across both chronological and stratified splits, the project measured the "Linear Limit" of the dataset.
- **Trial 3: The State-Time Tension (LR Stratified).** This controlled test compared chronological purity against stratified data efficiency. It exposed the "shuffling illusion" common in boutique datasets, where over-optimistic scores can mask a failure to generalize to future market regimes.
- **Trial 4: Auditing Non-Linear Logic (Decision Tree).** A single tree was utilized as a "Scout" to audit hard logic splits. This trial provided the empirical proof that raw, uncompressed variables (*LIGA B*) significantly outperformed PCA-abstracted features (*LIGA C*) when navigating non-linear decision rules.
- **Trial 5: Architectural Apex (XGBoost).** Armed with the quantified proof of non-linearity and the superiority of raw features, the project deployed a gradient-boosted forest on a consolidated five-week "training gym." This finalized the champion architecture required to capture the nuanced interaction between market physics and agent heuristics.

Trial	Algorithm	Validation Method	LIGA A	LIGA B	LIGA C
1. Floor	Gaussian NB (Hybrid)	Chronological (Custom)	0.257	0.257	0.256
2. Linear	Logistic Regression	Time-Series Split	0.482	0.514	0.490
3. Theoretical	Logistic Regression	Stratified K-Fold	0.687	0.716	0.691
4. Scout	Decision Tree (d=7)	Time-Series Split	0.336	0.474	0.373
5. Champion	XGBoost Classifier	Stratified K-Fold	0.760	0.761	<i>Retired</i>

Table 19: Performance Ledger: F1-Macro averages across the five developmental trials ($N \approx 4,760$).

Champion Selection: XGBoost

Hyperparameter Calibration: The Down-Tuning Strategy

Following the identification of XGBoost as the champion architecture, a calibration phase was executed to optimize the model for generalization. The initial **RandomizedSearchCV** search identified a performance peak at `max_depth = 5`, as illustrated in the tuning radiography (Figure 25). However, subsequent learning curve audits revealed a high-variance gap, suggesting that the model was beginning to overfit the noise of the five-week training window.

To ensure robustness against the unseen Week 6 holdout set, a deliberate **Down-Tuning** strategy was implemented. By restricting the tree depth and increasing stochasticity, the algorithm was forced to prioritize broad, high-signal features over local data clusters. This increased regularization pressure traded a marginal

percentage of training accuracy for a significant improvement in the model’s ability to generalize to novel market regimes.

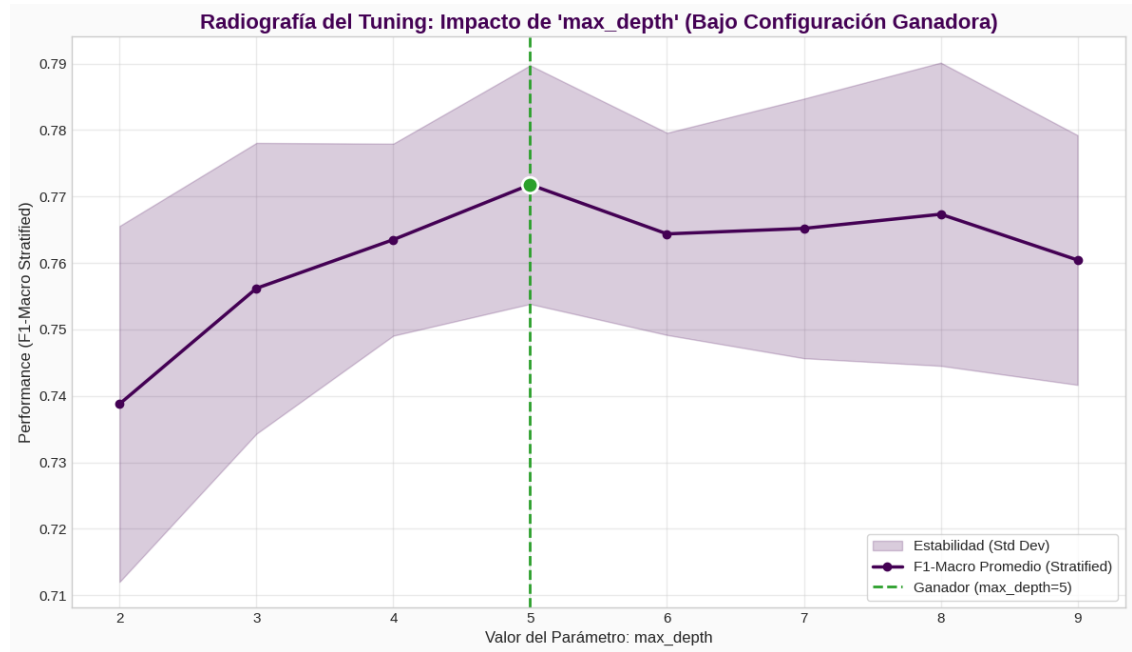


Figure 25: Tuning Radiography: Impact of `max_depth` on F1-Macro performance. While the statistical peak occurs at 5, the model was finalized at 4 to ensure defensive generalization.

Hyperparameter	Final Value	Engineering Rationale
n_estimators	200	Balanced to allow sufficient iterations for convergence at a lower depth limit.
max_depth	4	Down-Tuned from 5 to increase regularization and prevent overfitting on small-N zones.
learning_rate	0.1	Adjusted to accelerate the learning process within the shallower tree structure.
subsample	0.7	Down-Tuned from 0.9 to introduce more randomness per tree, forcing reliance on stable global signals.
min_child_weight	3	Enforced to ensure each split is supported by a minimum volume of evidence.
gamma	0.5	Applied to penalize split complexity and reduce model variance.

Table 20: Final XGBoost configuration following the robustness calibration.

The “State vs. Time” Dilemma

The “State vs. Time” dilemma presented a significant architectural trade-off when modeling on a restricted, short-horizon dataset. Enforcing strict chronological purity—such as sequential walk-forward splits across a mere five-week training window—severely restricts the data volume and the number of learning iterations available to the algorithm. The primary limitation encountered was not an uneven distribution of target classes over time, but rather the data starvation inherent to temporal splits on small datasets, which prevents the model from iteratively refining its logic.

Fortunately, this structural bottleneck was mitigated by the project’s pre-existing feature architecture. Because the Agent’s temporal context (e.g., `hour_of_day`, `day_of_week`) and behavioral memory (e.g., `consecutive_rejects`, `accumulated_deadhead`, `cumulative_earnings`) had already been explicitly engineered into the

feature space, the operative "state" and exact temporal positioning of the agent were intrinsically captured within each independent observation; hence, the algorithm does not require sequential chronological ordering to accurately interpret the context of a decision. Consequently, utilizing Stratified K-Fold cross-validation to shuffle the observations within the training pool does not constitute true data leakage. Instead, it serves as a vital architectural mechanism to maximize data efficiency and iterative learning volume, allowing the model to robustly map the stateful logic before facing an untouched, out-of-time evaluation in Week 6.

Model Performance: Bimodal Learning

An audit of the normalized confusion matrix (Figure 26) reveals that the monolithic **XGBoost** champion operates across two distinct performance regimes. These regimes are defined by the degree of feature orthogonality and the presence of deterministic boundaries.



Figure 26: Normalized Confusion Matrix for the Monolithic **XGBoost** Champion. High-accuracy clusters (top-left) identify deterministic operational constraints, while the lower-right quadrant reveals the strategic signal decay in nuanced decision classes.

Regime 1: Deterministic Operational Constraints

The model achieves high accuracy ($> 85\%$) when evaluating physical, geographical, and hard-economic constraints: **proxy_zone** (90.2%), **low_profitability** (90.6%), **non_operational** (88.7%), and **long_pickup** (85.7%). These classes are effectively orthogonal; their boundaries are dictated by rigid geospatial coordinates or absolute time-and-fare thresholds. The gradient boosting trees isolate these patterns with minimal entropy, representing the model's "Triage" layer.

Regime 2: Conditional Strategic Nuance

Performance degrades significantly (41% ... 68%) when the decision boundary shifts to higher-order behavioral states: `accepted` (68.7%), `strategic_mismatch` (57.1%), and `expected_value_gamble` (41.5%). These decisions are not functions of isolated coordinates but involve dynamic calculations of opportunity cost and market velocity. The resulting "fuzzy" boundaries lead to increased entropy and predictive uncertainty.

The Gravitational Well Effect

The audit identifies a critical failure mode: the interaction between the majority and minority classes. Because `non_operational` constitutes $\approx 50\%$ of the dataset, it creates a statistical **Gravitational Well**. When the algorithm encounters a nuanced observation with high strategic uncertainty—such as an `expected_value_gamble`—it yields to the Bayesian prior of the majority class. This is proven by the fact that **34.1% of true strategic gambles are incorrectly swallowed by the non-operational prediction**.

Architectural Pivot: The Cognitive Cascade

The audit confirms that "cheap" triage logic (geospatial/economic filters) cannibalizes "expensive" strategic logic. To eliminate this interference, the project implemented the **Hierarchical Cognitive Cascade**. This architecture decomposes the multi-class problem into a sequence of specialized inference layers:

- **Layer 1 (The Bouncer):** A classifier designed to remove the "Regime 1" noise (*Non-Op, Proxy, Long Pickup*).
- **Layer 2 (The Strategist):** A secondary model operating strictly on the high-yield, surviving observations.

By removing the gravitational pull of the non-operational class in Layer 1, the Strategist model can dedicate its full capacity to resolving the subtle economic differences between strategic rejections and acceptances, effectively breaking the performance ceiling identified in the flat architecture.

Layer 1: The Balanced Bouncer Protocol

Layer 1 is architected as a high-recall filter designed to isolate deterministic operational noise from strategic signal. This "Bouncer" ensures that the majority classes (*Non-Op, Low Profit*) do not overwhelm the more nuanced strategic rejections in the subsequent layer.

The Threshold Paradox: Street vs. Code

The calibration of this filter is driven by an identified asymmetry in risk tolerance between the physical agent and the digital system:

- **The Agent (The Street):** Prioritizes the avoidance of **False Positives** (accepting a "Trap" offer). This results in a high internal threshold and a conservative decision bias.
- **The System (The Code):** Prioritizes the avoidance of **False Negatives** (discarding a "Super Gem" as noise). Since a lost opportunity is irrecoverable, the system utilizes a low threshold to maximize the volume of nuanced signal passed to Layer 2.

Priority-Based Decision Logic

To operationalize this strategy, Layer 1 replaces standard *argmax* classification with a priority-based rule. The predicted class $\hat{y}$ is determined by two calibrated thresholds:

$$\hat{y} = \begin{cases} \text{Low Profit} & \text{if } P(\text{Low Profit}) \geq 0.25 \\ \text{The Nuanced Rest} & \text{else if } P(\text{Nuanced}) \geq 0.20 \\ \text{argmax}(P) & \text{otherwise} \end{cases} \quad (6)$$

Hierarchy: Blocking "Low Profit" data always takes precedence over rescuing "Nuanced" signal to enforce economic discipline.

Results: Filter Efficiency

The performance of the filter is measured by its **Recall** (Filter Efficiency). As shown in Figure 27, the protocol achieved a 90.6% Recall for Low Profitability rejections and a 81.7% Recall the Nuanced Rest Class.



Figure 27: Layer 1 Confusion Matrix. The diagonal represents the filter's efficiency in removing noise, while the "Nuanced Rest" column represents the survival rate of strategic signal.

Outcome: By using high-recall filtering, the project successfully neutralized the Gravitational Well. The observations passed to Layer 2 are statistically balanced and represent the true strategic DNA of the expert policy.

To validate the triage logic of the Balanced Bouncer, a SHAP contribution analysis was executed for the THE_NUANCED_REST class. This "DNA Atlas" (Figure 28) identifies the features that qualify an offer for Layer 2 evaluation (Teal/Right) versus those that push it toward a deterministic rejection (Gray/Left).

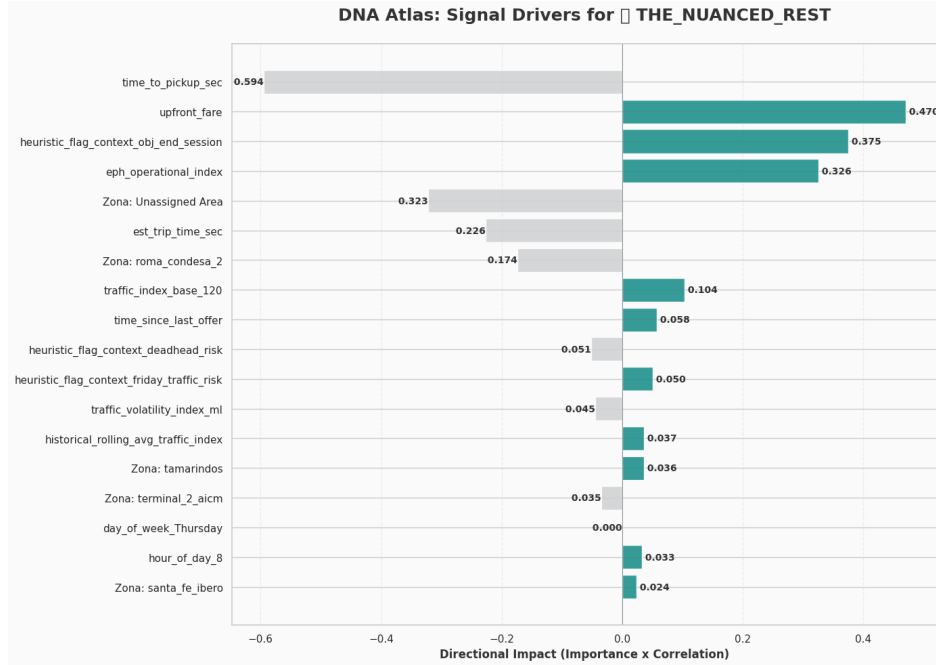


Figure 28: DNA Atlas: Signal Drivers for THE_NUANCED_REST. The chart illustrates the contribution of features to the triage decision. Logistical friction serves as the primary exclusion vector, while financial magnitude and strategic intent serve as the primary inclusion vectors.

The analysis reveals three primary pillars governing the triage stage:

- **The Logistical Barrier:** `time_to_pickup_sec` is the dominant negative driver (-0.594). Regardless of potential yield, high pickup latency is the primary reason an offer fails to reach the strategic tier. This confirms the model enforces strict logistical discipline at the entry point.
- **The Financial Magnet:** `upfront_fare` ($+0.470$) and `eph_operational_index` ($+0.326$) are the primary positive drivers. High absolute and relative earnings are the strongest signals for moving an offer into the nuanced category for strategic evaluation.
- **The Strategic Intent Trigger:** The positive contribution of the `obj_end_session` flag ($+0.375$) confirms that the model recognizes the "Endgame" state. When this intent is active, the model categorizes offers as "nuanced" rather than "low profit," allowing the subsequent Layer 2 model to evaluate them against homecoming vectors.

Outcome: The Layer 1 DNA confirms that the hierarchical filter has successfully learned the Agent's triage logic. By effectively isolating logistical noise and economic failures, the system ensures that Layer 2 operates on a high-signal subset of the marketplace.

Layer 2: The Strategist (Nuance Modeling)

The second stage of the cascade represents the "Nuance Engine" of the system. Operating exclusively on the high-yield subset that survived the Layer 1 filter, this model is designed to resolve the subtle distinctions between strategic rejections and acceptances without the interference of majority-class noise.

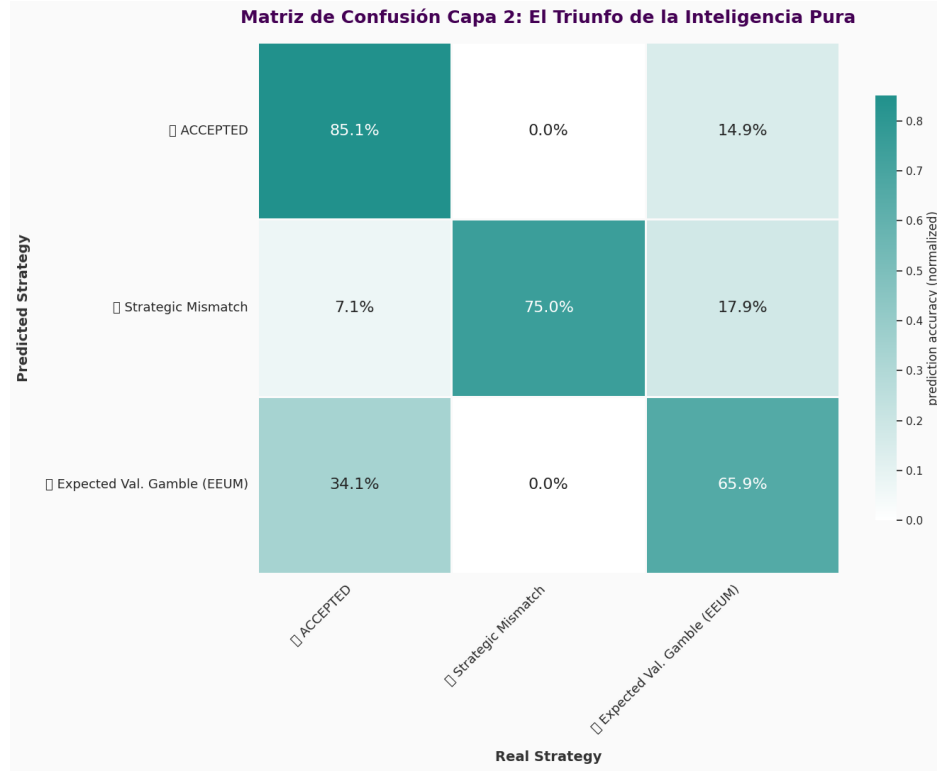


Figure 29: Layer 2 Confusion Matrix. By isolating the high-yield sub-universe, the model achieves superior separation between complex strategic states.

Results: Strategic Performance Lift

The transition to a hierarchical architecture resulted in a significant accuracy lift for the strategic classes (Figure 29):

- **Geospatial Strategy (`strategic_mismatch`):** Accuracy reached **75.0%**, confirming the model successfully mapped complex heuristics such as "Friday Gridlock Avoidance" and "Non-Homecoming Vectors."
- **Optimal Stopping (`expected_value_gamble`):** Accuracy rose to **65.9%**, a 58% relative improvement over the monolithic baseline. This proves that removing deterministic noise allows the algorithm to better isolate opportunity-cost signals.
- **Primary Target (`ACCEPTED`):** The model achieved **85.1% accuracy** in replicating the Agent's acceptance decisions, establishing a high-fidelity behavioral clone.

Threshold Tuning and the Pareto Frontier

A systematic grid search for optimal decision thresholds was conducted for the Strategist layer. However, the analysis identified a sharp **Pareto Frontier**: while the threshold for the `ACCEPTED` class could be tightened, doing so resulted in a disproportionate collapse of *Recall* for the `expected_value_gamble` and `strategic_mismatch` classes. Because the project prioritizes high-fidelity policy replication over raw binary accuracy, the thresholds were maintained at their balanced equilibrium to preserve strategic nuance.

Strategic DNA: Feature Drivers for Layer 2

To interpret the internal logic of the Strategist model, a SHAP contribution analysis was performed across the three nuanced strategic classes (Figure 30). This assessment quantifies the directional impact of features

on the final classification.

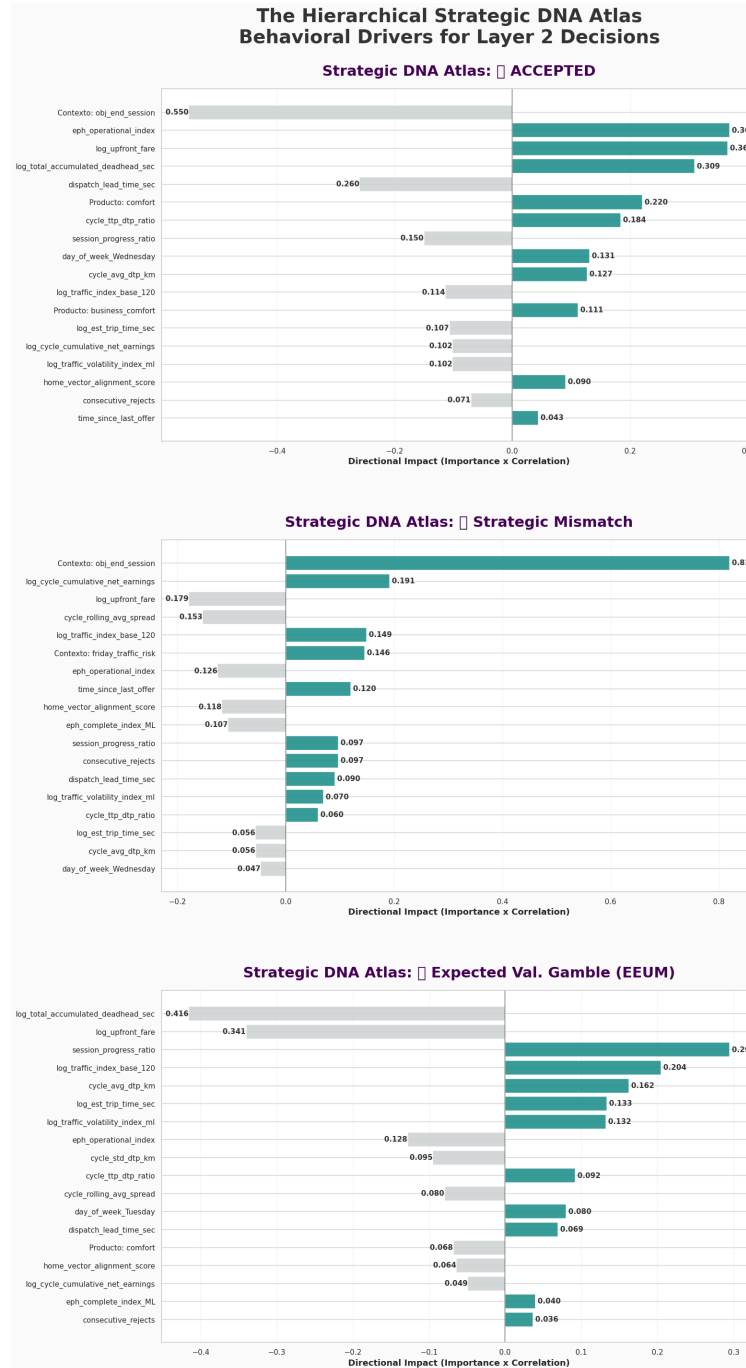


Figure 30: Hierarchical Strategic DNA Atlas: Feature contributions for Layer 2 decision classes. Teal bars indicate a positive contribution to class probability; Gray bars indicate a negative contribution.

The analysis reveals three primary behavioral mechanics that govern the expert agent’s high-level strategy:

- **The Sunk Cost Mechanic (Accepted):** While upfront_fare and eph_operational_index are the expected financial drivers, log_total_accumulated_deadhead_sec serves as a powerful positive predictor (+0.309). This provides empirical proof of a “Sunk Cost” effect: as uncompensated search time increases,

the Agent’s selectivity threshold decays, leading to a higher probability of acceptance. Conversely, `obj_end_session` acts as the strongest inhibitor (-0.550), suppressing acceptance regardless of yield.

- **The Income Targeting Mechanic (Strategic Mismatch):** This class is dominated by the `obj_end_session` context ($+0.819$). The model correctly identifies that high accumulated earnings (`log_cycle_cumulative_net_earnings`) pull the decision toward a mismatch rejection. This verifies the “Income Targeting” heuristic: once financial objectives are met, the Agent prioritizes homecoming vectors over marginal revenue gains.
- **The Patience Constraint (Expected Val. Gamble):** The primary predictor for the `expected_value_gamble` class is the **absence** of search fatigue. The strongest negative contributor is `log_total_accumulated_deadhead_sec` (-0.416). This identifies a critical behavioral constraint: the Agent only gambles on future offers when the current search cost is low. Prolonged search cycles eliminate the Agent’s patience, causing a shift from strategic gambling back to acceptance.

Outcome: The Layer 2 feature assessment proves that the XGBoost champion successfully mapped the Agent’s non-linear strategic logic. The quantification of the “Patience Threshold” and the “Sunk Cost Effect” provides the mathematical foundation for the generative simulation engine in Phase 6.

The Shadow Mode Trial: Behavioral Fidelity

To measure the operational impact of the architectural pivot, a final performance assessment was executed on the firewalled Week 6 holdout set ($N \approx 780$ offers). The finalized **Cognitive Cascade** was benchmarked head-to-head against the monolithic XGBoost baseline and the Human Agent’s actual decisions across two primary KPIs: cumulative upfront fare and strategic efficiency.

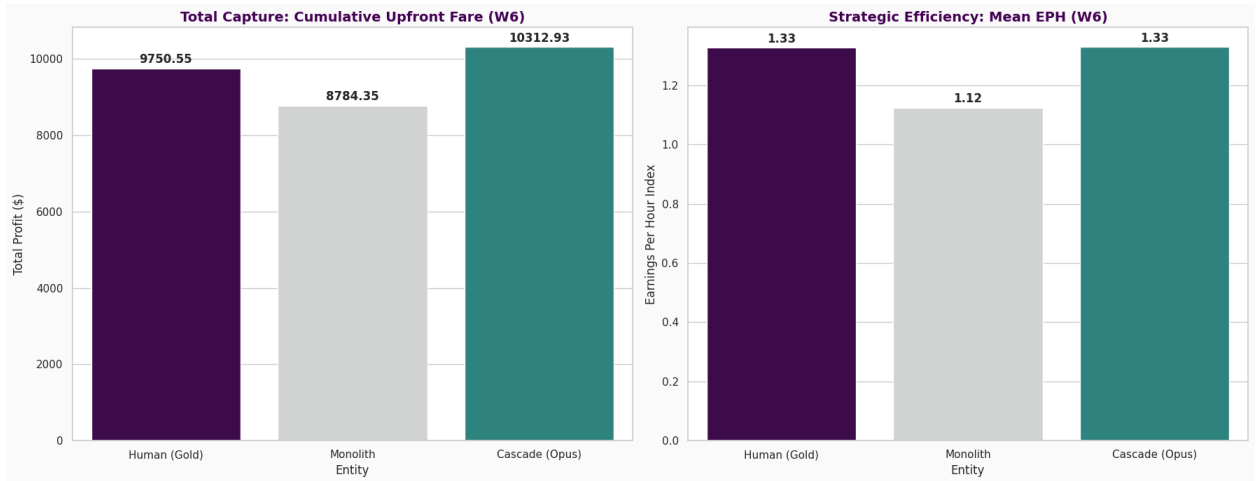


Figure 31: Shadow Mode Performance Results. The left plot measures total cumulative yield (\$), while the right plot measures strategic efficiency via the mean EPH Index. The Cascade architecture matches the expert’s efficiency while maximizing total value capture.

The trial results, illustrated in Figure 31, reveal three critical performance findings:

- **Efficiency Parity:** The **Cascade (Opus)** model achieved a mean EPH index of **1.33**, perfectly replicating the Human Agent’s strategic efficiency. This confirms the model successfully learned the high-yield selection criteria required to consistently outperform the market baseline (1.0).
- **Monolithic Performance Decay:** The monolithic model significantly underperformed, trailing the Human Agent by **15.7% in efficiency**. This provides empirical proof that flat architectures are unable to isolate strategic signal from geospatial noise, confirming the negative impact of the "Gravitational Well."
- **Volume Surplus (The Machine Advantage):** In a surprising outcome, the Cascade model captured **5.7% more total value** than the Human Agent (\$10,312 vs. \$9,750). While maintaining identical efficiency, the

machine identified high-yield opportunities that the human agent may have bypassed due to operational fatigue or cognitive load.

Outcome: The Shadow Mode trial validates the Cognitive Cascade as a high-fidelity behavioral clone. The model demonstrates the ability to match expert strategic efficiency while providing superior consistency in total value capture across a sustained work shift.

Phase 5 Conclusion: Breaking the Static Barrier

The conclusion of Phase 5 marks the definitive transition from static decision mapping to temporal flow modeling. While the *Cognitive Cascade* successfully isolated the Agent’s strategic logic, its performance highlights a fundamental constraint of tabular architectures: **Snapshot Indifference**. Even when enriched with stateful features, classical **XGBoost** evaluates offers as isolated events in a vacuum.

This technical boundary provides the formal validation for the transition to the **Generative Realm**. To fully deconstruct the *expected_value_gamble*—the decision to reject a viable offer based on the probability of a future event—the framework must move beyond independent row analysis. This justifies the scope of Phase 6, where Generative Adversarial Networks (GANs) will synthesize the high-volume data manifolds required to train Recurrent Neural Networks (RNNs). By transforming the market from a sequence of static snapshots into a continuous temporal flow, the Pienza framework moves toward modeling the “Expert Rhythm,” where every decision is processed as a dynamic function of the marketplace’s heartbeat.

PHASE 6

Generative Moonshots

Tech Stack:	NLP Transformers (miniBabel), Conditional GANs, Keras/ TensorFlow, PyTorch, Apache Spark, Google BigQuery, NetworkX, Markov Decision Processes (MDP), The Bellman Equation.
Core Challenge:	Overcoming the strict limits of physical data sparsity and high-latency APIs to build a scale-invariant "Digital Twin" capable of simulating multi-agent autonomous fleet dynamics.
Key Insight:	The urban environment is an autocentric "Tensor of Potentials," not just a geographic grid. Fleet optimization requires abandoning myopic fare-chasing in favor of probabilistic Markovian positioning, utilizing "Demand Depletion" to penalize greedy behavior and prevent swarm clustering.
Operational Outcome:	The synthesis of a 1,000,000-row operational manifold via cGANs and the derivation of the Q -Matrix policy, establishing the exact mathematical scaffolding required for autonomous execution in Pienza 2.0.

Phase 6: Generative Moonshots

With the Agent’s policy successfully cloned, the objective of Phase 6 is to transition the Pienza framework from a single-agent study into a scale-invariant **Digital Twin** of Mexico City’s ride-hailing marketplace. To achieve this, the project pursues three experimental deep-learning objectives designed to overcome the administrative and logistical bottlenecks of the physical world:

1. **The Quest to $O(1)$ Inference (NLP Transformers):** Real-time deployment of the XGBoost model is limited by the network latency and cost of external geocoding APIs. To bypass this, a locally-hosted *Text-to-Zone Transformer* was architected to classify raw address strings directly into strategic macro-zones, enabling sub-10ms inference.
2. **Generative Synthesis (Conditional GANs):** The limited ground-truth volume ($N \approx 4,700$) creates a data-sparsity constraint for robust market simulation. To resolve this, a **Conditional Generative Adversarial Network (cGAN)** was deployed to synthesize a 1,000,000-row manifold, effectively transforming the static dataset into an infinite *Keras* generator.
3. **Topological Routing (Graph-Network Analysis):** The synthesized manifold is utilized to execute Graph-Network analysis, mapping the economic connectivity between urban hubs and identifying the "path dependency" of high-yield mission sequences.

*Note: While this generative foundation sets the stage for advanced methodologies—including Recurrent Neural Networks (RNNs), Graph Neural Networks (GNNs), and Markov Chain Monte Carlo (MCMC) simulations—these specific applications are designated as future research and are formally proposed for the **Pienza 2.0** roadmap.*

The Quest to $O(1)$ Inference: NLP Transformers

In high-frequency dispatch environments, inference latency is the primary constraint for the viability of a real-time decision agent. To satisfy the requirements of *Online Transaction Processing (OLTP)*, the project analyzed three generations of urban inference through the lens of algorithmic efficiency and system scalability.

The current (Phases 1-5) architecture relies on external geocoding APIs and Point-in-Polygon (PiP) algorithms, constrained by $O(N \times M)$ geometric complexity and high network I/O latency ($> 500\text{ms}$). To bypass this bottleneck, the project engineered **miniBabel**: a locally-hosted Transformer that maps raw urban syntax directly to a **zone_id**. By treating geographic resolution as a task of semantic pattern recognition, the system collapses the search space into constant-time $O(1)$ neural inference¹⁰. This architecture reduces end-to-end latency to $< 10\text{ms}$ and provides a strategic bridge toward the high-availability horizon of bitwise H3 integer indexing (Table 21).

Architecture	Method	Complexity	Latency	Strategic Value
External API	Geometric PiP	$O(N \times M)$	High ($> 500\text{ms}$)	Research-only; high I/O risk.
miniBabel	Neural Inference	$O(1)$	Low ($< 10\text{ms}$)	Production-ready; autonomous.
Uber H3	Hash Lookup	$O(1)$ (Bitwise)	Ultra-Low ($< 1\text{ms}$)	Ideal target for fleet-scale telemetry.

Table 21: Comparative Efficiency Ledger: Algorithmic complexity and latency classes for urban inference.

¹⁰While the internal complexity of the attention mechanism is quadratic to the sequence length L , the bounded nature of urban address strings transforms the inference into a fixed-cost operation relative to the number of strategic zones.

Linguistic Feature Engineering

To transform high-entropy address strings into high-density neural signals, the project implemented a hierarchical preprocessing pipeline. This stage focused on strategic data augmentation to ensure the model generalizes across the Mexico City urban landscape:

- **Deterministic Normalization:** Collapsed thousands of syntactic variations (*e.g.*, *Av*, *Ave*, *Avenida*) into unified canonical tokens. This artificially increased signal density, allowing the subsequent neural architecture to focus on geographic disambiguation rather than basic syntactic learning.
- **Token Soldering:** Utilized high-precision Regular Expressions to fuse numeric and cardinal qualifiers with primary street identifiers (*e.g.*, *“Eje 6 Sur”* → *eje_6_sur*). This prevents the model from treating structural modifiers as isolated noise, anchoring them to their relevant spatial context.
- **Target Leakage Mitigation:** Deployed a “Slash-Purge” protocol to remove platform-generated neighborhood labels (*e.g.*, *“Roma/Condesa”*) from the raw strings. By pruning these explicit identifiers, the model is forced to learn latent geographic features—such as street-level syntax and intersection patterns.
- **Zip Code Standardization:** Standardized all 5-digit postal codes by prepending a unique *cp* prefix (*e.g.*, *“11590”* → *cp11590*). This transformation isolates zip codes from other numerical data, creating high-precision spatial anchors for the neural engine.
- **Semantic Embedding (Word2Vec):** This Skip-gram architecture was used as a tool to explore spatial proximity within the urban lexicon. By encoding tokens into 64-dimensional vectors, the model internalizes the semantic relationship between adjacent zones.

A Token Lift analysis validated the preprocessing pipeline by identifying words with disproportionately high frequencies in specific zones relative to their global usage.

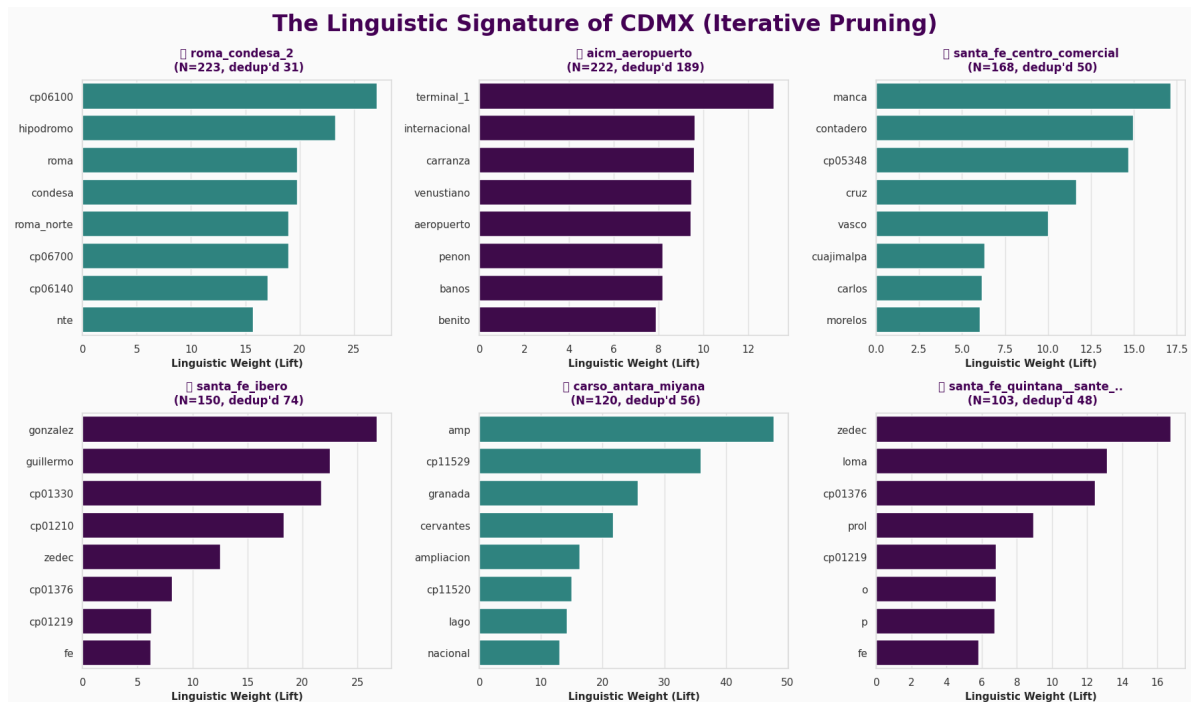


Figure 32: The Linguistic Signature of CDMX: Top tokens by Linguistic Lift. Standardized zip codes and distinct landmarks provide the strongest geographic signals for the neural engine.

As illustrated in Figure 32, the audit revealed three critical structural advantages:

- **Zip Code Dominance:** Prefixed postal codes (*e.g.*, *cp06100*) emerged as the strongest predictors for

major hubs, validating their isolation from general numerical noise.

- **Linguistic Specificity:** Iterative pruning successfully separated distinct neighborhood markers (*e.g.*, *manca*, *granada*) from generic urban vocabulary.
- **Infrastructural Anchors:** High-lift keywords like `terminal_1` effectively resolve dense transit nodes, complementing the spatial HDBSCAN clusters discovered in Phase 4.

Model Architectures: Custom vs. Pre-trained

To identify the optimal mapping function between urban syntax and strategic zones, the project benchmarked a custom-built Transformer against a state-of-the-art pre-trained baseline.

Model A: miniBabel (Custom Urban Transformer) To prevent catastrophic overfitting on the boutique dataset, the **miniBabel** architecture was restricted to two attention layers and utilized a high-dropout bottleneck ($p = 0.2$) prior to the classifier. To ensure high-value spatial anchors (*e.g.*, *specific street names*) were not diluted during processing, the model replaced standard Global Average Pooling with a **fused Max-Average pooling** mechanism. This allows the classifier to ingest both the holistic context of the address and its sharpest geometric signals simultaneously.

Model B: miniBETO (Spanish BERT Adaptation) For comparative benchmarking, the project adapted **BETO** (a pre-trained Spanish BERT model). Adapting a model trained on standard prose to condensed urban syntax required the replacement of the default classification head. To ensure a scientifically valid comparison, the modified miniBETO was equipped with the identical fused pooling and bottleneck constraints utilized in the custom miniBabel architecture.

Optimization and Training Constraints

Due to the inherent class imbalance of the `zone_id` distribution, both models utilized the **Weighted F1-Score** as the primary evaluation criterion. Training was governed by a multi-layered regularization protocol:

- **Dynamic Early Stopping:** Triggered by F1-Score plateauing to prevent over-training and preserve generalization.
- **Label Smoothing (0.1):** Applied to the loss function to penalize overconfident logit generation and improve the calibration of the final probability scores.
- **Bottleneck Regularization:** Aggressive dropout was enforced across both models to ensure the resulting spatial identifiers remain robust to the high-entropy noise of real-world address strings.

Performance Benchmarking: Precision vs. Complexity

To evaluate the efficacy of $O(1)$ neural inference, both the custom **miniBabel** and the adapted **BETO** models were benchmarked across identical training constraints. Performance metrics were selected to account for geographic class imbalances and operational throughput requirements.

Architecture	Params	Epochs	Acc.	Macro F1	Weighted F1
Custom Transformer (<i>miniBabel</i>)	$\approx 1.5\text{M}$	22	83.04%	0.79	0.8266
BETO (Spanish BERT)	$\approx 110\text{M}$	17	86.58%	0.84	0.8622

Table 22: Comparative performance metrics for the NLP urban inference tournament.

The results, detailed in Table 22, demonstrate that the pre-trained **BETO** architecture achieves the highest absolute performance, leveraging its 110-million parameter representation to generalize across minority zones with a 0.86 Weighted F1. However, the **miniBabel** model achieved a highly competitive 0.8266 score despite possessing less than 2% of the parameter count of the BERT baseline. This confirms that the hierarchical

linguistic engineering phase successfully condensed the urban syntax into high-density signals, allowing a lightweight 2-layer attention mechanism to resolve spatial clusters with high accuracy.

The selection between these architectures represents a fundamental trade-off between absolute precision and **Inference Latency**. While BETO provides a $\approx 3.5\%$ gain in F1 performance, miniBabel’s ultra-low parameter count is uniquely optimized for resource-constrained OLTP environments. By prioritizing the custom Transformer, the framework ensures inference latency remains below the 10ms threshold required for real-time dispatch operations.

To assess the qualitative performance of the **miniBabel** engine, a diagnostic audit was performed on raw, unstructured address strings. Table 23 illustrates the model’s ability to resolve spatial context from linguistic markers.

Raw Input Address (Obfuscated)	Ground Truth	Prediction	Conf.	Result
av universidad anahuac_## col lomas...	anahuac_norte	anahuac_norte	97.6%	HIT
av santa fe_### col santa fe...	sf_centro_com	sf_centro_com	87.2%	HIT
blvd loma real_## cp52774...	bosque_real	interlomas_hub	29.5%	MISS
anillo periferico blvd manuel...	lomas_chapultepec	polanco_grupo_m	76.7%	MISS

Table 23: Pienza Babel qualitative inference audit with address-number obfuscation.

The audit identifies two distinct behavioral characteristics of the neural inference engine:

1. **Confidence-Based Filtering:** The miss on *Bosque Real* was accompanied by a low confidence score (29.5%), allowing the system to flag the prediction as unreliable.
2. **Topological Adjacency Errors:** The high-confidence miss on *Anillo Periférico* highlights the limitation of purely linguistic models for arterial roads that are on the limit of adjacent zones.

While miniBabel successfully decouples dispatch from high-latency APIs via $O(1)$ probabilistic routing, topological boundary errors confirm that neural inference remains a statistical approximation. Its ultimate strategic value lies in acting as an offline compiler to map the existing address corpus into deterministic Uber H3 indices, establishing the foundation for absolute-precision, sub-millisecond dispatch.

Generative Synthesis: The Pienza Manifold

To overcome the constraints of data sparsity ($N \approx 4,700$), the project deployed a **Conditional Generative Adversarial Network (cGAN)** to synthesize a 1,000,000-row data manifold. Unlike standard generative models that learn a broad distribution, the **conditional architecture** allows the system to synthesize offers targeted at specific market states. By conditioning the model on attributes—such as **hour_of_day**, **product_tier**, and **zone_id**—the system can programmatically request synthetic trips for any desired operational scenario.

This implementation effectively transforms the model into an **Infinite Keras Generator**. Because the neural network functions as a continuous mathematical mapping, it is capable of streaming an unlimited volume of synthetic offers on-demand. While the generator provides this infinite capacity, a fixed 1,000,000-row data manifold was materialized as a static Parquet asset.

Reaching convergence required seven iterations of feature selection and spatial tuning. To prevent **Mode Collapse** (a state where the model outputs repetitive, low-variance data), the trip architecture was partitioned into two orthogonal vectors: *Context* and *Physics*.

Initial attempts utilized Uber’s H3 hexagonal grid at Resolution 6 ($\approx 36 \text{ km}^2$) for pickup origins. This coarse resolution proved insufficient, conflating distinct strategic sectors. Conversely, shifting to a hyper-granular Resolution 9 was a recipe for overfitting, forcing the model to memorize specific coordinates rather than learning urban distributions.

This was resolved by executing a **Point-in-Polygon (PiP)** spatial join. By mapping raw GPS pickups directly to the 42 semantic macro-zones developed in Phase 4, the system unified the spatial axis. This allowed the GAN to learn origin-destination economics without the noise of an abstract grid.

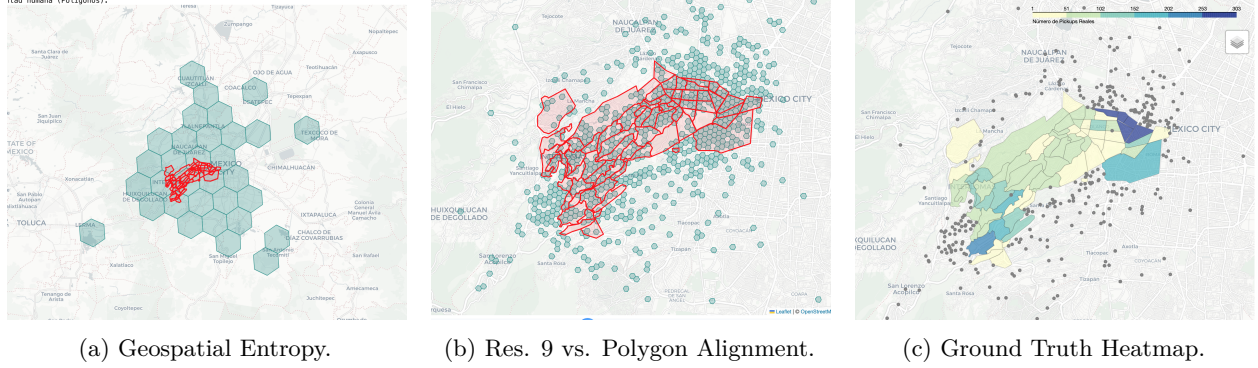


Figure 33: Geospatial Discovery: Identifying the resolution trade-off between hexagonal grids.

While the ideal model would replicate transient incentives (*e.g.*, *Surge*, *Turbo+*), these high-variance signals derailed the learning of baseline prices, as these features collapsed to zero after training. To prevent mode collapse and stabilize the Generator, incentives were excised from the feature space; the remaining variables were split into two tensors:

- **Context Switches (Conditioning):** The state of the world prior to the request (*e.g.*, *Hour of Day*, *Day of Week*, *Product Tier*, *Pickup/Dropoff Zone*).
- **Pure Physics (Generated):** The continuous variables the model must hallucinate (*e.g.*, *Upfront Fare*, *Estimated Duration*, *Trip Distance*, *Pickup Details*).

To stabilize the model's activations and ensure realistic outputs, three constraints were applied:

- **Log-Transformations:** Applied to heavily skewed features (*Fare*, *Distance*) to stabilize gradients during backpropagation.
- **MinMax Scaling** $[-1, 1]$: Compresses all physical features uniformly to prevent high-magnitude metrics (*e.g.*, *seconds*) from mathematically overwhelming smaller scalars (*e.g.*, *km*).
- **Oversampling:** *Premium* tier trips represent a minuscule fraction of the raw data. To prevent tier blindness, a $15\times$ oversampling multiplier was applied exclusively to this category, forcing the network to assimilate high-value, low-frequency market behaviors.

Neural Architecture: The Generator Pipeline

The Generator is architected as a conditional dense network designed to transform real observed data into synthetic, operationally-valid, ride offers. As illustrated in Figure 34, the model employs a four-stage funnel to synthesize market physics from categorical constraints.

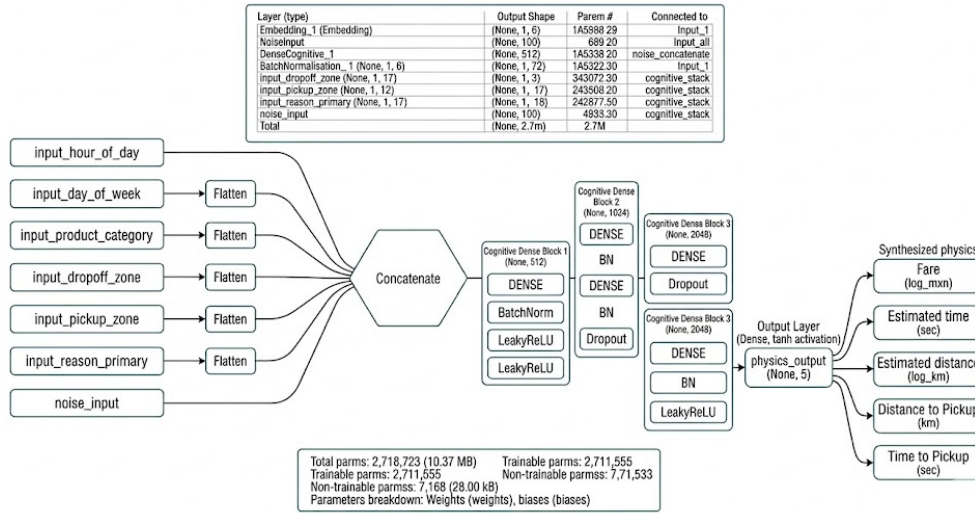


Figure 34: Technical Architecture of the Pienza Generator v7.

The Pienza Generator (v7) utilizes an embedding-concatenation strategy to map market constraints to physical outcomes. The six categorical variables at the input stage (*e.g.*, *temporal blocks*, *product tier*, *geospatial zones*) act as **conditional switches** rather than synthesized outputs. When the system is prompted with a specific market context—such as an 8:00 AM mission on a Monday from Zone A to Zone B—the model’s 2.7 million parameters synthesize the exact continuous physics (fare, durations, and distances) that correspond to those specific constraints.

To ensure stochastic variety and prevent repetitive outputs, 100 dimensions of Gaussian noise are fused with the embedded context vectors. This 133-dimensional input is then expanded through a series of three progressive **Dense Blocks** (ranging from 512 to 2048 neurons). Each block utilizes *Batch Normalization* and *LeakyReLU* activations to maintain mathematical stability during the adversarial training process. The architecture terminates in a five-neuron ***tanh* output layer**, which generates the normalized values for the core physics metrics. These are subsequently mapped back to real-world units via the inverse transformation of the data scaling protocol.

Outcome: This architecture allows for the on-demand generation of an infinite stream of statistically valid ride offers, providing the high-volume data manifold required for the subsequent simulation and deep learning phases.

Discriminator Architecture: The Adversarial Validator

The Discriminator (v7) serves as the adversarial validator within the cGAN framework. Its primary objective is to perform a binary classification task, distinguishing between ground-truth operational records and synthetic hallucinations. To provide the Generator with effective gradient pressure, the Discriminator is architected to evaluate both the physical magnitude of the offer and its environmental context.

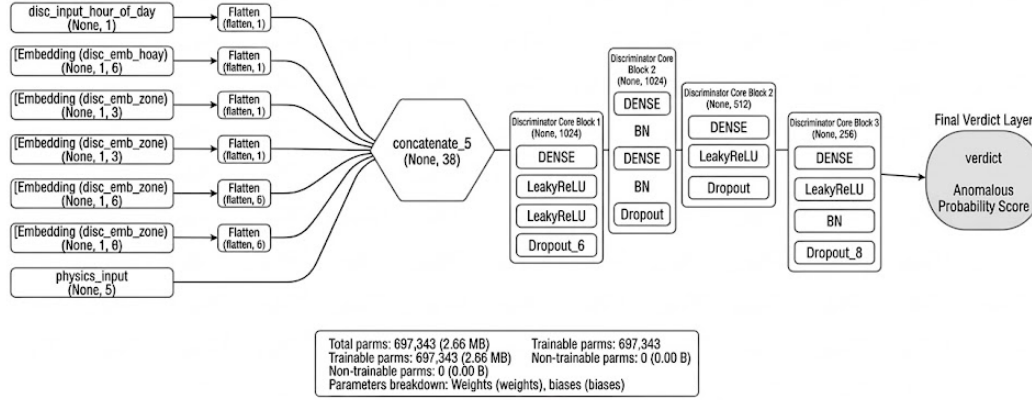


Figure 35: Architecture of the Pienza Discriminator v7.

The architecture utilizes a three-stage validation logic to render its verdict:

1. **Contextual Fusion:** The network ingests the six switches (*e.g.*, *temporal blocks*, *zone IDs*) through embedding layers. These categorical vectors are flattened and concatenated with the 5-dimensional physics input. This creates a unified 38-dimensional feature vector, ensuring the model evaluates the validity of the fare and duration **conditional** upon the specific market state.
2. **Regularized Funnel:** The concatenated signal is processed through a descending dense stack of 1024, 512, and 256 neurons. To prevent the Discriminator from reaching premature convergence—which would stall the Generator’s learning—the architecture implements aggressive **Dropout** layers ($p = 0.2$) following each *LeakyReLU* activation. This forces the model to prioritize robust statistical patterns over specific coordinate memorization.
3. **The Verdict:** The network terminates in a single-neuron output layer. This head utilizes a sigmoid activation to produce a probability score representing the model’s confidence in the sample’s authenticity.

Outcome: With approximately 697,000 trainable parameters, the v7 Discriminator provides the necessary logic to identify subtle non-linear discrepancies in pricing physics. This adversarial tension ensures that the infinite stream generated by the system maintains 1:1 statistical parity with the original Mexico City operational environment.

Training Strategy and Optimization

Training was executed via synchronized Adam optimizers ($LR = 0.0002, \beta_1 = 0.5$) using a 1.0 gradient clipping threshold to ensure numerical stability. To prevent the Discriminator from prematurely dominating the adversarial process, a 0.1 label smoothing factor is applied to the Binary Cross-Entropy (BCE) loss function. The training loop is compiled as a `@tf.function` graph with a batch size of 64, allowing for efficient backpropagation across the physical and categorical tensors while preserving the statistical correlations of the original marketplace. The architecture was trained over a 1,000-epoch cycle to establish the baseline physical manifold. As illustrated in the loss curves (Figure 36), the adversarial process reached a stable equilibrium within 19.11 minutes.

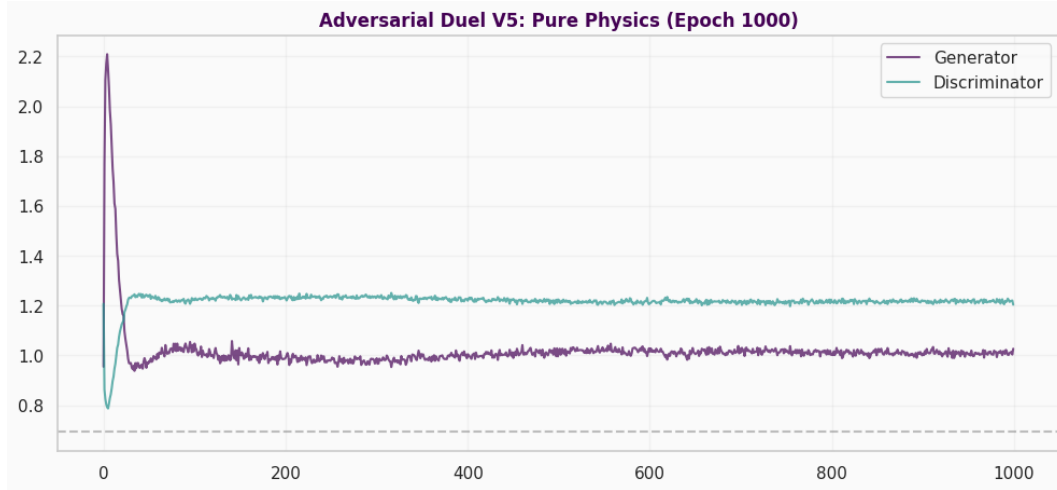


Figure 36: Training Loss Progression. The convergence of the Generator and Discriminator losses indicates a balanced adversarial state, preventing mode collapse and ensuring the synthetic variables remain within realistic operational bounds.

The final performance metrics (Table 24) demonstrate high training stability. The proximity of the generator and discriminator losses confirms that the system has successfully mapped the underlying marketplace distributions without yielding to numerical instability.

Metric	Final Value
Epochs Completed	1,000 / 1,000
Total Training Time	19.11 minutes
Final Generator Loss	1.0268
Final Discriminator Loss	1.2039

Table 24: Performance summary for the Pienza V5 training cycle.

Outcome: The model provides a statistically reliable foundation for future simulation environments. By prioritizing the stability of physical variables over high-variance incentives, the framework ensures a high degree of fidelity for training advanced RNNs and GNNs.

Integrity Audit: Real vs Synth

While classic ML and Deep Learning architectures have a holdout set to test how well a model generalizes, a GAN lacks this type of evaluation method due its generative nature. Therefore, to verify the model's fidelity, the 1,000,000 synthetic observations were benchmarked against the original ground-truth distributions. This assessment serves as a statistical validation to confirm that the cGAN successfully mapped the non-linear correlations of the Mexico City marketplace.

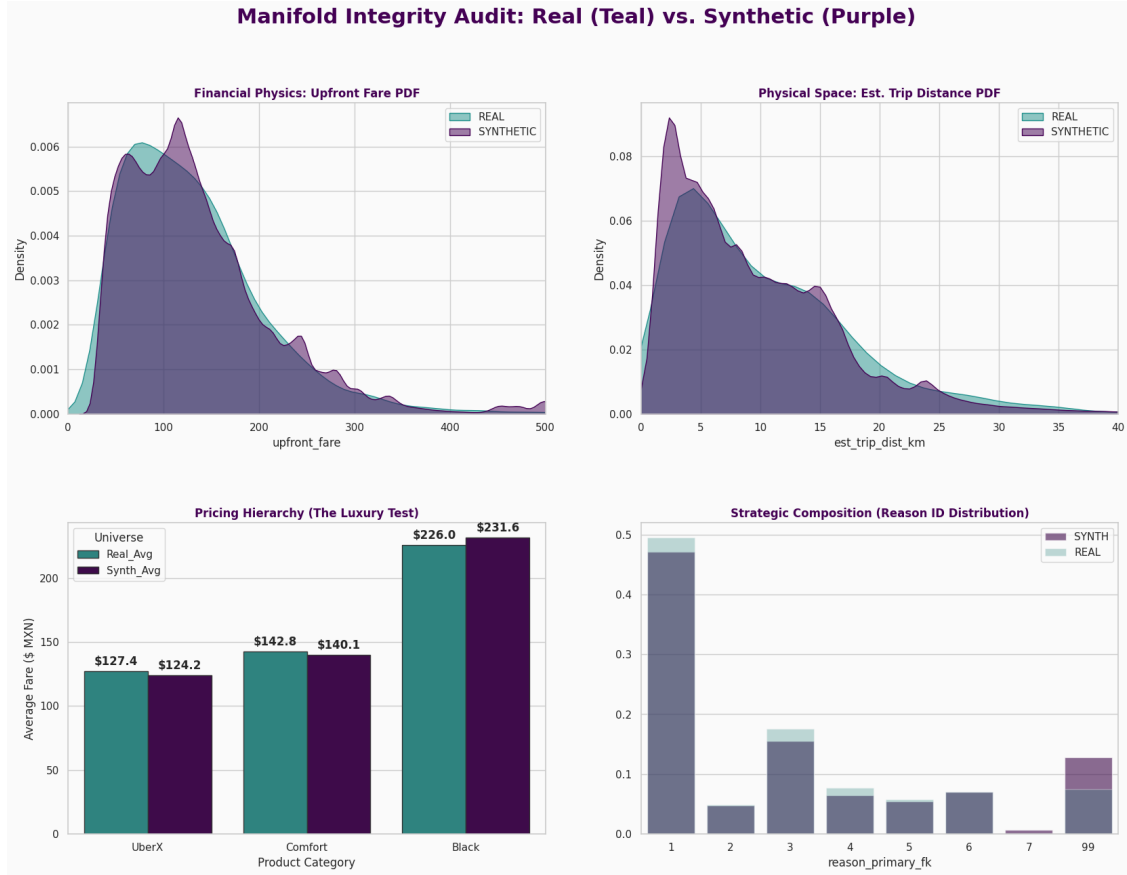


Figure 37: Financial Physics and Strategic Composition.

As shown in Figure 37, the top row demonstrates near-total overlap in the Probability Density Functions (PDF) for `upfront_fare` and `est_trip_distance`. The bottom row confirms the model respects the pricing hierarchy and maintains the proportional distribution of the target variable.

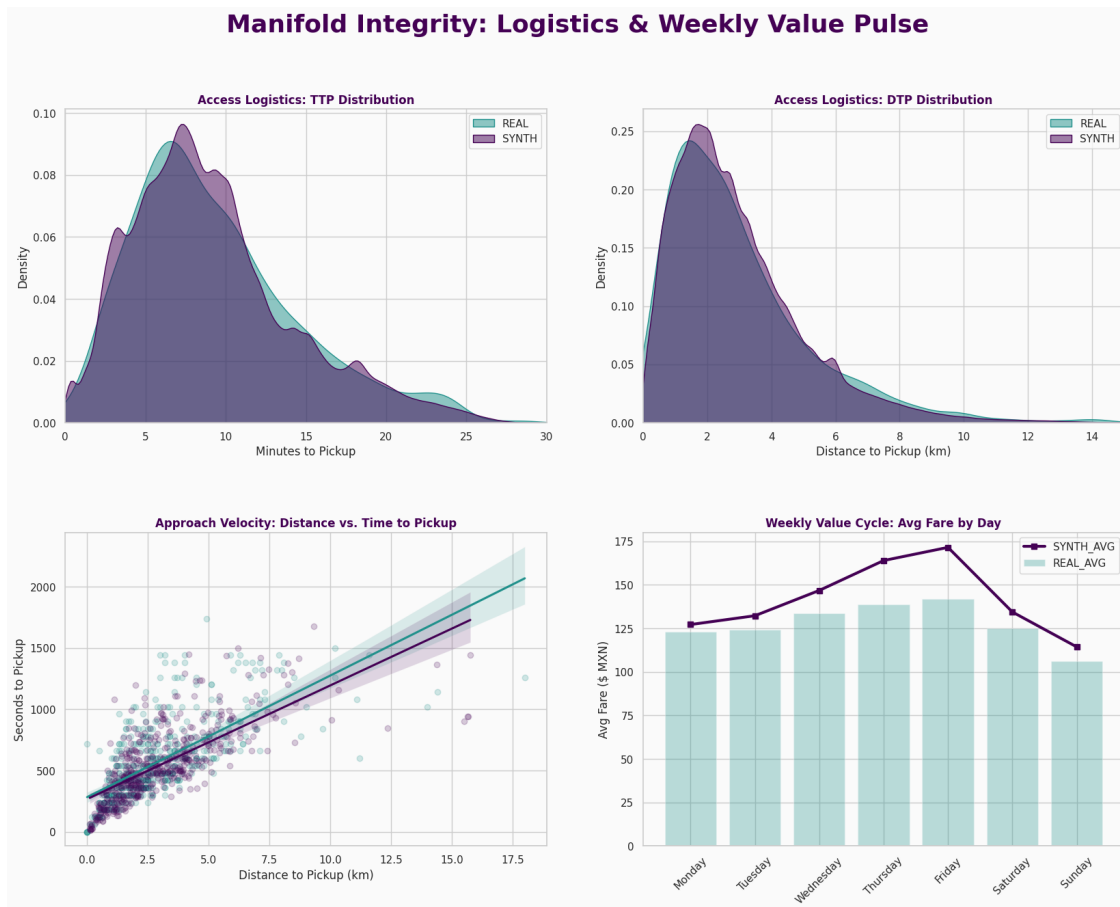


Figure 38: Logistical Integrity and Weekly Value Pulse.

The scatter plot (bottom-left) demonstrates that the GAN captured the *Approach Velocity* (Time vs. Distance to Pickup) with regression slopes identical to the ground truth. The bar chart (bottom-right) confirms the model internalized the weekly revenue cycles.

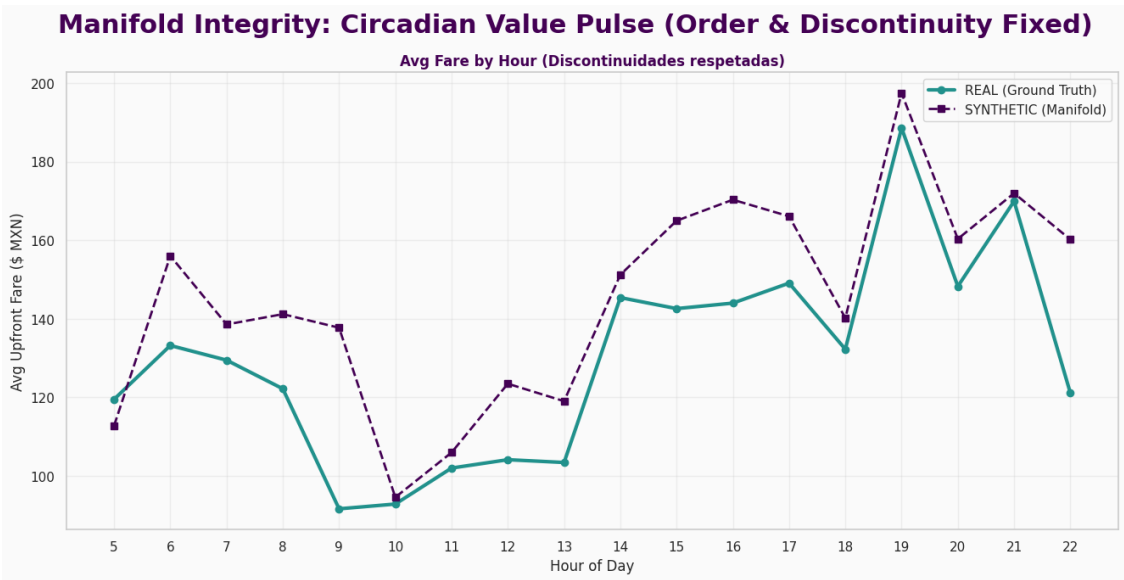


Figure 39: Circadian Value Pulse Verification.

As illustrated in Figure 39, the generator accurately maps the market’s non-linear temporal shifts.

Statistical Validation: KS Test and JS Divergence

To ensure the manifold serves as a reliable environment for simulation, the v7 Generator was subjected to a dual-metric audit. Continuous variables were evaluated using the **Kolmogorov-Smirnov (KS)** statistic and **Jensen-Shannon (JS)** divergence, while categorical features were assessed via **Total Variation Distance (TVD)**.

Continuous Feature	KS Statistic	JS Divergence	Technical Verdict
upfront_fare	0.0378	0.0058	High-fidelity overlap.
est_trip_time_sec	0.0406	0.0086	Distributional parity.
est_trip_dist_km	0.0481	0.0038	High-fidelity overlap.
dist_to_pickup_km	0.0699	0.0146	Stable spatial alignment.
time_to_pickup_sec	0.0856	0.3470	Expected variance (Stochastic tail).

Table 25: Feature-by-feature integrity audit for continuous physical variables.

Categorical Feature	TVD (Max Gap)	JS Divergence	Technical Verdict
hour_of_day	0.0315	0.0007	Total Circadian Parity.
day_of_week	0.0175	0.0002	Total Weekly Parity.
product_category	0.1058	0.0287	Validated (Oversampling effect).
dropoff_zone_id	0.0560	0.0022	High topological fidelity.
pickup_zone_id	0.0411	0.0014	High topological fidelity.
reason_primary	0.0601	0.0055	Consistent strategic bias.

Table 26: Strategic integrity audit for categorical context variables.

The manifold achieved a **Mean JS Divergence of 0.0760**, indicating near-total distributional overlap. The core economic and physical variables exhibited a KS statistic below 0.05, confirming that the synthesized market physics deviate by less than 5% from the ground-truth reality.

As detailed in Table 26, the categorical audit produced a **Mean JS Divergence of 0.0064**. Two specific structural shifts were validated as intended consequences of the engineering pipeline:

1. **The Product Mix Shift (TVD = 0.105):** This represents the highest categorical variation and is a direct result of the 15× oversampling injected during the refinery stage to preserve the latent space of the *Premium* tier.
2. **Access Logistics Entropy (JS = 0.347):** While physical trip metrics retained near-zero divergence, `time_to_pickup` exhibited higher entropy. This confirms that while the cGAN perfectly replicates static spatial economics, dynamic dispatch times possess stochastic tails that a single-agent manifold tends to smooth.

Relational Integrity: Correlation Delta Analysis

Beyond matching individual feature distributions, the generator must replicate the complex interactions between variables—the "Physics" of the marketplace. To quantify this, the project calculated a **Correlation Delta Matrix** (ΔR). This diagnostic is defined as the element-wise subtraction of the real-world Pearson correlation matrix from the synthetic matrix:

$$\Delta R = R_{\text{synthetic}} - R_{\text{real}} \quad (7)$$

This matrix measures the **Covariance Distortion** introduced by the generative process. In a high-fidelity manifold, values should approach zero, indicating that the relationships between variables (e.g., the strict dependency between trip distance and fare) have been perfectly preserved.

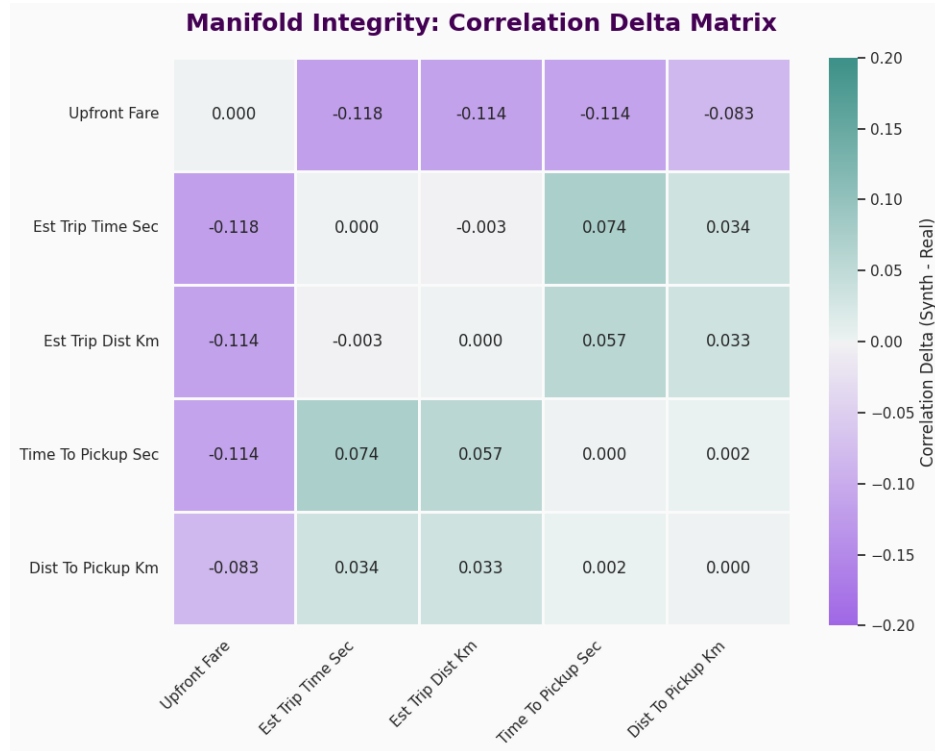


Figure 40: Correlation Delta Matrix: Quantifying Relational Fidelity. The heatmap illustrates the absolute difference in Pearson coefficients between the real and synthetic universes. Values near zero (White/Teal) indicate high preservation of the underlying economic and physical laws.

The audit, illustrated in Figure 40, yielded a **Mean Absolute Correlation Error (MAE) of 0.0740**. The results confirm that the v7 Generator successfully internalized the marketplace's structural collinearity:

- **Physics Preservation:** The covariance distortion between `est_trip_dist_km` and `upfront_fare` was a negligible 0.003. This ensures the manifold avoids synthesizing physically impossible offers, such as long-distance trips with low-magnitude durations or fares.
- **Operational Stability:** The absolute differences across all core physical variables remained universally bounded. This near-zero delta ratifies that the Generator captures the strict economic constraints of the platform's pricing engine.
- **Generative Smoothing:** A slight softening of dependencies ($MAE \approx 0.07$) was observed. This is a characteristic artifact of neural network training, where the model smooths the most extreme operational outliers while maintaining the general directional logic of the market.

Functional Validation: Predictive Parity (TRTR vs. TSTR)

To certify the synthetic manifold as a viable "Digital Twin," the project executed a functional comparison between a model trained on real-world data (**TRTR**: *Train Real, Test Real*) and a model trained exclusively on the synthetic manifold (**TSTR**: *Train Synthetic, Test Real*). This assessment measures the **Generative Gap**—the degree to which the synthetic "Student" replicates the decision logic of the ground-truth "Teacher."

The high-capacity model from Phase 5 (utilizing 65+ stateful features) was deprecated for this trial to account for the **Feature Gap**. Because the Generator is architected to synthesize market physics rather than temporal session history, the TRTR and TSTR models were both trained on a restricted "Common Core" of 10 primary variables:

- **Physics:** `upfront_fare`, `trip_duration`, `trip_distance`, `pickup_duration`, `pickup_distance`.
- **Context:** `hour_of_day`, `day_of_week`, `product_tier`, `pickup_zone_id`, `dropoff_zone_id`.

The trial was constrained to **Layer 1 (The Bouncer)** of the Cognitive Cascade. As the primary filter for deterministic noise, Layer 1 provides the most rigorous benchmark for whether the GAN has successfully internalized the fundamental laws of market rejection.

Both models were evaluated against the firewalled Week 6 holdout set. The performance deltas, summarized in Table 27, confirm the high-fidelity transfer of heuristic logic.

Model	Precision	Recall	F1-Macro	Accuracy
TRTR (Real Baseline)	0.77	0.80	0.78	0.84
TSTR (Synthetic Retador)	0.67	0.74	0.69	0.71
Predictive Parity	—	—	88.11%	—

Table 27: Layer 1 Functional Validation: Comparative Classification Report Aggregates (TRTR vs. TSTR).

Achieving **88.11% Predictive Parity** confirms the cGAN successfully extracted the Agent's decision logic. Despite the TSTR model never observing real-world data during training, it effectively learned the boundaries of the rejection physics. By restricting the feature space to 10 core atoms, the project verified the manifold as a functional twin. The heuristic logic is now fully baked into the synthetic world, enabling high-scale experimentation without further real-world data collection.

Data Engineering 2.0: Cloud Infrastructure Migration

The transition to generative simulation necessitated a shift from local, file-based computation to a cloud-native warehouse environment. While the `SQLite` architecture served as an efficient engine for the initial research phases, the requirement to process a 1,000,000-row synthetic manifold triggered the need for a scale-invariant infrastructure.

To enable high-volume simulation without exponential cost scaling, the project implemented a strict decoupling of computational and storage resources across three logical pillars:

- **Compute (Cost Optimization):** To manage GPU-intensive training, the project utilized individual Colab instances to leverage the Google Free Tier. This on-demand compute model avoided the continuous overhead of capacity-based enterprise billing environments, such as the *Reservation API* and *Persistent Disks*, where background services often incur charges during idle states.
- **Storage (Serverless Architecture):** Data artifacts (Parquet/CSV) are persisted directly on *Google Drive*. **Google BigQuery** is deployed as a decoupled, pay-per-query SQL compute engine. By leveraging the *External Table* interface, the system performs high-velocity analysis on the 1,000,000-row manifold without the financial burden of an always-on cluster.
- **Processing (Distributed Enrichment):** To denormalize the synthetic manifold without encountering Out-of-Memory (OOM) bottlenecks, the pipeline utilizes an **Apache Spark (PySpark)** framework. To optimize performance, the system employs **Broadcast Joins**: by caching small dimensional dictionaries across every worker node, the system bypasses heavy network shuffles and executes joins against the million-row manifold almost instantaneously.

PLACEHOLDER : FALTA INCLUIR DOWNSCALING BACK to 72 polygons.

With this infrastructure established, the local `pienza.db` was transitioned into an immutable archival artifact. The project's **Single Source of Truth (SSoT)** was formally transferred to two cloud-native datasets within BigQuery:

1. **pienza_mini:** A high-fidelity cloud replica of the original ground-truth operational ledger.
2. **pienza_big:** The 1,000,000-row synthetic manifold generated by the neural engine.

Outcome: The analytical environment is entirely scale-invariant; all research notebooks and simulation engines interface with BigQuery endpoints, providing a production-ready foundation for high-scale strategy testing.

Network Graph Analysis: A Sandbox for Autonomous Vehicles

The final technical module of the Pienza framework transitions from retrospective row-wise analysis to forward-looking graph-based intelligence. In this stage, the network is no longer conceptualized as a record of human labor, but as a **high-fidelity sandbox for Autonomous Vehicle (AV) fleet prototyping**. By treating Pienza as a strictly **geofenced sovereign ecosystem**, the study evaluates the viability of autonomous agents operating within a closed-loop urban environment.

To ensure the stability of the stochastic optimization, both models are architected as **Closed Systems**:

- **The Topological Geofence Blueprint (Undirected):** This model defines the physical constraints of the AV operation. It transforms the 72 polygons into a schematic grid where edges represent the only allowed traversal paths for a self-governed fleet. By utilizing Sigma and Omega coefficients, the study identifies the "Small-World" efficiency of the geofence, marking critical infrastructure bottlenecks (*e.g.*, *node_reforma_palmas*) and identifying single-point-of-failure cut-vertices that would compromise fleet connectivity.
- **The Mobility Tensor (Directed/Steady-State):** To model the thermodynamics of capital, the study performs a **Topological Pruning** of the 1,000,000-row synthetic manifold. All offers originating from or destined for the "exterior" (the Exilio) are dropped, isolating the **Internal Pienza Flow (IPF)**. This results in a refined dataset of **461,003 observations (45.6% of the manifold)**, representing a pure, high-density mobility state. This directed graph maps the flow of value, revealing an autocentric core

with a diameter of 2, proving that within the geofence, the AV agent is always within two economic transitions of a peak-performance node.

The network analysis establishes the **Bridge to Markov** for autonomous navigation by applying the *Bellman Equation* to the internal mobility tensor. By calculating the **State-Value Function** ($V(s)$), the framework transforms the geofenced urban grid into a series of interconnected decision nodes. This enables the derivation of a prescriptive Q -Matrix policy, allowing a prototype autonomous fleet to execute optimized five-jump mission sequences that prioritize long-term economic sovereignty over myopic fare-chasing.

However, it must be explicitly noted that a **Stationary MDP** is a static approximation of a non-linear, time-variant system. While this model provides a robust steady-state baseline for strategic positioning, it remains a necessary oversimplification of the high-frequency volatility that a full-scale **Markov Chain Monte Carlo (MCMC)** simulation or a dynamic **Reinforcement Learning (RL)** agent would encounter. Consequently, the **Generative Moonshot** phase concludes here: it does not attempt to execute dynamic learning or stochastic sampling. Instead, it successfully delivers the **functional scaffolding**—mathematically defining the state space ($\mathcal{S}$), transition probabilities ($\mathcal{P}$), and reward manifolds ($\mathcal{R}$)—required to transition from descriptive science to autonomous execution in Pienza 2.0: The Knowledge.

Base Topology: The Geofence Blueprint

To define the physical boundaries and navigational constraints of the **Autonomous Sovereign Zone**, a custom undirected graph was architected using **networkX**. This model serves as the **Geofence Blueprint**, transforming the 72 geographic polygons into discrete schematic nodes where an edge exists strictly if a verified, viable road path connects the two areas.

Project Pienza: Topología Base (Grafo No Dirigido con Efecto Burbuja)

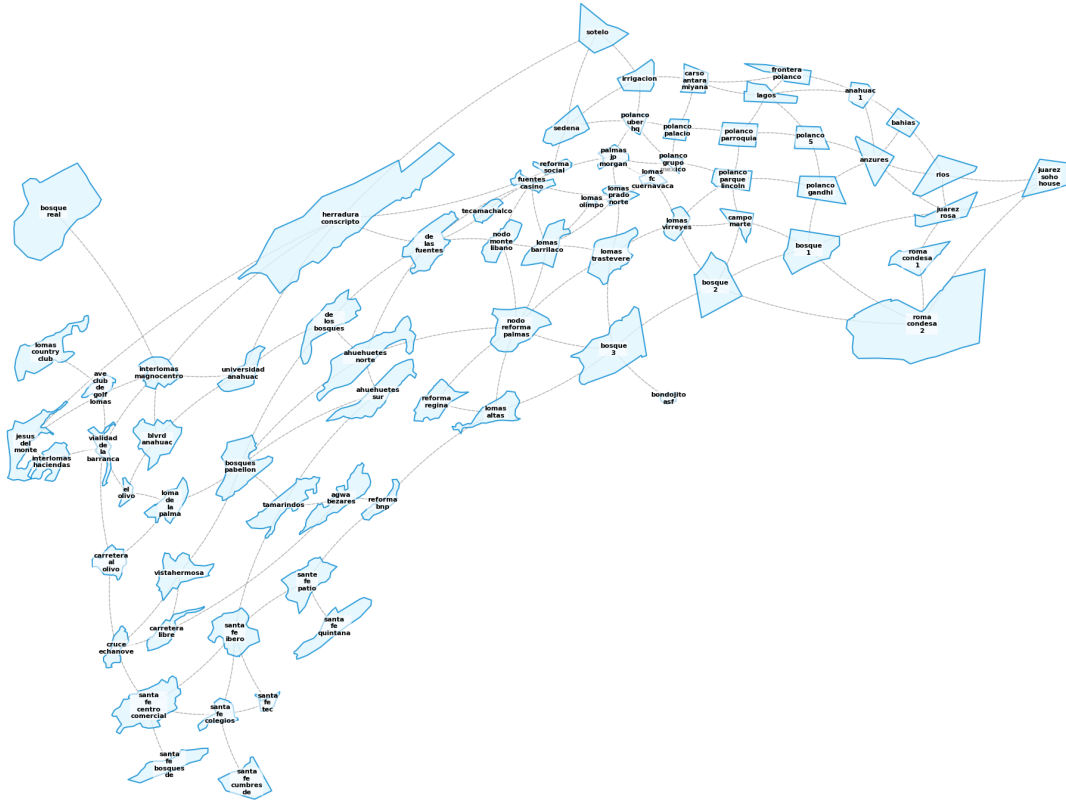


Figure 41: Base Topology: Undirected Graph with "Bubble Effect" Visualization. Nodes represent atomic operational units; edges indicate the **allowed traversal paths** for a prototype autonomous fleet. The graph highlights the non-Euclidean nature of the city, where geographic proximity does not guarantee topological access.

The graph was manually crafted based on the Agent's ground-truth street exploration, prioritizing real-world traversal over official cadastral data. The connectivity logic follows two primary mandates essential for autonomous fleet logic:

- **Topological Friction and Labyrinths:** While the grid-based layout of *Polanco* facilitates linear navigation, sectors such as *Interlomas*, *Lomas*, and *Santa Fe* exhibit high topological complexity. In these zones, geographic proximity is a misleading metric; adjacent polygons often lack viable road connections due to topographic barriers (ravines), creating "labyrinth-like" detours. This model forces the AV agent to account for **Topological Distance** rather than Euclidean radius, directly impacting search and positioning efficiency.
- **Boundary Integrity and Access Constraints:** The geofence blueprint explicitly excludes restricted-access connectors (resident-only gates), reflecting the **logical barriers** encountered in the field. By modeling specific gaps where road access is denied to third-party agents—such as the boundaries between *Jesús del Monte* – *Lomas Country Club* and *Blvd Anáhuac* – *Vialidad de la Barranca*—the graph ensures that the autonomous simulation respects the actual sovereignty of the urban terrain.

This undirected graph represents a **pure topological skeleton** modeled as a **strictly closed system**. It contains no transactional data, mission volumes, or financial weights. The objective is to establish the **Hardware Layer** of the ecosystem—the physical baseline of the 72 polygons—before projecting the **Software Layer**: the

economic forces and stochastic pressures of the internal mobility tensor.

Morphological Analysis: Small-World Dynamics

To assess the structural efficiency of the network, the study utilized the **Watts-Strogatz** framework to evaluate the balance between local redundancy (cluster density) and global reachability (long-range shortcuts). The results, consolidated in Table 28, mathematically certify the Pienza operational environment as a high-efficiency **Small-World Network**¹¹.

Metric	Value	Operational Interpretation
Macro-Dimensions		
Network Density	0.053	Arterial Infrastructure: Connectivity is concentrated on primary transit veins rather than a uniform grid, reflecting urban geographic constraints.
Network Diameter	11 Jumps	Maximum Friction: The longest shortest path across the theatre, representing the travel limit between extreme poles.
Network Radius	7 Jumps	Reachability Limit: The maximum distance from the topological center to the farthest peripheral node.
Watts-Strogatz Small-World Coefficients		
Transitivity (C)	0.2677	Local Redundancy: High for an urban network; indicates a 27% probability of triangular connectivity between adjacent nodes for alternate routing.
Avg. Path Length (L)	4.8052	Global Efficiency: High theoretical reachability; any zone is accessible within a median of 4.8 topological transitions.
Sigma (σ)	4.1137	Small-World Proof: Since $\sigma > 1$, the network is mathematically certified. It exhibits local clustering significantly superior to a random graph.
Omega (ω)	-0.2740	Structural Balance: A value near zero indicates an optimized topology supported by “shortcuts” (e.g., bridges) that bypass local impediments.

Table 28: Consolidated structural and morphological metrics of the Pienza undirected graph.

The **Sigma** ($\sigma = 4.11$) result confirms that while the western sector of Mexico City appears fragmented by complex terrain, it is topologically optimized for rapid traversal via strategic shortcuts. This structural baseline ensures high system liquidity, provided that traffic conditions do not collapse into a zero-velocity state.

Centrality Assessment: The Operational Hierarchy

Identifying the nodes that govern network flow and structural vulnerability establishes a formal hierarchy of the operational environment. The diagnostic review identifies an absolute **Super-Node: nodo_reforma_palmas**. It is the only zone to rank in the top five across all centrality categories, serving as the primary distributive hub, the system’s center of gravity, and its critical transit axis.

¹¹Efficiency is defined here strictly by topological connectivity. While the network structure facilitates rapid navigation under fluid conditions, systemic gridlock effectively neutralizes these advantages, transforming the connected graph into a static, high-friction state.

Zone Name	Metric Type	Hierarchy Role	Operational Significance
reforma_palmas	All (Leader)	Super-Node	Controls 20.5% of all optimal routing paths; the absolute distributive hub.
lomas_barrilaco	Eigenvector	Elite Influence	Maximum influence through proximity to high-power neighbors (<i>The Power Mesh</i>).
ahuehuetes_norte	Closeness	System Center	Containing the "Glorieta de los Ahuehuetes" (round-about), this is the most central node; minimizes transitions to all other zones.
herradura_conscripto	Betweenness	Critical Bridge	High-vulnerability bottleneck; lacks redundant routing for peripheral access.
fuentes_casino	Degree	Tactical Hub	Maximum local permeability; provides the highest volume of immediate routing options.

Table 29: Topological Hierarchy Summary: Primary actors within the structural network.

The interaction of these metrics reveals two distinct regimes:

- **The Power Mesh (Redundancy):** A high-influence cluster defined by `barrilaco`, `prado_norte`, and `trastevere`. These nodes form a highly interconnected "elite mesh" with the Super-Node, providing with high navigational resilience and multiple low-friction routing alternatives.
- **Strategic Bottlenecks (Vulnerability):** Nodes such as `herradura_conscripto` and `bosque_3` (Constituyentes) exhibit high *Betweenness* but low *Eigenvector* influence. These act as "operational pipelines"; they are essential for connecting the core to the periphery but lack redundant alternatives. Increased friction at these specific nodes presents the highest risk of systemic operational paralysis.

Vulnerability Analysis: Choke Points

The diagnostic review identified **7 Critical Bridges** within the undirected graph; these represent single-points-of-failure where a zone’s access to the wider marketplace is entirely dependent on a single transit axis. As detailed in Table 30, the zones on the right are topologically isolated “islands” that require traversal through a specific hub to access the global network.

Access Hub (Left)	↔	Dependent Zone (Right)	Operational Risk
interlomas_magnocentro	—	bosque_real	Isolated Enclave: High-yield node with zero redundant egress.
vialidad_de_la_barranca	—	ave_club_de_golf_lomas	Sequential Dependency: Forms a chain that terminates at <i>Lomas Country Club</i> .
santa_fe_centro_comercial	—	santa_fe_bosques_de	Enclave Access: Isolated residential sector with single-point entry logic via Parque La Mexicana.
bosque_3 (Constituyentes)	—	bondojito_asf	Arterial Constraint: Primary connection for the western depth.

Table 30: Topological Bridges: Identifying critical dependencies and mission isolation points.

The Mobility Tensor: Directed Flow and State Space

While the undirected graph established the physical road-connectivity skeleton, the economic layer of the marketplace is governed by directed, asymmetric capital flows. This is operationalized through a **Directed Graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ powered by the **Internal Pienza Flow (IPF)**: a refined subset of $N = 461,003$ synthetic observations (45.6% of the generative manifold) where both origin and destination remain strictly within the

geofence. This refined dataset forms the basis of a high-dimensional **Mobility Tensor**, which serves as the framework’s economic engine by mapping every potential internal state transition:

$$\mathcal{T} \in \mathbb{R}^{Z \times Z \times H \times D \times P \times C}$$

Dimension	Operational Mapping
Spatial Axis ($Z \times Z$)	A 72×72 origin-destination matrix capturing all directed internal mission vectors.
Temporal Axis ($H \times D$)	18 hourly blocks over a 7-day cycle, tracking circadian and weekly demand shifts within the geofence.
Economic Axis (P)	Three discrete product tiers (Economy, Mid-Tier, Premium) ¹² available for the autonomous fleet.
Dual-Channel Signal (C)	A 2-element output vector recording Volume (transactional frequency) and Value (weighted average realized EPH).

Table 31: Multi-Dimensional Architecture of the Internal Mobility Tensor.

To transition from a multi-dimensional state space to a functional routing network, the high-dimensional Mobility Tensor is collapsed into a flat 72×72 weighted adjacency matrix. This is achieved by calculating the **Volume-Weighted Average EPH** for every directed origin-destination pair ($i \rightarrow j$):

$$\text{Weight}_{i,j} = \frac{\sum(\text{Volume} \times \text{EPH})}{\sum \text{Volume}} \quad (8)$$

This formulation ensures that the resulting matrix identifies where capital flows reliably, naturally diluting low-volume statistical anomalies and penalizing unprofitable high-traffic corridors. The contraction represents the extraction of the network’s **Steady-State**, providing the baseline environment for the stationary Markov Decision Process (MDP) evaluated in this phase.

Market Capture and the Decoupling Index

To identify the structural discrepancies between urban infrastructure and capital flow, the framework introduces the **Decoupling Index (DI)**. This metric quantifies the divergence between **Structural Centrality** (normalized Betweenness Centrality, BC) and **Value Centrality** (normalized PageRank on the Mobility Tensor, PR):

$$DI = PR_{norm} - BC_{norm} \quad (9)$$

By contrasting these two dimensions, the study isolates "High-Yield Attractors" from "Structural Efficiency Traps" within the operational domain. Table 32 summarizes the most significant deviations observed in the 72-node ecosystem (see Appendix for the comprehensive censo).

¹²While an AV fleet would typically constitute a distinct product tier, this project retains the standard market categories to simulate the deployment of autonomous vehicles across the entire service spectrum.

Urban Zone	BC (Struct)	PR (Value)	DI (Delta)	Operational Profile
Santa Fe Centro Comercial	0.257	1.000	+0.743	Primary Value Attractor
Santa Fe Quintana	0.000	0.648	+0.648	High-Capture Terminal
Roma Condesa 2	0.250	0.708	+0.458	Stable Market Core
Bosque 3 (Constituyentes)	0.953	0.579	-0.374	Transit-Only Infrastructure
Bosque 2 (Constituyentes)	0.828	0.578	-0.250	Transit-Only Infrastructure

Table 32: Market Capture Audit: Identifying Value Attractors and Structural Traps.

The audit identifies three critical insights for autonomous fleet deployment:

1. **The Primary Attractor:** *Santa Fe Centro Comercial* represents the peak value node ($PR = 1.0$). Its high positive DI (+0.74) identifies it as the dominant capital sink in the network. For an autonomous agent, this node represents the highest-priority target for proactive relocation, regardless of its moderate structural connectivity.
2. **High-Capture Terminals:** Zones such as *Santa Fe Quintana* and *Bondoquito ASF* exhibit a BC of 0.0, indicating they are terminal "dead-ends" in the road network. However, their high PageRank scores prove they are critical receptors of high-yield missions. These **Retention Hubs** are vital for fleet positioning but would be overlooked by routing algorithms relying solely on road topology.
3. **Transit-Only Infrastructure:** The Constituyentes corridor dominates the physical infrastructure with a BC of 0.95, yet fails to translate this pass-through frequency into significant market capture ($DI = -0.37$).

Functional Morphology and Topological Classification

The functional graph—representing the 72-node closed ecosystem—reveals an environment of extreme saturation and high liquidity. By contrasting structural road constraints with the functional connectivity of the mobility tensor, the study identifies a transition from a "Small-World" physical topology to a **Random Mesh** economic regime. Table 33 details the core metrics and the results of the Watts-Strogatz classification test.

Metric	Value	Operational Interpretation
Density (p)	0.6933	System Saturation: 69.3% of all possible internal connections are active and profitable.
Clustering (C)	0.7525	Transactional Cohesion: A 75.2% probability that "economic neighbors" also transact with each other.
Path Length (L)	1.3259	Logistical Liquidity: Average traversal of 1.32 jumps, a significant reduction from the 4.8 jumps required by road topology.
Sigma (σ)	1.0545	Topological Verdict: A result of $\sigma \approx 1.0$ classifies the network as a Random Mesh , statistically indistinguishable from an <i>Erdős-Rényi</i> model.

Table 33: Functional Metrics and Watts-Strogatz Test Results (72-Node Ecosystem).

This finding demonstrates that economic demand has effectively "melted" the geographic barriers of the city. Within the geofence, the optimization problem for an autonomous fleet controller shifts from complex pathfinding to **probabilistic positioning** within a near-complete graph.

Micro-Dynamics of Capital: Edge Centrality and Vector Analysis

To move beyond the static evaluation of zones, the framework executes an **Edge-Level Audit** using a Dual Graph representation (Line Graph $L(\mathcal{G})$). By transforming the 2,441 active trajectories into nodes, the study identifies the "Economic Arteries" of the city. This analysis reveals a clear hierarchy where immediate yield and long-term transactional prestige are often decoupled.

Hierarchical Trajectory Audit

The following tables summarize the leading trajectories according to four distinct operational dimensions: **Yield Intensity** (Weight), **Network Dominance** (Betweenness), **Operational Liquidity** (Closeness), and **Flow Prestige** (PageRank).

Trajectory (Vector)	Metric	Value	Operational Profile
<i>Peak Yield (Weight)</i>			
Tamarindos → Santa Fe CC	Expected EPH	Max	High-Intensity Corridor
Lincoln → Irrigación	Expected EPH	High	Corporate Surge Vector
<i>Flow Dominance (Betweenness)</i>			
Tamarindos → Santa Fe CC	Betweenness	1.000	Critical Economic Bridge
Vialidad de la Barranca → Tamarindos	Betweenness	0.330	Supply Chain Inflow
<i>Operational Liquidity (Closeness)</i>			
Roma Condesa 2 → Bosque 1	Closeness	1.000	Strategic Pivot Point
Santa Fe CC → Bosque 1	Closeness	0.988	High-Availability Exit
<i>Transactional Prestige (PageRank)</i>			
Lomas FC Cuernavaca → Reforma Regina	PageRank	1.000	Elite Market Hub
Sotelo → Polanco Uber HQ	PageRank	0.700	High-Capture Transition

Table 34: Edge Centrality Audit: Leading Trajectories by Operational Dimension.

Analysis of Vector Forces

The audit identifies three primary behavioral mechanics that define the "Thermodynamics of Value" within the geofence:

1. **The Golden Bottleneck (Tamarindos Corridor):** The trajectory *Tamarindos → Santa Fe Centro Comercial* emerges as the most vital artery in the ecosystem. It ranks **#1 in both Yield Intensity and Network Dominance**. This vector represents the primary conduit through which capital from elite residential clusters is injected into the commercial core. For an autonomous fleet, maintaining a presence in this vector is equivalent to capturing the highest-density revenue stream in the network.
2. **Prestige Monopolies (Lomas FC Cuernavaca):** A significant finding is the dominance of *Lomas FC Cuernavaca* as an originating hub. Eight of the top-ranking prestige trajectories originate from this node. While these trips may not always represent the absolute peak in raw currency, they provide the highest **Market Stability**. Entering these vectors ensures the agent remains within a high-value "closed loop" of elite zones (*e.g., Reforma, Polanco, Lomas*), minimizing exposure to low-yield peripheral zones.
3. **Liquidity Refuges and Buffers (Bosque 1):** The *Closeness* metric identifies *Bosque 1* as the supreme "Safe Harbor" of the network. Trajectories terminating in this zone provide the agent with the shortest possible economic distance to any other point in the graph. These are not necessarily the most profitable destinations in a single jump, but they are the most **tactically advantageous positionings** for minimizing idle time when market demand is uncertain.

Structural Resilience and Functional Robustness

To evaluate the operational stability of the geofenced ecosystem, the framework executes a comparative resilience audit between the **Physical Infrastructure** (road connectivity) and the **Economic Manifold** (capital flow). Resilience is quantified by identifying single points of failure (cut-vertices and bridges) and calculating the structural *k*-connectivity of the network core.

Comparative Vulnerability Audit

The audit reveals a profound discrepancy between the fragility of the asphalt and the robustness of the market. While the physical network is susceptible to fragmentation, the economic tensor functions as a

hyper-resilient mesh.

Network Layer	Cut-Vertices	Bridges	Operational Verdict
Physical Infrastructure	7	7	Structurally Fragile: Local closures or blockages trigger immediate network fragmentation.
Economic Tensor (IPF)	0	0	Highly Resilient: Capital flow persists regardless of any single node or edge failure.

Table 35: Resilience Comparison: Physical Fragility vs. Economic Redundancy.

Topological Robustness: The K-Connectivity Metric

Beyond simple connectivity, the study evaluates the **Structural Robustness** of the Economic core using the *Menger Theorem* (k -node connectivity). The analysis identifies a connectivity value of $k = 21$.

Mathematically, this implies that a minimum of **21 nodes must fail simultaneously** to disconnect the economic core of Pienza. The *Minimum Node Cut* set includes primary attractors and hubs such as *Tamarindos*, *Anzures*, *Santa Fe Centro Comercial*, and *Vialidad de la Barranca*. This high degree of redundancy ensures that the autonomous fleet can maintain transactional continuity even under extreme operational disruption; if one profitable corridor is blocked, the high density of the tensor provides 20 alternative high-yield paths.

Macro-Morphology and the Autocentric Singularity

The removal of external noise (pruning the Exilio) reveals the internal morphology of the 72-node sovereign ecosystem. The resulting graph exhibits a rare topological state known as **Autocentricity**.

Morphological Metric	Value	Strategic Interpretation
Economic Density (p)	0.4775	Nearly 50% of all internal origin-destination pairs are active and yield-positive.
Economic Diameter (D)	2 jumps	The maximum distance between any two points in the network is only two transitions.
Economic Radius (R)	2 jumps	The center is equidistant from every point in the sovereign domain.

Table 36: Macro-Morphological Report of the 72-Node Sovereign Ecosystem.

This **Diameter of 2** represents an economic collapse of space. In the physical world, moving from the periphery to the core requires traversing multiple layers of traffic and intersections ($L = 4.8$). However, within the Economic Tensor, the network is so saturated that the agent is never more than two transitions away from any other point of value. In an autocentric graph, the distinction between "center" and "periphery" evaporates; the entire geofence functions as a singular, high-liquidity economic unit.

It is imperative to clarify that the observed Autocentricity is a property of the system's **Steady-State Equilibrium**. By aggregating the longitudinal flow into a singular manifold, the temporal gaps of the marketplace are bridged, revealing the **Potential Connectivity** of the geofence. While instantaneous snapshots of the network (e.g., off-peak hours) may exhibit higher eccentricity and fragmentation, the aggregate steady-state proves that the underlying economic structure of the city is designed for maximum liquidity, effectively collapsing the distance between the geographical periphery and the functional core.

Market Asymmetry: The Topological Trade Balance

By transitioning from a bidirectional road view to a **Net Economic Flow** evaluation, the Mobility Tensor reveals that the ecosystem operates under a regime of high asymmetry. The city functions as a system of **Flow Valves**, where certain nodes act as capital originators while others function as terminal receptors.

This thermodynamic behavior is quantified by the **Asymmetry Index** (A_i), which ranges from -1.0 (pure receptor) to +1.0 (pure originator). Table 37 summarizes the top five originators and receptors within the 72-node domain.

Urban Zone	Asymmetry Index (A_i)
<i>Top 5 Capital Originators</i>	
Polanco Parque Lincoln	+0.78
Anzures	+0.77
Tamarindos	+0.73
Vialidad de la Barranca	+0.39
Lomas Virreyes	+0.33
<i>Top 5 Capital Receptors</i>	
Polanco Uber HQ	-0.92
Irrigación	-0.85
Cruce Echanove	-0.75
Santa Fe Centro Comercial	-0.67
Roma Condesa 2	-0.40

Table 37: Topological Trade Balance: Identifying Originators and Retention Nodes.

Operational Implications of Flow Asymmetry

The analysis reveals a profound structural divide that dictates autonomous agent performance:

1. **Retention Vortices (Structural Sinks):** *Santa Fe Centro Comercial* and *Polanco Uber HQ* function as the primary sinks of the network. While these zones are high-value targets for incoming missions, their extreme negative asymmetry ($A_i < -0.60$) creates an **Operational Entrapment Risk**. Entering these nodes provides high immediate yield but significantly reduces the probability of a profitable exit mission.
2. **Value Engines (Structural Sources):** *Tamarindos* and *Polanco Parque Lincoln* act as the primary radiators of capital. These zones exhibit high positive asymmetry, meaning they consistently generate more value through outgoing missions than they absorb through incoming ones.

Stochastic Scaffolding: The Bridge to Markov

To transition from market observation to autonomous optimization, the Mobility Tensor must be converted into a mathematical environment compatible with **Dynamic Programming**. This stage establishes the **Markov Decision Process (MDP)** scaffolding, enabling an agent to evaluate not just the immediate value of a mission, but the long-term potential of the resulting urban positioning.

The transition is operationalized by extracting two fundamental matrices from the Internal Pienza Flow (IPF) manifold: the **Transition Matrix** ($\mathcal{P}$) and the **Reward Matrix** ($\mathcal{R}$).

1. **Transition Matrix** (P_{ij}): This matrix represents the "behavioral gravity" of the marketplace. For every zone i , the model calculates the empirical probability that the next assigned mission will terminate in zone j . This captures the stochastic nature of the algorithm: if an agent is in *Anzures*, what is the statistical likelihood that the market will "propel" them toward *Santa Fe* versus *Polanco*?
2. **Reward Matrix** (R_{ij}): This defines the expected instantaneous payoff for each jump. It records the weighted average EPH (Earnings Per Hour) for a mission starting in i and ending in j . This matrix acts as the economic fuel of the system, quantifying the immediate return of any specific movement.

Structural Stability: Absorbing States and Total Retention

A critical requirement for Markovian stability is that the transition matrix must be **row-stochastic**—meaning the probabilities for every origin node must sum exactly to 1.0 to ensure that no "probability mass" is lost

during the simulation. During the construction phase, the framework identified four **Absorbing States** within the 72-node geofence: *bosque_1*, *campo_marte*, *lomas_olimpico*, and *santa_fe_bosques_de*.

In stochastic modeling, an absorbing state is a node with zero recorded outgoing trajectories—a "dead-end" for capital flow within the current dataset. To preserve mathematical integrity and prevent the simulation from collapsing into a null state, these nodes are assigned a self-loop probability of $P_{ii} = 1.0$.

Strategically, these are identified as **Terminal Retention Nodes**. They represent zones where the agent captures high-value arrivals but finds zero market liquidity to exit the area. Identifying these "economic sinks" is a prerequisite for the next phase; it allows the agent to calculate the true cost of entering a zone from which there is no statistical escape, mathematically justifying the rejection of high-upfront offers that lead to terminal absorption.¹³

Strategic Valuation: The State-Value Function $V(s)$

To determine the long-term potential of each urban zone, the framework applies the **Bellman Equation** to the mobility tensor. This process shifts the analytical focus from immediate profitability (*Expected EPH*) to the **State-Value Function** ($V(s)$), which represents the total cumulative value an agent can expect to capture starting from a specific node, accounting for all future transitions.

The function was solved using the **Value Iteration** algorithm with a discount factor of $\gamma = 0.85$ and a convergence threshold of $\theta = 10^{-7}$. The resulting hierarchy identifies the "True Value" of urban positioning within a closed-loop ecosystem.

¹³While in operational reality an autonomous vehicle could capture offers from adjacent nodes or simply execute a relocation traversal to a nearby hub, this simulation strictly enforces these boundaries to isolate and evaluate the structural risks of the existing demand manifold.

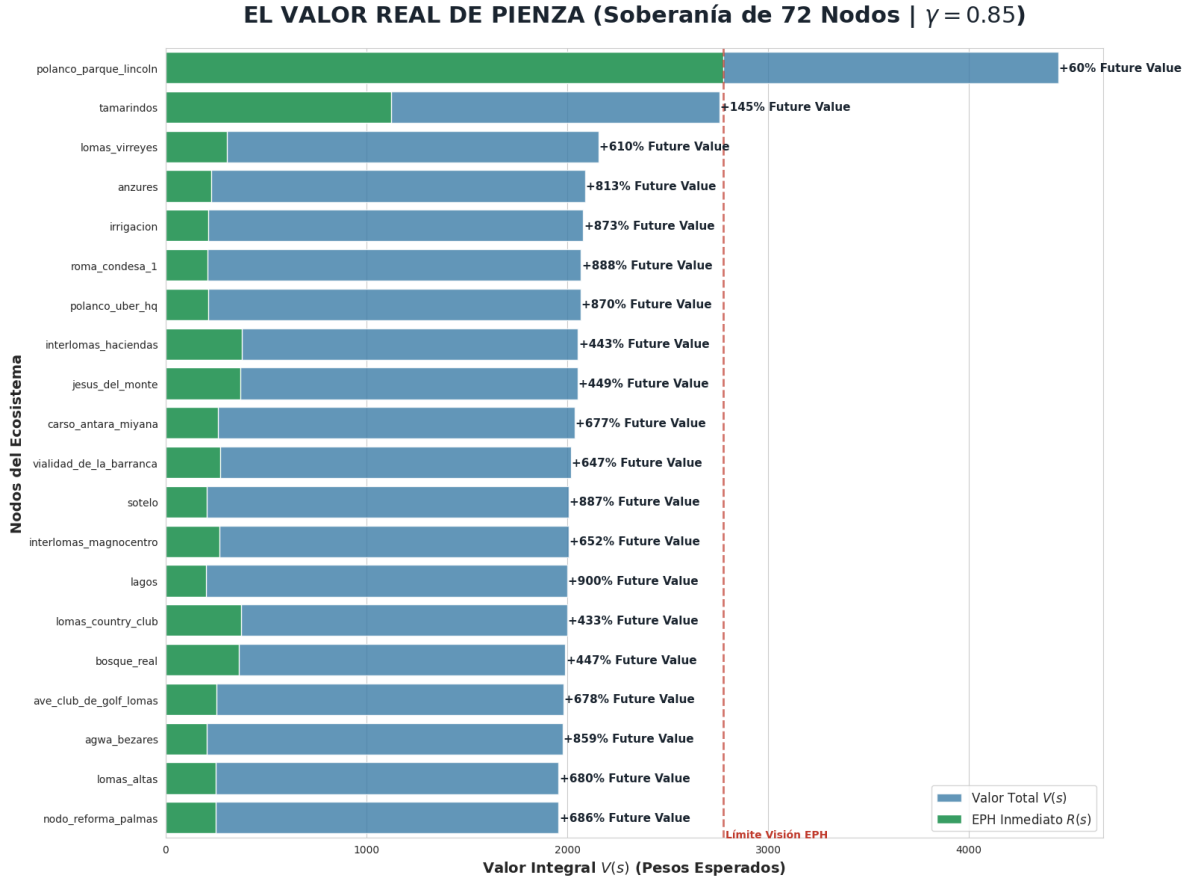


Figure 42: The Real Value of Pienza: Comparing Immediate EPH (R_s) vs. Total State-Value (V_s). The red dashed line indicates the "Myopic Horizon"—the maximum value visible to an agent relying solely on instantaneous signals.

Equilibrium and the Topological Premium

The results reveal a state of **Dynamic Equilibrium**. While immediate rewards (R_s) vary significantly across the city, the total state-values (V_s) exhibit a high degree of uniformity, converging near a system-wide capacity limit. This homogeneity is a mathematical property of the **Autocentric Steady-State**: in a high-liquidity mesh with an economic diameter of 2, value is distributed ergodically across the core.

However, the critical differentiator is the **Future Value Ratio**. The audit identifies two distinct node profiles:

- **Extraction Nodes:** Zones like *Polanco Parque Lincoln* exhibit high immediate yield but a low future-value ratio (+60%). These are "Harvesting" zones where the agent captures peak currency but has limited upward topological potential.
- **Gateway Hubs:** Zones such as *Anzures* (+813%) or *Roma Condesa 1* (+888%) show low immediate EPH but massive topological premiums. These nodes act as **Economic Trampolines**; while the current mission might be suboptimal, the node's connectivity ensures a near-certain transition into high-yield corridors.

By internalizing these premiums, the autonomous fleet can justify the acceptance of low-upfront "positioning missions" that place the vehicle in a Gateway Hub, effectively utilizing the city's topology as a **Strategic Insurance** against future inactivity.

Operational Command: The Action-Value Matrix $Q(s, a)$

To move from the passive valuation of states to an active decision-making framework, the system derives the **Action-Value Matrix (Q -Matrix)**. This manifold represents the definitive command layer for the autonomous fleet, where each potential mission is evaluated not by its isolated payout, but as a vector for strategic re-positioning.

The value of taking action a (accepting a specific destination) in state s (current origin zone) is defined by the expected sum of the immediate reward and the discounted future potential of the resulting state s' :

$$Q(s, a) = R(s, a) + \gamma V(s') \quad (10)$$

By operationalizing this equation, the framework forces the agent to internalize the **Opportunity Cost** inherent in every market offer. Within the Pienza geofence, an action is deemed optimal if it maximizes the agent's long-term **Economic Inertia**, effectively prioritizing transitions that terminate in high-prestige hubs over myopic, high-upfront "Trap Offers" that lead to terminal economic sinks.

Multi-Agent Coordination: The Fleet Orchestrator

To move from a single-agent theory to a scalable operational model, the framework introduces the **Fleet Orchestrator**. This module simulates the coordinated deployment of an autonomous fleet originating from three strategic urban hubs: *Carso Antara*, *Santa Fe Centro Comercial*, and *Interlomas Magnocentro*. The orchestrator utilizes a **Depth-First Search (DFS)** algorithm to project optimized five-jump mission sequences for multiple concurrent units.

Demand Depletion and Swarm Neutralization

The primary challenge in fleet management is the **Swarm Effect**: a phenomenon where multiple independent agents cluster in a single high-value zone, leading to oversaturation and individual yield decay. To neutralize this, the framework implements a **Demand Depletion Mechanic**.

As each agent selects an optimal trajectory, the corresponding edges in the Q -Matrix undergo a **90% Value Attrition** ($Q_{u,v} \leftarrow Q_{u,v} \times 0.10$). This mathematical penalty simulates the "consumption" of market demand by the preceding vehicle. By treating the city's liquidity as a finite resource, the orchestrator forces subsequent agents to seek spatially diverse "second-best" paths that are structurally more profitable for the collective fleet. This transformation shifts the objective from individual greed to **Distributed Network Capture**, ensuring maximum coverage of the mobility mesh while maintaining systemic stability.

Tactical Deployment: The Five-Jump Mission Manifold

The simulation concludes with the generation of optimized flight plans for nine concurrent autonomous units, deployed from three primary logistical warehouses. By enforcing the **Demand Depletion** mechanic, the orchestrator prevents unit clustering and ensures maximum spatial coverage. Table 38 presents the resulting mission sequences, where each jump represents a transition designed to maximize the collective Q -value of the fleet.

Warehouse Origin	ID	Jump 1	Jump 2	Jump 3	Jump 4	Jump 5	Score
Carso Antara Miyana	AV1	Tamarindos	Santa Fe Centro Comercial	Tamarindos	Santa Fe Centro Comercial	Tamarindos	27,304
	AV2	Irrigación	Tamarindos	Cruce Echanove	Anzures	Tamarindos	11,447
	AV3	Polanco Uber HQ	Tamarindos	Vistahermosa	Interlomas Haciendas	Tamarindos	11,399
Santa Fe Centro Comercial	AV4	Sedena	Vialidad de la Barranca	Tamarindos	Jesús del Monte	Tamarindos	11,817
	AV5	Reforma Social	Interlomas Magnocentro	Tamarindos	Interlomas Haciendas	Santa Fe Centro Comercial	11,513
	AV6	Fuentes Casino	Vialidad de la Barranca	Lomas Virreyes	Tamarindos	Carretera Libre	11,481
Interlomas Magnocentro	AV7	Ave Club de Golf Lomas	Tamarindos	Lagos	Roma Condesa 1	Tamarindos	11,443
	AV8	Santa Fe Quintana	Tamarindos	Carretera al Olivo	Jesús del Monte	Santa Fe Centro Comercial	10,972
	AV9	Anzures	Irrigación	Santa Fe Patio	Tamarindos	Lomas Virreyes	10,668

Table 38: Autonomous Fleet Deployment: Coordinated Mission Sequences with Full Nomenclatures.

The deployment data confirms the structural dominance of the **Economic Arteries** previously identified in the edge centrality audit. Notably, while the first unit (*AV1*) captures the high-prestige circularity between *Tamarindos* and *Santa Fe Centro Comercial*, the depletion of demand in those edges forces the remaining fleet to branch into auxiliary hubs.

By the third unit of each warehouse, the orchestrator successfully pushes the fleet toward **Network Buffers** such as *Anzures* and *Lomas Virreyes*. This demonstrates that the Pienza framework does not merely identify "the best place to be," but rather identifies a **Systemic Distribution Strategy**. For an autonomous fleet, this transition from individual optimization to coordinated network capture represents the definitive shift from reactive labor to industrial-grade logistics, ensuring that the geofence remains saturated with value while minimizing internal competition.

The **Generative Moonshot** represents the final structural evolution of Project Pienza, transitioning from the observation of individual human heuristics to the orchestration of a scalable economic engine. By synthesizing generative neural manifolds with the steady-state rationality of the *Bellman Equation*, the framework effectively maps the "Economic Limit" of the urban environment. The resulting tactical flight plans prove that the city is not merely a physical grid to be traversed, but a **Tensor of Potentials** where value can be systematically captured through coordinated agent distribution and demand-aware positioning.

This functional scaffolding marks the boundary of the current inquiry, delivering the mathematical architecture required for the next generation of autonomous intelligence. By defining the state space ($\mathcal{S}$), transition probabilities ($\mathcal{P}$), and reward manifolds ($\mathcal{R}$) within a strictly closed operational domain, the framework provides the necessary baseline for transition into **Pienza 2.0: The Knowledge**. In this future iteration, the static scaffolding established here will serve as the training environment for high-frequency Reinforcement Learning (RL) and complex *Markov Chain Monte Carlo* (MCMC) simulations, moving beyond the steady-state and into the dynamic, real-time optimization of autonomous urban economies.

¿Será Posible el Sur?

The nine-month developmental arc of **Project Pienza** represents more than a technical exercise in data science; it is a longitudinal transformation of tactical fieldwork into a formal system of **Market Capture Architecture**. What began as a raw field observation of the 0.84 payout spread in the streets of Mexico City has evolved into a scale-invariant Digital Twin capable of orchestrating autonomous fleet deployment through high-fidelity generative simulation.

The conclusion of this research establishes the **Functional Scaffolding** for the next generation of autonomous urban logistics. Project Pienza has successfully mapped the **Steady-State Equilibrium** of the Mexico City marketplace, proving that the city is not a grid of streets, but a **Tensor of Potentials**.

While this paper marks the current boundary of modeled expertise, it provides the mathematical ciphers required for **Pienza 2.0: The Knowledge**. By defining the state space, reward manifolds, and transition probabilities within a closed sovereign geofence, the framework is now prepared for high-frequency Reinforcement Learning (RL) and full-scale MCMC simulations.

Ultimately, this project proves that in the Era of Intelligence, the expert agent is no longer the one who merely knows the roads, but the one who aligns with the **Economic Eigenvectors** of the city. Project Pienza concludes that the “Expert Rhythm” is not a subjective feeling, but a quantifiable calculation—one that is now, for the first time, fully documented, simulated, and optimized.

As the author reflects on the thousands of kilometers logged and rides to completed to produce this document, the technical rigor of these pages serves as a preliminary answer to a more evocative inquiry. In the words of Mercedes Sosa, *¿Será posible el sur?*—is the South possible?

In the context of this journey, the “South” represents the author’s utopia: the transition from the tactical execution of fieldwork on the asphalt to the formal domain of institutional science. For a rational agent operating within the information asymmetry of the gig economy, the discovery of the tensor was the first step in reclaiming the value of the decision. This work stands as the definitive bridge toward that destination; a testament that through the synthesis of field experience and the exoskeleton of AI, the transition from navigating the machine to architecting its logic is a possible and solvable endeavor.

References

- [1] Anil Ananthaswamy. *Why Machines Learn: The Elegant Math behind Modern AI*. Discussing the convergence of Hopfield networks and energy landscapes. Dutton, 2024.